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Project Document for nationally implemented projects
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Project title: Promoting better access to modern energy services through sustainable mini-grids and low-carbon bioenergy technologies among Guinea-Bissau's forest-dependent communities.		
Country: Guinea Bissau	**Implementing Partner:** Ministry of Environment and Biodiversity	**Execution Modality:** National Implementation Modality (Full NIM)
UNDAF/Country Programme Outcome: Outcome 2.2 outcome 4: "National institutions implement appropriate gender-sensitive policies and strategies to promote sustainable economic development" and outcome 5: "the most vulnerable populations in the poorest regions have greater access to economic opportunities".		
UNDP Social and Environmental Screening Category: **Moderate Risk**	**UNDP Gender Marker:** GEN2 *(It is designed to contribute significantly to gender equality. The different needs of women/girls and men/boys have been analysed and integrated well in the activities and outcomes*	
Quantum Award ID: 1185178 **Quantum TRAC award:** 00130599.1	**Quantum Project ID: 00116217**	
UNDP PIMS ID number: 5885	**GEF ID number: 9561**	
LPAC meeting date August 13, 2020		
Last Possible date to submit to GEF: November 30, 2018		
Last Possible CEO Endorsement date: June 28, 2020		
Project duration in Months: 60		
Planned start date: December 15, 2023	**Planned completion date: December 14, 2028**	
Excepted date of Mid-Term Review (MTR) Submission to GEF: June 15, 2026	**Expected date of Terminal evaluation (TE) Submission to: June 15, 2028**	
Expected Operational Closure Date: 14 September 2029	**Expected Financial Closure Date: 14 March 2030**	
Brief project description: The project aims to promote investment and sustainable business models in both solar mini-grids and low-carbon bioenergy technologies. The project will strengthen and further build up an enabling		

political, institutional and administrative environment for the management of solar mini-grids; particularly in rural forested areas. It will also put in place an enabling environment to secure supply and stimulate demand for energy-efficient stoves and firewood/charcoal is put in place. By providing sustainable energy solutions, the project will improve the livelihoods of marginalized forest communities living in isolated areas.		
FINANCING PLAN		
GEF Trust Fund	USD 2,912,702	
UNDP TRAC resources	USD 500,000	
(1) Total Budget administered by UNDP	**USD 3,412,702**	
PARALLEL CO-FINANCING		
Government	USD 10,000,000	
Private sector	USD 4,000,000	
(2) Total co-financing	**USD 14,000,000**	
(3) Grand-Total Project Financing (1)+(2)	**USD 17,412,702**	
SIGNATURES		
Signature:	**Agreed by Government**	**Date/Month/Year:** *within 6 months of GEF CEO endorsement*
Signature:	**Agreed by Implementing Partner**	**Date/Month/Year:** *within 6 months of GEF CEO endorsement*
Signature	**Agreed by UNDP**	**Date/Month/Year:** *within 6 months of GEF CEO endorsement*

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II. DEVELOPMENT CHALLENGE

2.1. Global environmental challenges and their root causes

Poverty in modern energy and pollution from extensive biomass use, mainly for cooking and heating, affect men and women, boys and girls, the old and marginalized populations in rural and remote areas all over the developing world. Indoor pollution from firewood and charcoal counts for in numerous deaths per year, mainly among women and children.

Ongoing deforestation for commercial purpose and the cutting of firewood for cooking and charcoal production are one of the hugest environmental challenges, also in Guinea-Bissau, where 95% of energy production builds on biomass and the coastal mangrove-based ecosystem is threatened to diminish dramatically due to population's energy needs. Lack of modern infrastructures (education, health, good rural roads and accessible rural markets) and lack of electricity for processing of agricultural goods hinder economic progress and further aggravate dependence on biomass and the poverty cycle.

It is the time to improve the energy access of impoverished rural communities with sustainable clean (renewable) energy where beneficiaries are recognized as men and women with different energy needs and priorities. Access to modern energy is a key enabler for women's empowerment (SDG 5). As rural women and girls are predominantly responsible for the bulk of household work, access to modern clean energy makes a significant difference to their health and well-being.

Although renewable energy and improved use of biomass by itself do not necessarily guarantee gender equality, it has a catalytic effect as it would go a long way on this path through providing women and girls with time and better technologies for income-generating opportunities and education (see UNDP Energy 2016, Seney Habtezion).

2.2. Capacity barriers

Guinea-Bissau is a West African country and one of the poorest countries in the world. According to the 2015 UNDP Human Development Report (2017), Guinea-Bissau ranks 178th on the Human Development Index out of 188 countries assessed. Guinea-Bissau is classified as Small Island Developing States (SIDS) and Least Developed Countries (LDCs). The country has experienced several civil wars and political instability.

The country's area is approximately 36,125 km2 and its total population is estimated at 1.7 million (2014). Part of the country is an archipelago, with about 88 islands and islets in the Atlantic Ocean. About 25% of the population lives in urban areas compared to 75% in rural areas.

Status of the energy sector

Guinea-Bissau has one of the lowest electrification rates and the highest electricity service costs in Africa. The country is totally dependent on petroleum products despite its high energy potential, particularly with regard to hydropower. Energy consumption in Guinea-Bissau is about 0.3 toe per person per year, one of the lowest in the world. National final energy consumption is characterized by the predominance of traditional biomass use (up to 87.8%), followed by 11.7% of petroleum products and only 0.5% of electricity. Wood is the dominant fuel, with demand exceeding 500 000 tons per year,

followed by charcoal, which is the most used fuel in the capital, Bissau. The amount of biomass used is estimated at about 738 000 tons per year.

The situation in the electricity sector is characterized by a structurally deficient service, both in quantity and quality, due to the obsolescence of production, the intermittency of electricity production and the high dependence imported oil in a context of limited financial resources. Only a small proportion of the population has access to electricity: the national electrification rate is estimated at 11.5%, although this average hides huge disparities between Bissau (29%) and the other major cities of the country (with an average of only 4%) electrification) and rural areas with an electrification rate of less than 1%. With 75% of the population living in rural areas, rural electrification is almost non-existent in Guinea-Bissau. Due to almost continuous political instability, the available power in Guinea-Bissau has fallen by more than 80% in recent years (from 13 MW in 2003 to 2 MW in 2013). Today, the available power is between 2 and 8 MW while the demand is estimated at 30 MW and increasing.

Guinea-Bissau has a national electric utility, EAGB, created in 1983 and wholly owned by the State (public enterprise). The Public Electricity and Water Company of Guinea Bissau (Empresa Publica de Electricidade e Aguas da Guine Bissau) is responsible for the production, transmission and distribution of electricity. Its electricity production cost is US $ 0.51 / kWh while the average tariff applied is US $ 0.34 / kWh (subsidized in part).

The 75% of the population living in rural areas depend directly on natural resources to live. Forests cover 73% of the total land area; especially in the south and south-east. The mainland includes the forest belt, a region characterized by dense forests and open forests, closely linked to woodland. Increasingly unsustainable clearing for shifting cultivation, use of biomass energy, illegal logging, expansion of cashew nut plantations and mining have all been cited as contributing factors to deforestation and forest degradation. Guinea-Bissau currently has only one protected forest and some remaining indigenous forest areas. Between 1990 and 2005, Guinea-Bissau lost 6.5 percent of its forest cover, or about 144,000 hectares. Recently, since 2007, the rate of deforestation is estimated at 1% per year. The country faces the interrelated challenges of access to energy, energy security and climate change mitigation and adaptation to climate change simultaneously. The chronic energy crisis hampers the social, economic and industrial development of Guinea-Bissau. The need for sustainable, reliable and affordable energy services is important at all levels. The lack of access to affordable and reliable energy services is linked to a variety of economic, social, environmental and political problems in the country.

The efficiency of using traditional stoves through the "three stone stoves" and "metal braziers" is very low, between 5 and 7% and between 12 and 15% respectively. In addition, traditional kilns for charcoal production have a very low efficiency (33%). Thus, the biomass sector in Guinea-Bissau could improve significantly in terms of overall efficiency through increased efficiency in the production and consumption of biomass fuels.

Potential of renewable energy in the country

The country has a range of renewable energy sources, with different development potential. Biomass, hydroelectricity, solar and wind are available.

Biomass: Guinea Bissau has forest resources (estimated at about 2 million hectares) and agricultural

residues (rice husks, cotton stalks, cashew nuts, sugar cane residues, etc.). The amount of biomass used is estimated at about 738 000 tonnes per year. But biomass is not currently exploited in a sustainable or renewable way.

Hydropower: The hydroelectric potential of Guinea-Bissau is estimated at 33 MW. This potential would be achieved with the implementation of two hydroelectric dams, Cusselinta (13 MW) and Saltinho (14 MW), and several small / micro hydro systems, according to studies conducted in the 1980s.

Solar: Solar energy is the most abundant source of renewable energy in the country. Annual irradiation ranges from 4.79 kWh / m^2 / day to 6.96 kWh / m^2 / day, with an annual average of 5.87 kWh / m^2 / day. So far, the solar resource has been used to a limited extent for the production of electricity with photovoltaic (PV) systems in rural households, schools, offices, hospitals and health centers; for solar pumping of water; for solar telecommunication systems; and for specific applications of solar water heater. But these were essentially donor-driven and unsustainable initiatives.

Wind: There are no appropriate studies on the wind potential of Guinea-Bissau. Roughly, it is estimated that the wind resources are of average level, with a wind speed (measured at 50 meters) higher than 4 m / s. Wind energy can be exploited throughout the year on the mainland coast or between the islands and in the Boe region (east), where the altitude exceeds 200 m. There is no current exploitation of the wind resource.

Main barriers to the development of sustainable mini-grids and low-carbon bioenergy technologies

Legal, regulatory and institutional framework: The current legal framework constitutes an obstacle to the development of renewable energies in Guinea-Bissau. So far, there are no regulations, incentives or legislative framework conditions to support the implementation of renewable energies in Guinea-Bissau. The government plans to create regulations allowing private operators to become involved in rural electrification, thus offering opportunities for the use of renewable energies. There are also plans, programs and strategies to increase the use of renewable energy, which is a very important "first step" in Guinea-Bissau for a more sustainable development scenario in the future.
Stakeholders (government, institutions, and national agencies) do not have sufficient capacity to formulate and implement policy and regulatory frameworks on biomass resources in an integrated manner, especially the use of firewood, charcoal and improved stoves that are sustainable, improved and more efficient.

Technological supply chain: The technological supply chain for renewable energies in Guinea-Bissau is at a very nascent stage. There are some local SMEs capable of assembling simple RE installations based on imported machines and turbines, but they lack the technical and technical capabilities to ensure optimal system design, installation and maintenance. In rural areas, there is only very limited local technical expertise in how to properly manage, operate, and manage RE-based mini-grids. Low quality and low numbers of skilled and skilled workers in the electricity sector add additional risks and increase the cost of operating mini-grids due to the need for costly imported services, even for repairs and basic maintenance. The technology supply chain for the production and use of bioenergy is very basic and lacks coordination and optimization among farmers, artisans, retailers and end-users.

Sustainable O&M Model: Lack of experience and demonstration in the sustainable operation, maintenance and management of RE-based mini-grids is a major bottleneck and the reason for the

failure of donor-funded projects. The barrier is aggravated by the fact that Guinea-Bissau is a post-conflict society and that political instability is still sporadic. Technical and management capacity is extremely weak at the local level, especially in provincial and rural areas. The same problem exists with local businesses: the ranks of experienced managers and skilled technicians, already rare in provincial and rural areas, have declined further due to instability.
The main missing aspects of a sustainable operation and maintenance model should be: (i) technical supervision of plant operations and responsibility for the repair of defective equipment such as cracked solar panels; (ii) an efficient and effective rate structure that covers both start-up costs and operating and maintenance costs; (iii) a strong and efficient system of financial management, billing and payment collection; (iv) community mobilization, customer relations and dispute resolution procedures (e.g. in case of illegal connections or theft), the engagement of productive end-users, etc. O&M is less of a problem for bioenergy technologies, but still needs to be put in place for kilns, stoves and alternative solutions such as biogas plants.

Access to capital: Significant initial investment needs remain an obstacle to the implementation of many projects. RE projects are capital-intensive and require large investments that generally exceed the capacity of local businesses or communities. In addition, the local banking sector is not sufficiently capitalized to facilitate the financing of RE projects with a longer return on investment and substantial risks.

Investor Awareness and Risk Perception: Information on the potential and benefits of renewable energies (especially small hydropower plants) for the development of provincial and rural electrification is rare in Guinea-Bissau. There is very little data on potential sites and their hydrological, climatic and other characteristics. Even when such studies exist, they are often not accessible to the public. Basically, there is no single point of information where a potential developer can receive the advice and data required to make an informed investment decision. The government is not able to gather these guidelines / data due to limited budgetary resources, staff capacity, lack of prior experience and a global vision of promoting RE-based mini-grids and private sector investment. While the national energy strategy recognizes the importance of developing renewable energy in the fight against energy deficits in secondary cities of Guinea Bissau, the government has so far focused on addressing the energy deficit of the capital Bissau and to facilitate hydroelectric and photovoltaic projects with public financing and IFI. Promoting solar PV mini-grids requires a different approach, more focused on the private sector and local communities, and requiring open and transparent access to information for investors. The scarcity of successful and sustainable RE projects limits the opportunities for awareness and trust of local communities, project promoters and investors, and is in itself a major obstacle to market development.

Nascent private sector interest: with political instability, there are limited private investors and all projects are donor driven and largely unsustainable. Private sector investors consider Guinea-Bissau a high-risk environment to invest in renewable energy projects. In the 2018 Doing Business data of the World Bank and the IFC, Guinea-Bissau ranks 175th out of 190 countries.

Sensitivity to information / culture: There is still a lack of knowledge and negative perceptions (lack of social or cultural acceptance) of the use of new technologies for cooking.

2.3. Consistency with National priorities (national strategies, plans and reports)

In latest years, Guinea-Bissau has adopted various policy and management tools for different sectors where development priorities are set up. Such priorities are likely completed and reinforced through actions defined not only at sub-regional and regional level but also at international level.

In its Document of Strategy for Poverty Reduction II (2011-2015), the Government of Guinea-Bissau aware of importance of energy in promoting economic growth (*Axis 3: Promoting inclusive and durable economic development – support to growth carrier sectors*). As support to the energy sector, DENARP II aimed at accelerating the implementation of electrification projects in certain interior regions (rural areas) in view to attract more investment in sectors of transformation of agricultural products and tourism. A target of 100 MW increase in national electricity production capacity was set up as concrete action for the achievement of that objective.

Regarding environment sector, DENARP II concern goes in integrating climate change in planning policy and investment in sectors that are growth carrier in order to make them more climate resilient as taking into account climate risk would allow development of agricultural zones, housing, tourism and coastal zones, etc. Mainstreaming climate change would include also aspects related to reduction GHG emissions, allowing also for prioritizing energy efficiency and renewable energies in all political decisions and investments. Development strategies of different sectors of growth should be transformed and aligned with the low-carbon emissions development and to be climate resilient to promote sustainable development.

Although this target could not be reached until end of 2016, DENARP II influenced afterward the 2015-2030 Strategic Operational Plan "Terra Ranka" (SOP 2015-2030) which targeted an increase in power production for a capacity up to 90 MW. This SOP considered energy sector as building bloc and component of Catalytic Sectors which is one of the three pillars of a house named "Guinea-Bissau". For the achievement of the projected electricity production capacity, "Terra Ranka" calls for the development and the utilization of local natural resources, particularly renewables sources.

Guinea-Bissau also produced it National Determined Contribution (NDC) in 2015 in the track of Paris Agreement where energy and AFOLU sectors are taken as top priority for intervention in term of policies, strategies, plans and direct action, encompassing main objectives of National Environment Management Plan.

Guinea-Bissau is also Party to the "Rio de Janeiro" Generation Conventions: UNFCCC (ratified in 27/10/1999), the UNCBD (ratified in 27/10/1995) and the UNCCD (ratified in 27/10/1995). This country has also ratified the Protocol of Kyoto, ratified in 18/11/2005. In the framework of UNFCCC implementation, the country has produced its Initial/First (INC), Second (SNC) and Third National Communications (TNC), which GHG Inventories have produced, respectively, following figures of emissions:

INC, 2004 (Ref. Year: 1994) ----------1, 359, 88 Gg CO_2 (Energy sector emissions: 13% of total)

SNC, 2008 (Ref. Year: 2000) ---------- 3,780.81 Gg CO_2 (Energy sector emissions: 4% of total)

TNC, 2018 (Ref. Year: 2015) ---------- 6,899.36 Gg CO_2 (Energy sector emissions: 63%)

Currently, the execution of project of the first biannual update report (BUR 1) is underway and the preparation of the Fourth National Communication of UNFCCC implementation will start shortly.

At sub-regional and regional level, Guinea-Bissau has made adopted all major priorities set up for sustainable development in the West African Economic and Monetary Union (WAEMU/UEMOA), such as the Common Policy for the Improvement of Environment in WAEMU region, and the ECOWAS White Paper on Energy.

At the institutional level, it can be noted that in the energy sector of Guinea-Bissau there are several institutions and entities that protect the sector. Although responsibilities are defined theoretically by Decree-Laws No. 2/2007 and No. 3/2007, which mainly determine the role of the MEI and the DGE, it is still not always clear which direction and at what level is defined strategy for the implementation and follow-up of energy policies. This gap is not even more evident in the regions of the interior of the country, where ERDs do not have the same information as general management in Bissau, nor are they always aware of national strategies.

In addition, coordination between the above entities and other government entities that have an indirect role in the sector, such as the Ministries Energy and Natural Resources, Environment and Sustainable Development, Finance and Economy, Public Works, Construction and Urban Development, Direction General of Industry, and INITA is quite weak. In the course of the research and interviews, besides the actors directly involved in the sector, few actors are aware of the policies, although they are impacted or involved in them. Therefore, better articulation and coordination between government entities, both at MEI level and among Ministries, is recommended. The clear definition of the responsibilities of each entity is also important to ensure the follow-up of the policies by the actors.

According to PANER, the PANEE and the SE4ALL documents, to ensure the monitoring of policies and targets for renewable energy, energy efficiency and access to energy, plans foresee the creation of a Rural Electrification Agency. The creation of this agency can contribute to the monitoring and evaluation of plans, but a clear definition of its responsibilities and its perimeter, as well as its articulation with existing entities is imperative.

The creation of the new regulatory body can also strengthen coordination and supervision of the sector. Its responsibilities are defined by Decree-Law No. 3/2007, but have not yet been published and the actors in the sector are not involved in its creation. It would be expected to set up working groups to consult and involve the actors who know the energy sector. In addition, it is recommended to disassociate electricity and water services, with weak links between entities and the experience of the EAGB and several projects shows the difficulty of managing the two sectors in parallel.

Similar to the low level of knowledge of energy policies and targets among government entities, the other actors in the energy sector interviewed, particularly from the private sector, are not familiar with these plans. Although the plans are available online and have been presented at several conferences, the information collected demonstrates that communication and dissemination actions are necessary. Rather than re-circulating the documents, these actions can take the form of simpler and shorter communication actions, through pamphlets, newsletters, press articles or explanatory videos that will summarize the main points.

At the level of the legal framework, one of the main barriers is the unclear procedures for the implementation of a renewable energy project in Guinea-Bissau. In the course of the research it was identified that each project implemented followed a different process, thus creating different conditions for each entity proposing and executing, and to a certain extent conditions of unfair competition in the market. The clarification of the process of implementing a project in the energy sector, determining each step in a clear and transparent way for all actors, is imperative. The environmental licensing process that is clearly determined by Law 10 of 24/2010 and supplemented by an AAAC guide is an example to follow in the energy sector.

In parallel, the creation of the new electricity supply service concessions regime will clarify the procedures and can guarantee transparency in the sector. At present there is no legal framework and each actor (promoter/manager of energy projects, independent and autonomous producers) operates in a different way. The establishment of a fair and equitable system of concessions for all, which will clearly determine the obligations and responsibilities of operators in an appropriate manner in the national context, will facilitate the functioning of the sector both in terms of supervision and in the implementation of new projects.

Regarding tariff regimes, gaps have been identified in the definition of tariffs applied by operators, which are not harmonized, do not reflect operating costs and do not correspond to the socio-economic reality of the population. In the Bissau area, tariffs are subsidized and do not allow for the sustainability of the plants, while in the interior of the country the highest tariffs are objects of complaints by the population that cannot access the service. Therefore, it is crucial to implement a socio-economic approach in the definition of tariffs that have to be ratified prior to the implementation of the projects and with fair criteria by the authorities to allow access to the service and the sustainability of the systems.

At the level of investment policies, the procedures are not known and clear to the actors, and thus constitute a barrier for investors. According to the interviews, the processes in practice are slow and not very fluid, requiring a long time to obtain the exemptions and the expected benefits. Facilitation of procedures and ratification of the proposed Renewable Energy Act (No 2.013-21), which provides for exemptions for sub-sector projects, can help boost the sector and attract more investment.

Below is a *resumé* of main gaps in off-grid policy and regulatory framework.

Table 1 : Gaps in the Off-Grid Policy and Regulatory Framework

Indicator	Policy/Regulatory/Market Gaps	Recommended Intervention
Specific National Policies, Laws and Programs	**Lack of National Electricity / Electrification Policy** No policy exists for rural electrification Main focus of policy is on national grid extension only Government is subsidizing fossil fuel electricity production	Support Government to establish a clear Rural Electrification Policy which encourages least cost, integrated planning for all options Support Government to develop a comprehensive, fully integrated electrification plan with geo-spatial modelling on least cost planning to consider where extension is the most efficient and sustainable approach to increasing energy access vs. development of the off-grid sector – mini-grids and stand-alone systems powered by local renewable resources Support Government to analyze where fossil fuel subsidies serve as an impediment to development of safe, clean energy access alternatives
	Lack of Integrated National Electrification Plan No integrated plan exists Insufficient focus on or understanding of framework to support private sector participation	Support Government to develop a comprehensive, least cost, integrated plan for all rural electrification options Support Government to improve the planning framework to encourage private participation in mini-grid and stand-alone solar system options
	Lack of Energy and Electricity Law No specific Energy or Electricity Law with off-grid provisions exists	Help Government develop new legal framework that is flexible and helps create appropriate incentives for private sector participation in off-grid market development (e.g. to initiate process of unbundling / electricity market liberalization)
	Lack of national policies, laws, programs and/or action plans targeting off-grid market development No specific Off-Grid Policy, Law, or Action Plan in place No Lead Agency	Help Government establish the medium-long term rural electrification strategy in the country through development and implementation of a rural electrification Master Plan

Indicator	Policy/Regulatory/Market Gaps	Recommended Intervention
	Insufficient focus on or understanding of framework to support private sector participation	Help Government establish a lead agency / rural electrification agency that has a clear mandate to coordinate activities with the private sector, donor community and at national and local level in order to accelerate off-grid market growth to achieve energy access objectives Help Government improve off-grid framework to create appropriate incentives for private sector participation
Financial Incentives (waivers on import duty, taxes, etc.)	**Insufficiently supportive financial incentives / tax regime**	Help Government develop appropriate VAT and tariff policies covering the entire off-grid / stand-alone solar product supply chain (including batteries, inverters or other system components) that would provide necessary support to the industry[1] Help Government establish a Special Task Force to mitigate potential difficulties in customs clearance and import logistics Help Government introduce appropriate grant and subsidy schemes which require private funding matches and are predictable and not overly bureaucratic Help Government create PPP schemes to share high project development and market entry costs particularly with developers in remote areas Help Government analyze where subsidies or exemptions for non-renewable energy sources provide unfair advantage for fossil-fuels and impede development of clean energy solutions
Standards and Quality	**Unclear / lack of quality standards**	Help Government establish international quality standards for off-grid stand-alone solar products, including minimum technical standards (IEC Technical Specifications), warranties, required availability of and cost guidelines for post-sale services/O&M, etc.

[1] The GoGB currently has incentives in place for solar panels only, but not for other components of a solar kit / SHS

Indicator	Policy/Regulatory/Market Gaps	Recommended Intervention
		Help Government integrate standards with appropriate oversight agencies to ensure quality-verification procedures are in place Help Government implement a legal framework that enables companies or public authorities to prosecute those caught distributing counterfeit or poor-quality products that are not up to promulgated standards
	Lack of capacity of local technical sector (solar PV technicians, installers, services providers etc.)	Support establishment of technical certification and vocational training programs through government, private sector, and/or academia for installation and maintenance of stand-alone solar systems[2] Support development of database of best practices / information sharing services to ensure skills transfer from international, local and regional initiatives (e.g. through MEIRN)
	Insufficient attention of private companies to environmental/social standards and community engagement	Assist private sector and/or civil society organizations to ensure environmental/social standards are in place Assist in development of strategies encouraging inclusive gender participation Support with the implementation of a repair and recycling framework for off-grid solar systems and equipment
	Insufficient public awareness	Support Government, trade associations and civic society organizations to develop and implement consumer awareness/marketing/education programs on the benefits of off-grid solar products and the existence of related national programs Support development and implementation of programs to educate consumers, retailers and distributors on the benefits of quality certified solar products (vs. 'non-standard' products)

[2] Acção para o Desenvolvimento is a local NGO that runs a technical school that provides solar PV training

Indicator	Policy/Regulatory/Market Gaps	Recommended Intervention
Concession Contracts and Schemes	**Lack of clear and transparent licensing and permitting procedures** Unclear procedures Insufficient communication and streamlining	Help Government develop clear licensing and permitting procedures Help Government develop improved systems for sharing and disseminating information to project developers and key stakeholders, including establishment of a "one-stop-shop" for national level permits and approvals and expediting of local permits
	Lack of experience/understanding of emerging concession and energy services schemes for off-grid providers Need for understanding of different SHS concession schemes Need for understanding of emerging models for 'Integrated Private Utilities' or 'Energy Companies of the Future' Public procurement or public finance/budget laws that hamper deployment of energy services models for public facilities Lack of standardized contracts for energy services provided by private system operators to public facilities Insufficient protection for stranded investments	Help Government to understand and develop approaches to facilitate pilots of 'Integrated Private Utility' or 'Energy Company of the Future' schemes.[3] Help Government develop procurement and public finance laws that will facilitate stand-alone solar system investment for public facilities (schools, health care facilities, etc.) Help Government develop proper procedures and guidelines to protect against stranded investments from competition among all on-grid and off-grid rural electrification approaches[4]

[3] Innovative models are emerging for entire geographic areas to be concessioned to integrated private energy services operators who may offer an appropriate mix of solutions within their franchised area (i.e. a mix of SHS, rooftop solar, specialized systems for productive use, mini-grids and micro-grids). This is being piloted by the Shell Foundation in several countries.

[4] As the off-grid sector becomes populated by a variety of different approaches, all private operators are subject to potential stranded investments "when the grid arrives" and even SHS providers can have their assets and revenues threatened when the mini-grid arrives.

Indicator	Policy/Regulatory/Market Gaps	Recommended Intervention
Business Model Regulation	Lack of understanding about different pricing schemes and business models offered by stand-alone solar system developers	Support Government and off-grid enterprises to collaborate specifically on developing pricing schemes for productive use market segment[5] Support off-grid entrepreneurs and telecom companies in building the capacity of and fostering linkages between telecom companies / mobile money providers and off-grid solar companies to help roll out technology platforms and PAYG business models

2.4. Baseline scenario and associated baseline projects

The Government of Guinea recognizes that lack of access to energy in provincial and rural areas is a major detrimental factor to the country's economic development, social stability and environmental sustainability. To solve the problem, the government intends to create a rural electrification agency, a national agency for rural electrification, under the auspices of the Ministry of Environment and Biodiversity. It is also planned to create another agency dedicated to the promotion of renewable energies. None of these agencies has materialized yet.

The government has also launched an ambitious program (Terra Ranka) to improve the country's energy infrastructure. This program includes major investment projects in power generation and transmission lines, including the construction of a 55 MW thermal power plant at Bôr (with an initial phase of 15 MW), financed by BOAD (West African Development Bank). On the renewable side, there are two hydroelectric dams, Cusselinta (13 MW) and Saltinho (14 MW). But that would require more than 200 million USD, which has not yet been obtained. There is also a project for a 10 MW photovoltaic solar plant in Gardette, which would require $ 30 million, which has not yet been exploited. All these plans are projects connected to the grid. Even if they materialize, they will be limited to the capital Bissau and the big cities and will certainly not serve rural areas where the main electricity grid is non-existent.

There are, however, some initiatives for off-grid electricity in rural areas. The most important, already running, is the Bambadinca project developed by TESE, an international NGO. It is a mini solar photovoltaic power plant of 312 kW. The mini-grid provides electricity to about 450 homes in Bambadinca. The overall project cost $ 2 million. The Bambadinca project is rich in lessons, from legal aspects to business model challenges, which will guide the design of the GEF-funded project.

In the area of clean cooking, many initiatives have been launched in the past, but these were mostly done by donors or pilot projects and did not result in a real transformation of the sector. Past initiatives

[5] The productive use segment is brand new with SHS providers, mini-grid operators and vendors specialized on a single type of SME or agricultural productive use (i.e. grain mills, water pumps, cocoa processing etc.) all grappling to arrive at attractive approaches to billing for energy services. This is a particular area where TA support is much needed to help all stakeholders sort out fair and practical approaches.

have included: (i) the rational use of forest resources - an FAO / EU funded project - to protect forests through the rational use of wood products, the dissemination of improved carbonization techniques, and kilns; and (ii) a biogas project funded by the Chinese government.

III. STRATEGY

3.1. Selected scenario

Guinea Bissau faces the interrelated challenges of access to energy, energy security and climate change mitigation and adaptation to climate change simultaneously. The chronic energy crisis hampers the social, economic and industrial development of Guinea-Bissau. The need for sustainable, reliable and affordable energy services is important at all levels. The lack of access to affordable and reliable energy services is linked to a variety of economic, social, environmental and political problems in the country. For example, the main goals set by SE4ALL AA by 2030 are aligned with PANER (National Renewable Energy Action Program) and PANEE (National Energy Efficiency Action Program), with a focus on the percentage of the population with access to electricity (at least 80%) and the percentage of the population with access to modern cooking options (at least 75%). The priority areas of action are:

- access to energy;
- renewable energies in the electricity sector;
- energy efficiency.

With this GEF-funded project, the additional 2 MW targeted will contribute about 6% of the country's renewable energy share by 2030. In addition, the project will create a supportive legal environment and market transformation that will contribute indirectly to the achievement of INDC's target capacity (Nationally Determined Contribution).

Institutions in charge of environment and energy issues.

In the field of the environment, it mainly has the Ministry of Environment and Sustainable Development / State Secretariat for the Environment. This ministry is responsible for the competent Environmental Assessment Authority (AAAC) and the Directorate General for the Environment (DGE), among others. The Ministry of Environment and Biodiversity, sometimes called the State Secretariat for the Environment (SSE), is responsible for environmental policy and the preservation of biodiversity. These institutions represent and respond to the main international agreements and conventions that bind the country in the field of environment and sustainable development. They therefore participate in the implementation, monitoring and evaluation of energy policy through environmental licensing and the management of environmental and social impact mitigation plans through the Environmental Protection Authority. They also collaborate in planning, financing energy and, thanks to their know-how in fiduciary management procedures for projects financed by the international community on various environmental issues such as the fight against climate change, preservation of biodiversity and soil degradation. Thus, MASD has been an important national technical partner in the design and implementation of projects (National Execution). In the context of the Global Environment Facility (GEF), MASD / SSA represents and serves as the GEF's focal point and operational focal point.

In the energy sector, the Ministry of Energy and Industry / Ministry of Energy, Industry and Natural Resources (MEI / MEIRN) is in charge of energy issues. It is structured as shown in Figure 1:

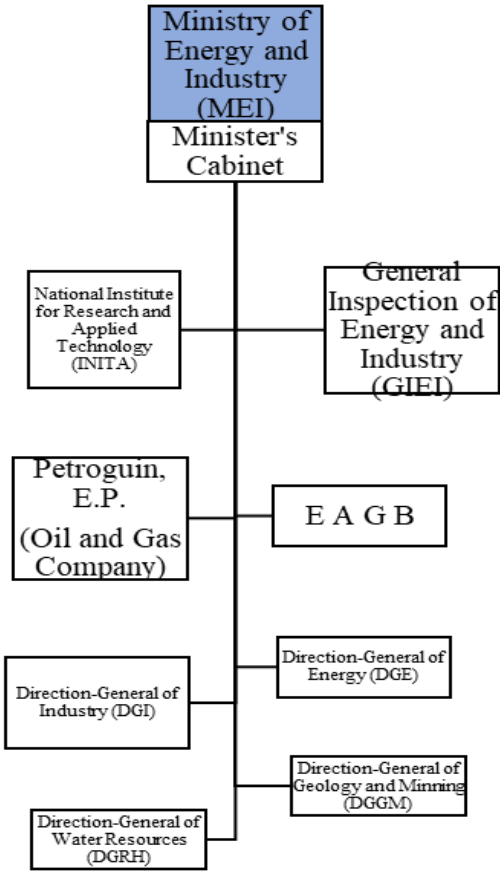


Figure 1 : Organization chart of the MEI / MEIRN (Source: MEIRN, DGE, 2018)

The CEO is responsible for implementing the national energy policies defined by the MEI / MEIRN, accompanying its execution and supervising all activities related to the production, distribution and sale of electricity and fuels. More specifically, according to Legislative Decree no. 2/2007, the Chief Electoral Officer must, in particular, ensure:

(a) monitor the maintenance of energy supplies, including fuel stocks;

(b) the development of programs for the protection of the environment;

(c) the proposal for programs for the efficient use of energy;

(d) the introduction and validation of energy standards and the regulation of activities in the energy sector, including technical, quality, environmental and personnel standards and regulations.

To ensure its responsibilities at the national level, the DGE is deployed at the regional level in the form of regional energy delegations (ERDs), which represent the DGE in different regions of the country. ERDs ensure the implementation of the national policy at the regional level and are supervised by the various directorates of the DGE according to the technologies existing in the perimeter of the region. However, it appears that, in practice, ERDs have focused almost exclusively on the context of conventional electricity, because of the isolated know-how accumulated over the years and the very low consumption of renewable energy. DGE is the entity responsible for coordinating and monitoring the implementation of PANER, PANEE and SEA4LL AA & IP, through its Renewable Energies and Rural Electrification Directorate. Both institutions act in synergy with other stakeholders. Nevertheless, a concrete

monitoring framework remains to be created at the time of the creation of the Rural Electrification Agency.

The EAGB has the character of a public company, having as sole shareholder the State of Guinea-Bissau and operating under the supervision of MEIRN. It has administrative and financial autonomy and, in accordance with its statutes, its corporate purpose and its attributions include operations throughout the country, directly or indirectly related to the establishment and operation of public production services, transmission, distribution and import and export of electricity in urban and rural areas, as well as the establishment and operation of public water collection, supply, storage and distribution urban areas. That is to say, the statutes give the EAGB the monopoly of the activities related to its objective. There is no independent regulator in the energy sector in Guinea-Bissau. Legislative Decree No. 3/2007 provides for the creation of a regulatory body for the electricity sub-sector with the following responsibilities:

a. the monitoring of the technical, economic and financial performance of undertakings by their specifications and regulations relating to the efficient use of energy and the protection of the environment;
b. the control of prices and tariffs;
c. Supervision of the implementation of State property by the institution responsible for the property and the enterprise or operating company;
d. Arbitration between operators and between operators and consumers, protection of consumers and promotion of fair competition.
e. Inspection of imported equipment for the purpose of issuing certificates of conformity.

The proposed project is in line with the following national plans and strategies:

- TERRA RANKA - Strategic and Operational Plan 2015-2020: Terra Ranka is an ambitious development plan, conceived after the 2014 parliamentary elections. It is based on several pillars, the following of which are relevant to this project: (I) sustainable ecosystem management (II); stimulate human capital (III); and build a strong private sector (IV).
- PANER (National Renewable Energy Action Plan) and PANEE (National Energy Efficiency Action Plan) for 2015-2030: both plans were developed as part of Sustainable Energy for All (SEE4ALL), with the support of the ECOWAS Regional Center for Renewable Energy and Energy Efficiency (ECREEE).
- Second National Communication to the UNFCCC (2011): The CNS highlights that the main sources of GHG emissions are land-use change and forestry (96%) and energy (4%). The report identifies four priority mitigation measures: (i) promotion of improved stoves and kilns; (ii) promotion of hydroelectricity; (iii) rural electrification based on solar photovoltaic energy; and (iv) large scale renewable energies connected to the grid. The first three measures are consistent with the proposed project.
- The Intended nationally Determined Contribution (INDC) 2015 of Guinea-Bissau provides for an increased target

The component (Political and Financial Instruments and Incentive Program for Renewable Energy Mini-Grids and Low-Carbon Bioenergy Technologies) envisages the preparation and adoption of a comprehensive policy framework for the promotion of rural electrification based on renewable energies. The framework will complement existing policies on energy sector development and rural electrification with a focus on decentralized renewable energy technologies and more favourable conditions. The policy framework will include specific timelines, targets and deployment plans for the development of mini-grids. The decentralized policy framework for REs will also establish a fundamental

policy instrument (e.g. a financially viable tariff for RE-based mini-grids) and related policies and regulations, including, but not limited to, regimes. Harmonized and simplified concession agreements, standardized PPPs and land and water rights for mini-grid projects. To support the implementation of the proposed policy framework, a capacity building program will be provided to relevant national agencies and directorates, potential private sector PPIs and community groups.

Establishing financially sustainable rates for the right price of energy is one of the most important factors in ensuring the sustainability of renewable energy mini-grids. Under Component 1, the project will help the Ministry of Energy to develop and introduce new regulations for decentralized renewable energy tariffs. It is proposed that the mini-grid pricing system be accompanied by a progressive toll regime, similar to the system in place for the grid. This will allow the rates to be better proportioned to the customer's ability to pay. As most planned mini-grids will be relatively small in terms of users, there will be natural constraints on the differentiation of rights levels. In addition, since most customers are poor, it is envisaged that productive uses (businesses) will bear a disproportionate burden of recovering electricity costs. The identification and inclusion of financially strong anchor clients will therefore be a central element of the mini-grid deployment planning as part of the "Installing RE-based mini-grids and deployment of bioenergy technologies".

IV. RESULTS AND PARTNERSHIPS

4.1 Expected Results:

This energy deficit situation prompted the Government of Guinea-Bissau to collaborate with UNDP to develop and submit to the GEF secretariat a project concept entitled "Promoting Better Access to Modern Energy Services through Solar mini-grids and low carbon bioenergy technologies amongst Guinea Bissau's forest communities". The project focuses on the specific obstacles related to investment in decentralized mini-grids and low-carbon bioenergy technologies.

The project objective is "To promote investment and sustainable business models in both solar mini-grids and low-carbon bioenergy technologies". The project is structured into 3 components which are:

- Component 1: Policy and financial instruments and incentive scheme for solar mini-grids and low-carbon bioenergy technologies;
- Component 2: Capacity building for mini-grid and low-carbon bioenergy;
- Component 3: Mini-grids and low-carbon bioenergy technologies roll-out.

Investment and business models are at the center of the project design. For mini-grids, the project's approach recognizes both Guinea-Bissau's challenging general private sector investment environment, as well as the specific early-stage and nascent status of solar mini-grids. As such it is envisaged that three main business models will be explored in this context:

- A utility-based business model, where the public utility, EAGB, will build, own and operate the solar mini-grids
- A PPP business model, for example where the capital costs of the mini-grid are donor-funded, and the ongoing O&M/servicing is performed by an NGO and/or private sector. This model was pursued for the successful Bambadinca baseline mini-grid investment.
- A private-sector business model, where private developers, either domestic or international, will build, own and operate solar mini-grids.

It is envisaged that a mix of these three business models can be explored, with the project and its activities flexible to accommodate these different approaches. Over time, as the market evolves, the intent will be to increasingly move to a full private sector business model. Varying degrees of targeted financial support may be required for each of these models. Financial support will be designed with the objective to be phased out over time, as the market evolves on a path to financial sustainability.

With regard to engineering specifications, the project will promote lowest cost approaches for system design and sizing. Innovations in productive use will be encouraged, to increase demand over time and improve the mini-grid's financial viability. Designs will include both solar-battery mini-grids, and solar-battery-diesel hybrid mini-grids (both green-field and brown-field sites)

- **Component 1: Policy and financial instruments and incentive scheme for solar mini-grids and low-carbon bioenergy technologies**

This component envisages the preparation and adoption of a comprehensive policy framework for the promotion of solar mini-grids and low carbon bioenergy.

Outcome 1: Institutional and financial sustainability of solar mini-grids and low-carbon bioenergy strengthened.

Output 1.1: Policy package to operate Solar mini-grids developed.

A simplified policy and an established and operational legal / regulatory framework for electricity generation by the private sector to provide mini-grids isolated by renewable energies is defined. The project will examine Decree-Law no. 2/2007 and no. 3/2007 which govern the energy sector and the NAPER "National Renewable Energy Action Plan" proposed to identify the problems that hinder the private sector and play a role in the decentralized generation of electricity from REs in the country. Subsequently, the project will develop a policy document outlining the necessary corrective measures, including the procedures / regulations accompanying the Electricity Framework Law, and propose a legal / regulatory framework that will encourage private sector investment in the development of mini-grids. The project will then seek government approval to operationalize this set of documents. Particular attention will be paid to the rapid implementation of all accompanying procedures (implementing texts) to ensure that the political and legal / regulatory framework does not suffer from delays in implementation.

This output will have a 3 layer of interventions: (i) on higher level intervention with the entire off-grid sector; (ii) on specific interventions explicitly on Mini-grid policies and regulation; and (iii) on tariff setting, and design of financial support.

(i) Higher level intervention with the entire off-grid sector:

Table 1 on Gaps in the off-grid policy and regulatory framework detailed the gaps (barriers) and provided some recommended interventions. As per the off-grid sector, the project will follow the recommendations by supporting the Government to:

- Establish a clear Rural Electrification Policy which encourages least cost, integrated planning for all options;

- Develop a medium-long term rural electrification strategy in the country through development and implementation of a rural electrification Master Plan. This includes establishing a lead agency / rural electrification agency that has a clear mandate to coordinate activities with the private sector, donor community and at national and local level in order to accelerate off-grid market growth to achieve energy access objectives;

- Develop a comprehensive, fully integrated electrification plan with geo-spatial modelling on least cost planning to consider where extension is the most efficient and sustainable approach

to increasing energy access vs. development of the off-grid sector – mini-grids and stand-alone systems powered by local renewable resources

- Analyze where fossil fuel subsidies serve as an impediment to development of safe, clean energy access alternatives
-

(ii) Specific interventions explicitly on Mini-grid policies and regulation

As per the solar mini-grid specificities, the project will follow the recommendations by supporting the Government to:

- improve the planning framework to encourage private participation in mini-grid systems;

- establish international quality standards for solar products, including minimum technical standards, warranties, required availability of and cost guidelines for post-sale services/O&M, etc.;

- Support with the implementation of a repair and recycling framework for solar systems and equipments;
-
- establish a Special Task Force to mitigate potential difficulties in customs clearance and import logistics

- develop improved systems for sharing and disseminating information to project developers and key stakeholders, including establishment of a “one-stop-shop” for national level permits and approvals and expediting of local permits.
-

(iii) Tariff setting, and design of financial support

As per the tariff setting, and design of financial support, the project will follow the recommendations from table 1 by supporting the Government to:

- Analyze and then propose design for a public financial viability mechanism which will aim to improve the financial viability of private sector models for mini-grids, and set them on a path to sustainable financial viability (this can be in the form of concessional finance, upfront capital subsidies, etc.).

- Develop appropriate VAT and tariff policies covering the solar mini-grid supply chain, noting that the project will work on putting in place the best criteria for tariff settings and not specifically setting the tariffs. Tariff flexibility and innovative approaches by the private sector on tariffs will be encouraged.

- introduce appropriate grant and subsidy schemes which require private funding matches and are predictable and not overly bureaucratic.

- Support mini-grid entrepreneurs and telecom companies in building the capacity of and fostering linkages between telecom companies / mobile money providers and solar companies to help roll-out technology platforms and PAYG business models. In addition, the project will

explore financing models like “Pay as you save” which enable users to take out loans, the payback of which, will be less than payments made for business-as-usual energy solutions (in this case diesel-powered generators).

Output 1.2: Financial incentives and market mechanisms to secure supply and stimulate demand for energy-efficient stoves and firewood/charcoal developed

Incentives and market mechanisms will be designed to support private investment (craftsmen) in high-efficiency stoves and kilns, which may include the subsidy of equipment used in the realization of kilns and improved stoves. In addition, the project will collaborate with loan guarantee funds and commercial banks to facilitate promoters' access to financing, provide advice when formulating the business plan and loan applications, and identify other sources of financing for promoters.

There are essentially 5 steps in the charcoal value chain (Figure below). In the light of discussions with key stakeholders in Guinea-Bissau, the project proposes a series of integrated interventions focusing mainly on stages 2 (charcoal production) and 5 (charcoal consumption) in the chain of production. This also applies to the production and use of fuelwood.

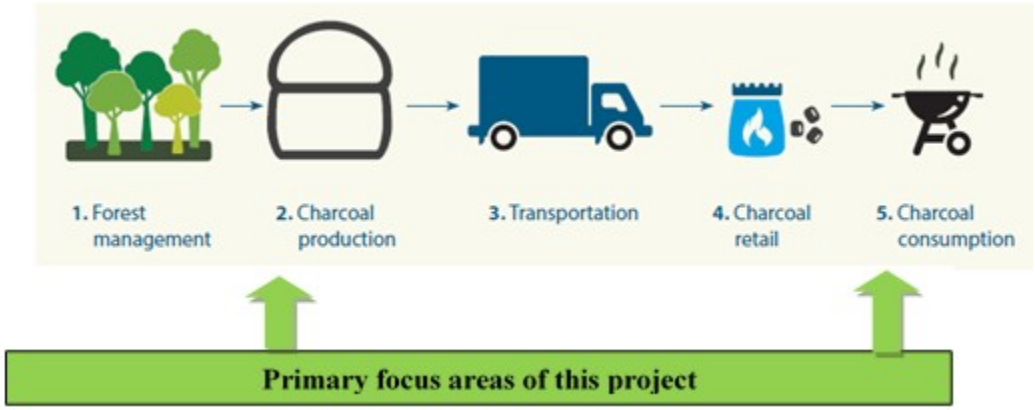


Figure 2 : Charcoal Value Chain – Interventions by Stage

The improved kilns/kilns of Stage 2 will mainly consist of Casamance prototypes. Improved stoves in Step 5 will include prototypes using charcoal or firewood. Through these activities, a viable business model for improving the production and distribution of kilns and stoves will be designed and implemented, focusing on the deployment of microfinance and loan guarantee programs. Sustainable alternatives to wood energy will also be explored and assessed.

Output 1.3: Developed gender-sensitive capacity development and modules for the production and utilization of improved kilns and cookstoves

Women are the final users of firewood and charcoal. Therefore, they are essential for the adoption of new cooking techniques with improved cook stoves and they can also play a determining role for the use of energy efficient kilns. These women will be informed, sensitized and trained on this news. They will be supported for the creation of association or GIE who will be in charge of the production, the marketing and the after-sales service of the improved stoves. While working in putting in place an

enabling environment for improved energy-efficient stoves and charcoal, the project will ensure that there is a gender-sensitive participation in the decision-making process.

Output 1.4: Alternative sustainable solutions to wood energy explored and assessed

Biomass is the main household energy used in Guinea Bissau. Output 1.1 should include measures for alternatives to wood energy. For example, with the electrification of villages, firewood can be gradually replaced by hot plates (electric stoves). Rural areas are areas with a high concentration of forest residues but also breeding areas. Biogas can be used as an alternative, but social, cultural and religious barriers must be overcome through awareness raising and training. NGOs will be privileged partners for the success of this product.

- **Component 2: Capacity building for mini-grid and low-carbon bioenergy**

Through this component, the project will work on a market structure that should has the right balance between international and local firms, in order to develop and build skills and expertise and the country level for mini-grids and low carbon bioenergy development.

Outcome 2: Capacity to deliver turn-key solutions and quality O&M services for mini-grids and low carbon bioenergy technologies developed.

Output 2.1: Tailored capacity building programme delivered to public sector and civil society.

Relevant national stakeholders for this output include Government bodies but also local community-based organizations. Capacities developed within the MEI / MEIRN, EAGB, Agencies, local banks and key national actors such as the Ministry of Agriculture and Rural Development and the Ministry of the Interior on best practices and opportunities for models of decentralized village electrification in forest areas (off-grid). This will include developing capabilities to familiarize them with suitable public derisking instruments, system sizing and optimization tools to evaluate system design options, including how to use the criteria and guidelines established to technically evaluate projects, determine the amount of the grant to be provided to project developers and decide on the appropriate fee that a given developer can charge to consumers. Finally, community-based organizations in selected locations (women's groups, local NGOs and SMEs / productive users) will receive training, assistance and advice on potential income-generating activities and security aspects related to the use of electricity.

Output 2.2: Various support delivered to and established for the private sector

The project will provide various support to the private sector, via a help desk, clearing house, guidebook, targeted financial sector assistance, and a public relations and investment promotion campaign:

- *Help desk:* A "Help Desk" will be established to provide commercial and technical advisory services to potential mini-grid developers. This Help Desk will be housed within the Ministry of Renewable Energy (Rural Electrification Agency) and will have trained staff to provide quick and targeted responses to requests for assistance and / or advice to developers on specific issues related to the fundamental aspects of project development, including preparation of feasibility / business plans and interpretation of tripartite contracts / agreements involving them as developers, rural electrification agencies and EAGB. It will also undertake individual project reviews prepared by the promoters for their technical and financial soundness prior to

submission to credit institutions. Support to be provided by the help desk will be billed; this will ensure its financial viability beyond the project period.

- *Clearing house:* Set up a one-stop shop (established in accordance with the Investment Charter under the supervision of the Ministry of Trade and Industry) by setting up a special office for the issuance of licenses and building permits to small project promoters' mini grids. At present, the one-stop shop does not exist. The window dedicated to the one-stop shop will be the repository of all the information that a potential developer will need before making a request, all the application forms and the required documentation must be submitted in support of an application, the fees to be paid, notify proponents if additional documents are required and provide a final decision on the outcome of an application. This will prevent the proponent from personally visiting multiple government offices to obtain the necessary approvals and speed up the approval process.
- *Guidebook:* This guide will provide a step-by-step approach to implementing mini-grids and serve as a tool for system designers, installers and operators to design, build, operate and properly manage mini-grids and grids. Assist all stakeholders to improve their common understanding and commitment to mini-grids. It will also aim to facilitate discussions between prefecture / sub-prefecture community groups and the private sector and will serve to demonstrate how mini-grids can foster economic and social growth, achieving development imperatives, while minimizing impacts, negative social, cultural and environmental impacts in villages. Finally, it will contain standard application forms and provide information / guidelines on the documentation required for the issuance of building licenses and permits to potential promoters, as well as associated costs.
- *Targeted financial sector assistance*. These activities will reach out to the nascent domestic financial sector, both publicly owned and private/commercial banks, to help inform domestic financiers with regard to the upcoming investment opportunities around solar mini-grids and low-carbon bioenergy. The aim is to build comfort with these domestic financial actors so that they begin to lend to the private sector in these areas: Issues to be addressed will include: typical financial profile, cash-flows, capital structure of these investment types and credit-worthiness of counterparts; investment assessment and appraisal of potential mini-grid and bioenergy activities; typical track-record profile of investments and international case-studies.
- *Public relations and investment promotion campaign:* National plan to implement awareness raising / promotional activities targeting national and international investors. This will include the preparation of promotional material, information sessions with investors already active in the energy / renewable energy sector in the country, local companies that have an interest in expanding their activities to include energy in the energy sector, rural areas and, potentially, possibly organizing road shows to attract foreign investors to establish consortia with local businesses to provide rural areas with modern energy services.

Output 2.3: On-the-job training programme for 100 solar mini-grid installers delivered.

The project will train private sector installers to enhance knowledge and understanding of different aspects of renewable energy development for electricity generation and distribution / consumer sales, including identification of potential sites, evaluation pre-feasibility and preparation of feasibility studies / business plans that will necessarily include plant design, equipment selection and assembly, construction, operation and maintenance. Training modules will be designed and implemented for the main beneficiaries (male and female promoters, component producers, system designers / installers, service technicians and consumers) and capacity building for general skills development and

implementation of projects. In this regard, capacity building activities will include issues related to the potentially negative impacts that the development of solar mini grids such as batteries can have on land use, water rights, biodiversity, etc.

Output 2.4: Capacity building for the fuelwood value chain delivered

This output is designed to enhance the fuelwood value chain by conducting capacity building and training activities to strengthen the technical and commercial capacities of the chain actors.

Capacities developed within the IBAP, artisans, farmers, loggers, and key national actors such as the Ministry of Agriculture and Rural Development and the Ministry in charge of natural resources on best practices and opportunities for the sustainable use of firewood. This will take into account the awareness of deforestation and the sustainable use of forests. This capacity building will include the development of competence in the cutting and use of wood. This use will be based on the difference in efficiency between traditional stoves and improved cookstoves, as well as improved stove making techniques. Emphasis will be placed on entrepreneurship to enable the establishment of a structure capable of developing viable business models for the production, marketing and maintenance of improved stoves.

On-the-job capacity building for 100 craftsmen, farmers and other actors in the fuelwood value chain will be conducted.

Finally, an awareness and information campaign on the impact of deforestation and climate change will help rural communities understand the notion of sustainability of the source. During this campaign, alternatives will also be proposed. The organizations (associations, producers) will then be trained on the methodology of manufacturing and operating improved stoves and kilns. Women's organizations will preferably be formed. They will then be accompanied to set up a local company producing improved stoves. Legal devices will frame them and allow them to market their production. The project will also propose a detailed study of the wood industry in Guinea Bissau. This study will provide a better understanding of the areas of harvest, coal production, actors, quantities produced and their end uses.

Component 3: Mini-grid and low-carbon bioenergy technologies roll-out

The expected outcome of this component is increased confidence of communities, developers and potential investors in the technical and economic viability of renewable energy mini-grids for rural electrification and local socio-economic development. This is seen as an alternative approach to centralized systems of grid extension and diffusion of low carbon technologies by bioenergy. This goal will be achieved through the implementation of pilot investments, exploring business models that enable sustainability and replication. It is expected that the successful presentation of electrified villages will act as a precursor to the implementation of similar solar mini-grids to harness the country's solar potential for rural electrification. It is also expected that the successful presentation of villages using improved kilns and stoves will act in the same way. NGOs will support the project, including TESE who developed and implemented the management strategy of the Bambadinca plant.

Through the implementation of investment projects under this component, the relevance of risk reduction instruments and proposed policies will be demonstrated (Component 1). The solar plant and

the deployment of improved kilns and stoves will also contribute to the demand for skills and capacity building (Components 2).

Outcome 3: A functioning business model is demonstrated for the technical and financial viability of mini-grids.

Output 3.1: Seven pilot sites totaling 1.2 MW for mini-grids identified and assessed, and business model defined.

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At the PPG stage, initial engineering specifications have been developed for seven hybrid solar-battery-diesel mini-grids based on a lowest-cost tecno-economic modelling approach. Hybrid systems will be a mixture of green and brown-field sites. For any hybrid systems, GEF INV support will be targeted to brown-field sites, and to the low-carbon components of green-field sites.

Under each of the seven sites, the project will expressly seek to explore and establish track records for each of the three different business models (utility, PPP, private sector). Targeted support and assistance will be provided, and lessons-learnt extracted and disseminated to the broader mini-grid market community.

For each of the seven sites, a generator sizing will be determined for each village according to socio-economic activities. The capacity of the battery is dimensioned for a range of 8 hours. This duration makes it possible to optimize the number of charge-discharge cycle of the batteries. It also significantly reduces the size of the battery and therefore the initial investment and replacement costs. Loads affected by subscriber category for network sizing are:

- Power per ordinary subscriber (household): 500 w
- Power per concession: 1500 w
- School: 1000 w
- Multifunctional platform: 15000 w
- Drilling: 10000 w
- Dispensary, Heath center: 2000 w
- Health case: 500 w
- Shop: 500 w
- Public lighting lamp: 18 w
- Mosque / church: 500 w
- Socio-educational home: 1000 w
- Other workshops: 4000 w

Table 1 presents the assumptions used to estimate energy requirements

Table 2: assumptions for estimating energy requirements

Household	Mill	Lighting	Pumping
1,000 Wh per household	10 Wh per person	5 Wh per person	Depending on water requirements and HMT if existing drilling otherwise 5 Wh / person

Figure 2 shows the set of site load curves selected. It is obtained by considering the different types of energy use and their probable time of use. The load profile thus determined will make it possible to size the power of the generating set to be used because the latter will have to be able to feed the tip (maximum power). It will also determine the operating scenario of the plant for a typical day. This day is taken for the least Sunny month.



Figure 3 : Load curves per villages

The following criteria were set for the site selections:

- Availability of the resource
- Sustainability of the resource
- Population density
- Socio-economic activities
- Accessibility of the site
- Geographical distribution of sites

For the villages visited, the exploitable energy potential for mini-power stations is solar energy. The hydroelectric potential is either too high for a micro hydroelectric run-of-river, or the river dries up for a certain period of the year, or the river is dependent on the phenomena of floods and major recession. After the visit of the sites and the various meetings with the stakeholders and the authorities, the following sites were selected:

- In the Gabu area, Pitche was chosen for a mini pilot solar plant. This locality is administrative chief town. It has a local distribution network that was powered by a 175 kVA generator. In addition to this generator, the village has a multifunctional platform run by the women's association. The multifunctional platform was used to peel the rice and grind the other cereals. The main activity of the village is agriculture. The village is heavily populated. The Tchetche site was also selected. This locality is a crossroads by its position (just at the exit of the river). Tchetche has a solar pumping system over the sun with low capacity (financed by JICA). It also has a military cantonment, a school with an improved fireplace used for its school canteen. This school is supported by the World Food Program (WFP) and UNICEF. At this village, it is planned to review their pumping system and install a mini-grid.

- In the Buba area, the Lamane site was chosen. It will be dimensioned a cold room for the conservation of the halieutic products but also fruits and a mini solar power station. Lamane is a village of sinners. It has also capitalized many successful projects with the Institute of Biodiversity and Protected Areas (IBAP). These projects were executed with the village association. In this same area of Buba, Dutadjara which is another very organized village with successful projects, is chosen for a mini-solar power station. By their organization, the populations have two rice husking machines and a tractor for field work. This village has two women's associations.

- In the Cantanhez area, the Jamberem site is retained. It will be a question of a cold room for the conservation of the halieutic products and fruits but also of mini-solar power station. It will be the pilot site for this region. It is also the site of Cacine.

- In the region of Cacheu, Cacheu is selected as the only site. Cacheu is non-electrified and is a commune that has several community-based organizations. It is also a tourist site and the income generating activity that is developed there is fishing. A mini-grid will be installed with an anchor client as a cold room for the preservation of fishery products.

For the villages involved in the project, it is decided to make a water supply system because they are all faced with the problem of access to water. For the same villages, it will be question of development of improved hearths and kilns. The improved stoves are intended for households and women's associations that smoke fish. The kilns for their part, are intended for producers of charcoal.

Installation of 1.2 MW of Solar PV-based power generation capacity installed

During the implementation of the activities related to this component, the project will sensitize and train energy managers at the national and prefectural / sub-prefectural levels on best practices and opportunities for decentralized rural electrification models through mini-grids. It will also work with the Government's Bureau of Standards to ensure that only high-quality products / equipment associated with the development of renewable energies including solar PV that meet approved standards are allowed for import and installation in the country.

Finally, during implementation, the project will monitor new developments in decentralized production that could be applied in the context of Guinea Bissau when they become commercially available. Characteristics of the sites

Figure 3 shows the operation of the various power sources of the plant. It is given for a typical day of the month the sunny month. It shows how energy needs are met by different sources. The solar PV is used as a first option, the batteries are then used, and the generator does not contribute when the demand is higher than the PV production and the stored energy or when the battery level is low and PV production n is not enough. The maximum operating time of the groups is estimated at 3 hours when all the loads are used. This time may be at least null during the first five (05) years of operation of the plant. The peak power to be installed is 1,155 MWp for the energy needs of the 7 selected sites. Villages in protected areas may benefit from drinking water supply. The summary of the peak power required for each village is presented in Table 5 (for an annual output of 53.6 MWh). This peak power to install in the 7 selected sites takes into account a central and a water supply for all the sites but also a cold production according to the sites (Tchetche, Lamane, Jemberem, Cacine and Cacheu). The following table summarizes the peak powers for each type of use.

Table 3: Summary of the peak power to be installed for the type of plant chosen (hybrid)

Site	Number of modules for the Central	Number of modules for the pumping	Number of modules for the cold room	Number of total module	Total Peak Power (kWp)	Peak power of a module (Wc)	Annual production of PV (MWH)	annual needs (MWh)	Supply of Diesel by year (MWh)
Pitche	540	34		574	172,2		240	308,06	68,06
Tchetche	160	17	51	228	68,4		95,1	106,58	11,48
Lamane	240	4	68	312	93,6		128	137,5612	9,56
Duta Djara	240	4		244	73,2	300	100	113,88	13,88
Jemberem	700	16	51	767	230,1		303	326,99255	23,99
Cacine	300	17	68	385	115,5		155	161,79355	6,79
Cacheu	1220	51	68	1340	402		546	625,70125	79,70
Total	**3,400,00**	**143,00**	**306.00**	**3,850.00**	**1,155.00**		**1,567.10**	**1,780.57**	**213.47**

TABLE 4 PRESENTS THE SUMMARY AND THE ESTIMATED PRODUCTION FOR THE OPTION OF A PURE SOLAR POWER PLANT.

TABLE 4: SUMMARY OF PEAK POWER FOR A PURE SOLAR OPTION

Site	Number of modules for the Central	Number of modules for the pumping	Number of modules for the cold room	Number of total modules	Total Peak Power (kWp)	Peak power of a module (Wc)	Annual production of PV (MWH)	annual needs (MWh)
Pitche	760	34		794	238,2	300	332	308,06
Tchetche	240	17	51	308	92,4		128	106,58
Lamane	340	4	68	412	123,6		169	137,5612
Duta Djara	340	4		344	103,2		141	113,88
Jemberem	1,000	16	51	1067	320,1		421	326,99
Cacine	420	17	68	505	151,5		204	161,7936
Cacheu	1740	51	68	1859	557,7		757	625,7013
Total	**4,840.00**	**143.00**	**306.00**	**5,289.00**	**1,586.70**	**300.00**	**2,152.00**	**1,780.57**

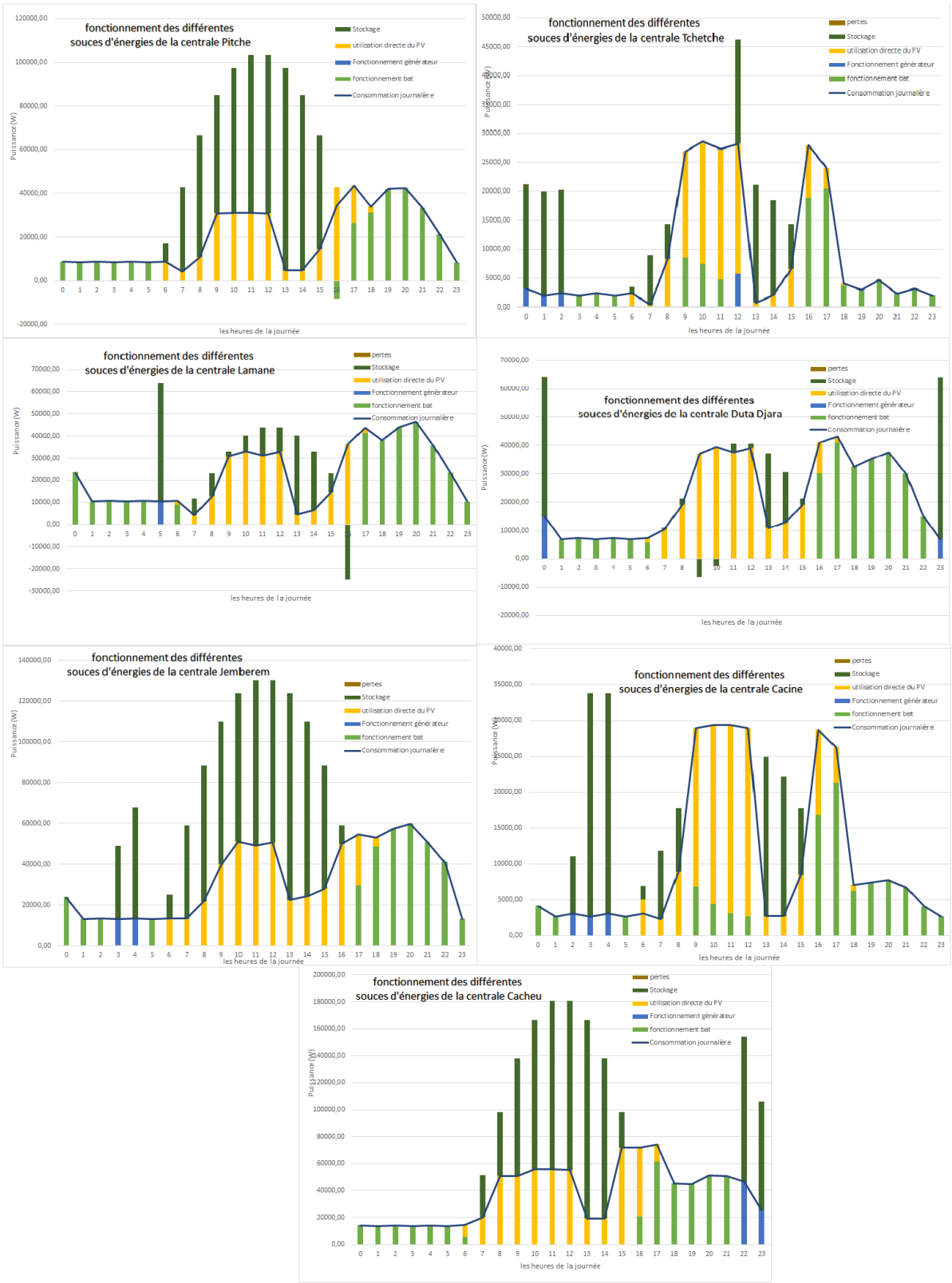


Figure 4 : Operation of the various power sources of the power stations

Table 5 presents the characteristics of the 7 selected sites.

Table 5 : Characteristics of the project sites

N°	Region	Department	Site and proposed power	Number of households	Population[6]	Women	Daily energy requirements	Electricity services to be used for:
1	Gabu		Pitche 172.2 kWc	602	6021	3071	844 kWh	Lighting Community needs Income generating activities Pumping
2	Gabu		Tchetche 68.4 kWc	42	422	215	292 kWh	Lighting Community needs Income generating activities Pumping Cold room
3	Buba		Lamane 93.6 kWc	47	471	240	376.88 kWh	Lighting Community needs Income generating activities Pumping Cold room
4	Buba		Duta Djara 73.2 kWc	38	376	192	312 kWh	Lighting Community needs Income generating activities Pumping Cold room
5	Cantanhez		Jemberem 230.1 kWc	99	985	503	895.87 kWh	Lighting Community needs Income generating activities Pumping Cold room
6	Cantanhez		Cacine 116 kWc	129	1,289	657	443.27 kWh	Lighting Community needs Income generating activities Pumping Pumping Cold room

[6] INEC, 2009; WB, 2016 (official figures), growth rate 2.5%, female-to-male ratio 51%

7	Cacheu	Cacheu	Cacheu 402 kWc	713	7,126	3,640	1,714.25 kWh	Lighting Community needs Income generating activities Pumping Cold room

Table 6: Summary of peak powers for pumping and per village

Location/ Project Site	Village Name	Total Population	Female Population	Pumping (Wc)
Cacheu	No village at distance less than 5 km	0	0	
Cacine	Bonia	**237**	121	1,200
	Tabanca Nova	**99**	50	500
	Cucuma	NF	NF	
DutaDjara	Bacar Conte	**243**	124	1,200
	Dandu	**223**	114	1,000
	Gamalique	51	26	250
	Gandjai	NF	NF	
	Salemina	59	30	100
	Darsalame	66	34	350
	Tchadanca	**241**	123	1,200
	Gã Quinda	NF	NF	
	Ponta Nova	**733**	374	3,600
	Gã Gregório	**109**	56	500
	Gã Toure	**792**	404	4,000
	Gancodia Balanta	NF	NF	
	Incassol	**170**	87	1,000
	Nhala	**936**	477	4,600
	Indjassane	**621**	317	3,000
Jemberem	Candjatra	**103**	53	500
	Amedalai	**134**	68	660
	Camecote	33	17	
	Madina de Cantanhez	**202**	103	1,000
	Catchamba Balanta	**85**	43	500
Lamane	Ba Mela Intchingue	NF	NF	
	Buba Intchingue	75	38	400
	Bani	**213**	109	1,000

	Gambapor	**122**	62	600
	Gamperto	**229**	117	1,100
	Dada	30	15	150
	Minna	NF	NF	
Pitche	Uarbe	NF	NF	
	Iancor	**328**	167	1,600
	Sutuco	**184**	94	1,000
	Dembacunda	**237**	121	1,200
Tchetche	Bugafali	30	15	150
	Queue II	**227**	116	1,200
	Dande Coli	**88**	45	500
	Maria	**83**	42	500
	Vendu Tcham	**112**	57	600
	Bilonco	**385**	196	2,000
	Ore Tchom	**122**	62	600
TOTAL	**All VILLAGES**	**7,602**	**3,877**	**37,760**

Output 3.2: Productive use promoted to increase electricity demand in the targeted sites.

The project will emphasize the importance of productive use, both to improve the financial viability of mini-grids (which benefit greatly from economies of scale due to higher loads), and to improve the livelihoods and human development impact of end-users. Activities will include piloting and exploring innovative approaches to productive use in the business models supported by the project, including the opportunity for lease-back and third-party ownership arrangements, where the mini-grid developer absorbs the upfront cost of productive use equipment, and then spreads this cost over time to the end-user.

Capacity building of the rural population, especially women, to undertake income-generating activities using electricity will be conducted. Access to electricity services in rural areas opens up opportunities for income-generating activities associated with the processing of agricultural and fruit products such as rice, maize, sorghum, millet, mahogany, mango, banana. Electricity also allows the conservation and processing of fish products. This will add value to crops by allowing farmers to obtain higher prices through their sale in processed form rather than as raw products. The availability of electricity can and will also promote labor activities such as sewing, metal carpentry...

As productive activities increase the disposable income of some rural households and the availability of electricity, it may be wise to control household appliances for comfort and / or to robotise certain works dedicated to women. Electrical energy can also provide an alternative to using wood and charcoal for cooking with hotplates. Induction cookers are compact, portable, lightweight, efficient and affordable, with a selling price of 30,000 FCFA / unit (US $ 54.5). Cooking with induction cookers does not generate smoke or soot and keeps the cooking surface clean. Induction cookers have the potential to move some households away from using wood or charcoal for cooking, reducing pressure on forests and reducing deforestation.

Output 3.3: 5,000 improved cookstoves commercialized and 50 improved kilns disseminated

The output of the project will work on the roll-out of the bioenergy technologies, namely the energy-efficient stoves and kilns. Activities under this output will includes:

- Prototypes for stoves and kilns: Clean cooking success resides on tailor-made approaches. A thorough consultation with beneficiaries, craftsmen and local entrepreneurs will be conducted to select the appropriate prototypes for improved cookstoves and for kilns. If the “Casamance” prototype is particularly relevant in this region, stoves prototypes need to be determined. The prototypes will then be tested before any scale up.
- Production: Local entrepreneurs and craftsmen will be supported to invest in energy-efficient stoves and kilns. A traditional stove cost around $2 while energy-efficient stoves can cost between $7 to $15, depending on the prototype. A small portion of the production will be supported by the project as a subsidy, allowing to lower the overall capital cost of local entrepreneurs, thus being more affordable for the beneficiaries. International stoves manufacturers will also be contacted to explorer further partnerships. The same approach will be adopted for improved kilns.
- Dissemination: In order to overcome the capacity to pay of households, several financial options will be explored, including the “money box” and the “exchange”. The “Money Box” is a tin with a slot. The local entrepreneurs give its customers the stoves on credit, and they pay out of the savings they make from reduced charcoal consumption. The principle in common in West Africa. For example, in Ghana where the money box experience started, 3 of 4 customers used this money box mechanism. The “exchange” mechanism is allowing the consumer to exchange his/her traditional and improved cook stoves.

4.2 Partnerships:

The project will work together with a range of projects aimed at providing the rural population of Guinea-Bissau with renewable energy and improved use (efficiency) of biomass. The objective is to and learn from their experiences. The project will particularly incorporate best management practices and lessons learned through the “Bambadinca Sta Claro” Project, the first off grid Hybrid Photovoltaic Power Plant in Guinea-Bissau, implementing a public-communitarian management model. This model should facilitate the appropriation of the project through local authorities and clients/ inhabitants of Bambadinca and improve the socio-economic effects of energy supply to Bambadinca and surroundings. The implementation period ended in 2015 and the plant is running since then without major technical or administrative problems.

Its evaluations showed that it successfully achieved the expected goals and results and produced a positive impact on the community (Interview NGO TESE with its Energy Sector Chief). In order to deal with lack of energy and poverty, based on socioeconomic studies, the Portuguese NGO TESE conceived and implemented the Community Program for Access to Renewable Energy -Bambadinca StaClaro, which consisted in creating a community energy service capable to manage a hybrid PV mini-grid. The project was executed from October 2011 to March 2015 and financed by European Union (ACP-EU Energy Facility); Portuguese cooperation; UNIDO (GEF project). It established a successful partnership with public organizations (DGE -National Energy Direction / DREB -Regional Energy Office (GB) as well as with local partners, the Bambadinca Community Development Association (ACDB) and national NGOS.

Furthermore, the project will coordinate actions and draw lessons learned from the following three projects related to renewable energy and will contribute to the improvement of livelihood of rural population in Guinea-Bissau:

One of the strategic areas of the national NGO ADPP (Action for Development from People to people) is to combat and mitigate the effects of Global Warming and Climate Change through promotion of renewable energy use and sustainable production. ADPP has installed RE off grid solutions for the development of rural communities in the region of Bissorã in the frame of the project “Energia Renovavel para o Desenvolvimento Local em Bissora” (Renewable Energy for Local Development in Bissorã), between 2011 and 2016. Financed by the European Union, the project serves plenty of communities which have been completely energy poor before project implementation. The project establishes so called “Farmer Clubs” in the communities which largely consist of small-scale women farmers who aim at diversifying agricultural production with the help of irrigation by renewable energy (solar) which is also installed for domestic use, school lightening and water-pumping. This ADPP project and upcoming follow-up projects of ADPP in the region of Cacheu and Empada (SE4ALL GB IP, both with 350.000 kwh capacity, start: I: 2017 and II: 2019)) are of special interest due to the community-based bottom-up approach and the prominent role women play in the project conception.

The main objectives of ADPP´s projects are very similar to the present project and allow for cooperation and of experiences sharing: increase supply of solar sustainable energy in rural areas; increase human capacity to encourage the absorption of renewable energy in rural areas; reduce environmental impact by promoting sustainable sources of energy.

The UNIDO project “Promoting investments in small to medium scale renewable energy technologies in the electricity sector” (UNIDO Project ID 130012); running from October 2014-October 2019, is funded

by the Global Environment Facility and the EURO Trust Funds. It comes very close to the objectives of this project in promoting energy investment into small / medium scale renewable energy technologies. The project will assist Guinea-Bissau in the transformational change of the electricity sector to a sustainable low-carbon development path. It combines activities in the areas of technology demonstration, policy support as well as capacity building. It is an important baseline project and as it is in its last stage of implementation, it has numerous lessons learned and practical experiences for the current GEF funded project.

There are other projects in the pipeline focusing on the installation of RE off grid solutions in other parts of Guinea-Bissau, such as projects of TESE in Bolama and Bijagos for which technical feasibility studies have been finalized, and the recently installed 500 mpi plant in Bissorã.

In general, there are two types of partner projects (baseline):

1. Those investing in RE off grid solutions in other parts of GB, such as projects of TESE for Bolama and Bijagos; the recently installed 500 mpi plant in Bissorã, ADPP;

2. Different types of projects tackling rural development in GB which may include RE components or activities for cleaner cooking, combat deforestation, climate change mitigation and adaptation. There is a huge number of projects from a wide range of donors/partners with different approaches.

- **Main Projects related to the current GEF-UNDP- PROJECT**

Table 2 : Main projects in Guinea Bissau

Project Name	Agency / Funding	Financing Amount (USD)	Geographical Region (intervention zone
Recently Executed Projects			
Adaptation to climate change and the coastal line in West Africa (ACCC), 2009-2012, Environment Sector/ Ministry of Environment	UNDP-GEF	423,000	Northwest: Varela
Consolidation of Terrestrial Protected Area systems, 2012-2016 – Environment Sector / Ministry of Environment	UNDP-GEF	3,000,000	Southeast: PA Natural Complex Dulombi, Tchetche e Boé
Reinforce the resilience and adaptation capacity of the agricultural sector and hydro-resource/ Ministry of Environments 2012-2017	UNDP-GEF	5,000,000	Gabu Region: Pitche, Bajocuda, Canquelifá, Buruntuma, etc
Bambadinca Sta Claro – Energy	TESE - EU	1,000,000	Bambadinca

Sector - Photovoltaic Hybrid Offgrid system, 322 MW			
Mini-grid Photovoltaic Hybrid system of Contuboel 2015-2017 – Energy Sector	FRES	1,000,000	Contuboel
Projects Currently under Execution			
Protected Areas and Resilience to Climate Change (Global Climate Change Alliance Initiative, GCCA) - is a mitigation project, which addresses aspects of energy efficiency in protected areas and afforestation 2014-2019 - Environment Sector / Ministry of Environment	EU	4,500,000	Protected Areas National
Biannual Update Report (UNFCCC) is an enabling activity project with no actions on the ground 2017-2019 - Environment Sector / Ministry of the Environment	UNEP-GEF	200,000	
Economic development program in the Southern regions (PADES) - 2016-2020 – Ministry of Agriculture	FIDA	15,000,000	Regions of Quinara/Tombali and Bolama/Bijagos
Project AGIR: Improving resilience and restoration of degraded agricultural land – Ministry of Agriculture	EU/BAD	2,000,000	ver o Kaoussou Diombera /
Regional Program for the Development of Renewable Energies in the UEMOA Region Phase II (PRODERE II) - solar power plant and construction of a mini-grid system (central solar of 500 kW) in Bissorã 2016-2017 – Energy Sector	SABER/UEMOA	2,000,000	Bissau/Bissorã
Improving the electricity distribution system of Bissau city (includes the installation of 500 PV solar street lighting systems)	AfDB	5,000,000	Bissau
Promoting investments in small and medium scale renewable technologies in the Energy Sector of Guinea-Bissau, 2015-2019 – Ministry of Energy /Ministry of Environment – project is similar	UNIDO-GEF	1,700,000	All Guinea-Bissau

to the UNDP-GEF project (installation of 2 MW of Renewable Energy all over Guinea-Bissau)			
Projects in Preparation (Pipeline)			
Managing mangroves and production landscapes in line with climate change mitigation - implementation started by end of 2018 - covering major energy efficiency initiatives in the protected areas (domestic energy) and afforestation - Environment Sector	UICN-GEF	3,200,000	Protected Areas National
Restoration of degraded landscapes – under design- Environment Sector	UICN-GCF	50,000,000	
Strengthening the resilience of vulnerable coastal areas and communities to climate change in Guinea Bissau. PRODOC recently endorsed.	UNDP-GEF / LDCF	12,000,000	Protected Areas: Cacheu, Varela, Mansoa, Cufada Region, Bijagos PA
Building the climate forecast and creation of the early warning system - Ministry of Environment / Ministry of Transport and Communications (National Institute of Meteorology)	UNDP-GEF	5,000,000	All GB
Construction of Hybrid Photovoltaic/diesel Power Plant of 15 MW in Gardette/Bissau	BOAD	35,000,0000 (in 3 phases) Phase 1	Bissau
Construction of Hybrid Photovoltaic/diesel Power Plant of 2 MW in Gabu	BOAD	35,000,0000 (in 3 phases) Phase 2	Gabu
Construction of Hybrid Photovoltaic/diesel Power Plant of 2 MW in Canchungo	BOAD	35,000,0000 (in 3 phases) Phase 3	Canchungo
Construction of Hybrid Photovoltaic/diesel Minigrid of 500 kW in Bolama	EU	500,000	Bolama
Construction of Hybrid Photovoltaic/diesel Minigrid of 500 kW in Bubaque	EU	500,000	Bubaque

Data sources:

- Results and detailed notes from stakeholder consultation meetings during the mission to Guinea-Bissau. Refer to PPG Reports: Inception Workshop Report, Communities visited; Baseline Study; interviews with NGOs, such as ADPP; AD and TESE
- Extensive web research, which focused on UNDP's as well as partners' sites, of the Energy Sector/ renewable energies: UNIDO, ECREE, SE4ALL; and rural development/livelihood/CC projects, research at corporate sites for WB, IFAD, FAO, AfDB and EC

4.3 Risks and Assumptions:

The risks and assumptions are described in the "project management" section.

4.4 Stakeholder engagement plan:

Stakeholder Analysis and Engagement[7]

The successful implementation of the project will largely depend on the effective communication and coordination with the multiple project stakeholders and the implementation of mechanisms to ensure stakeholders' participation and commitment to the project. The project's detailed Stakeholder Engagement Plan is unfolded in Annex F.

The formulation of the stakeholder engagement plan has the following objectives: a) to clearly identify the basic roles and responsibilities of the main participants in this Project; b) to ensure full knowledge of those involved concerning the progress and obstacles in project development and to take advantage of the experience and skills of the participants to enhance project activities; and c) to identify key instances in the project cycle where stakeholder involvement will occur.

The ultimate purpose of the stakeholder engagement plan will be the long-term sustainability of the project achievements, based on transparency and the effective participation of the key stakeholders. The broad consultation process and mobilization of national ownership aim at obtaining, at the end, a Project Document which once submitted to the GEF, corresponds to the reality and is aligned with the priorities of Guinea-Bissau, in which this aim will only be achieved with the shrewd and competent technical participation of all and public-private partnerships.

During project implementation, it will be ensured that implementation approaches are well targeted to meet 'end users' needs (i.e. the beneficiaries) and to establish strong ownership of project outcomes by national partners and beneficiaries. A strong emphasis will be placed on consultation with the local forest dependent communities to assist the path to clean energy and its management at local level and assess the impact of project support in meeting the communities` needs. The participatory decentralized approach to project implementation will help to ensure that each project zone and locality has ownership of the transformative process to renewable energy and its applications.

The main project stakeholders, including the government, its directorates, services and agencies, research institutions, National and International Non-Governmental Organizations (NGOs) as well as local communities of the targeted sites and community-based organizations, and private sector representatives, mainly from the energy sector have been profiled. During the communities visits

[7] For the requirements see "GEF Policy for Stakeholder Engagement, November 2017", https://www.thegef.org/sites/default/files/council-meeting-documents/EN_GEF.C.53.05.Rev_.01_Stakeholder_Policy_4.pdf

participatory assessments with local communities, focusing on energy needs, gender perspectives and the promotion of socio-economic rural development through clean energy have taken place.

For all stakeholders (Government, NGO, Private Sector, Local Communities), an inventory of their formal mandates and roles and responsibilities has been carried out. In the renewable energy sector, national government institutes and ministries play a very important role. The Ministry of Environment and Sustainable Development and the Ministry of Energy, Industry and Natural Resources, in particular, are not just the project's proponent and "owner", but also have a wide mandate and varied attributions (responsibilities). At the same time, interaction with other institutions and the private sector and local community based organizations is a pre-requisite for achieving the goals of sustainable long-term RE off grid supply, stimulation and increase of socio-economic development, the broad dissemination of clean cook stoves and efficient charcoal production to manage the energy sector in an integrated way, taking the reduction of GHG emissions, forest protection, climate vulnerability and risks fully into account.
The overall analysis of the stakeholders that are relevant for the project and its implementation is based on

- consultations and interviews with the known actors, such as heads of institutions, leading employees in the relevant ministries, project managers of NGOs, the participants conveyed to the Inception Workshop on 18 July 2018,
- the detailed stakeholder analysis in the baseline study
- other existing data in documents that serve as baseline or previous intervention in the renewable energy sector of Guinea-Bissau`s forest regions (SE4ALL GB IP, National RE Energy Plan (PANEE), http://www.aler-renovaveis.org/en/communication/news/guinea-bissau-towards-new-and-renewable-energies/28 of July 2017)

Stakeholder Selection Process:
An assessment of relevance of different public and private stakeholders has been made, including NGOs and local associations, in relation to the following parameters:

- Mandate and/or legitimacy
  - ✓ Does the stakeholder have an influential position?
  - ✓ If a public entity, does they have a strong legitimacy?
  - ✓ Is management interested in renewable energy solutions and low carbon biotechnologies?
- Sector (social function, production functions, infrastructure functions, cross-cutting functions)
  - ✓ Social functions: housing, health, education, other (sports, religions, etc.);
  - ✓ Production functions: agriculture, livestock, fisheries, industry, mining, trade and tourism;
  - ✓ Infrastructure functions: public services (water, electricity, gas): transport (port, roads and others),
  - ✓ Cross-cutting functions: environment (ecosystems, pollution), knowledge and research
- Role / Network of Influence
- Type of Stakeholder
- Proximity and importance for project
- Gender sensitivity screening

The stakeholders can be classified along the criteria "influence" and "interest"

- **Main stakeholders with high influence and high interests**: necessary for the process;
- **Non-key stakeholders with low influence**, but high interests: necessary for the process because these include the local communities, women groups and fishermen associations, the main

beneficiaries, local traditional/non-state authorities; teachers and public health personnel at the intervention sites

- **Other stakeholders with great influence and little interest**: necessary to bring together powerful stakeholder who have the capacity to thwart a process, because they are influential. They are likely not very aware of technics and benefits of decentralized mini-grids and low-carbon bioenergy technologies and small hydro power, and therefore not immediately interested. It is therefore necessary to ensure strategic communication with this group.
- **Non-key actors** with little influence and little interest: not of interest.

Mapping of project's key stakeholders and their envisaged role and responsibilities in the project

After the thorough stakeholder analysis, understanding their interests, needs and perception about the project as well as their responsibilities a matrix with key project stakeholders has been produced. It relates each stakeholder`s mandate and function and their prospective roles in the project; i.e. in which activities they engage and the engagement methods/means. The stakeholder inventory includes the technical ministries (Group 1); other government agencies (Group 2); NGO's and private sector parties (Group 3) and local communities (Group 4)

Table 3 : key stakeholders and their envisaged role and responsibilities in the project

Stakeholders	Mandate and Functions	Prospective role in the project
GROUP 1 **The Ministry of Environment and Sustainable Development (MADS))** Currently: Secretary of State of Environment (S E: Ms. Quita Djata)	Mandate: Law No. 1/2011 approving the Basic Legislation on Environment. Law No. 10/2010 on Environmental Impact Assessment Regulation. (The AAAC is the Competent Environment Assessment Authority Regulation No. 5/2017 of Public Participation Regulation No. 6/ 2017 The Environmental Fund Regulations No. 7/2017 Regulation of Environmental Impact Study Regulations No. 8/2017 Environmental Licensing Decree No. 9/2017 Environmental Audit Regulation Decree No. 10/2017 Environmental Inspection Regulation The MADS is responsible for environmental policy and the preservation of the biodiversity.	Leading Executing Partner for the project Coordination of the overall project preparation and activities Ensure consistency of the project Co-participate in the identification of pilot sites Environmental and social impacts assessment Resources assessment for pilot projects Ensure the Monitoring GHG emission reductions Investment support and promotion, including from international climate finance Monitoring and evaluation

Stakeholders	Mandate and Functions	Prospective role in the project
	• exercises its mandate based on the implementation of environmental policy, based on the National Environmental Management Plan (1994), in compliance with international and regional commitments • represents and responds to the main international agreements and conventions that bind the country in the field of environment and sustainable development. • represents and serves as the GEF Policy Focal Point and Operational Focal Point. MADS has special services in different environmental topics, such as General Inspection of Environment (AAAC); Directorate-General for Environment (DGA); Directorate General for Sustainable Development (DGSD), Institute for Biodiversity and Protected Areas (IBAP).	
GROUP 1 **Ministry of Energy, Industry and Natural Resources** (in short Ministry of Energy and Industry - MEI) Minister: Antonio Serifo Embalo Secretary of State of Energy: João Saad	The MEI is responsible for defining policies, laws, regulations, sector supervision, import / export licenses, granting of concession schemes and definition of electric energy tariff structures (Decree-Law No. 2/2007) It represents Guinea-Bissau in all national, regional and international energy forums; develops sectoral policies; develops structural, investment and fiscal policies; fixes energy prices where these are not fully liberalized; creates stable climate that attracts investors; definition of policy and management of national energy resources • creating a favorable climate	Ensure the integration of proposed mini-grid related policies in the national policy and institutional framework for power sector reform Co-participate in the identification of pilot sites Technical advisory minigrid Plan activities related to transfer and development of domestic supply chain and O&M models Facilitate investment promotion, support for mini-grids, and issuance of co-financing letters Run awareness campaigns and co-participate in training on RE

Stakeholders	Mandate and Functions	Prospective role in the project
	for the development of traditional and renewable energy resources; • granting of concessions, licenses and certification conditions for energy and business activities; • acceleration and expansion of national energy production and distribution. The MEI has the following services: General inspection of Energy and industry (GIEE); Directorate- General of Energy (DGE), of Industry (DGI), of Water Resources (DGRH), of Geology and Mining (DGGM), Directorate of Renewable Energy (DER); EAGB (National Electricity Company), Petroguin E.P (Oil and Gas Company); INITA (National Institute for Research and Applied Technology) The DGE is responsible for implementing the national energy policies defined by the MEI, accompanying its execution and supervising all activities related to the production, distribution and sale of electricity and fuels. It also protects consumers and verifies companies operating in the energy sector.	
GROUP 1 **Ministry of Finance and Economics and Planning**	The Ministry of Finance and Economics and the Planning Department control public investment funds and the implementation of all energy sector projects, having direct relations with donors, donors and international funds.	Provide guidance on the design of appropriate financial mechanisms and to ensure the programming and the disbursement of the co-financing (cash) committed by the Government
GROUP 1 **Further stakeholders** - Ministry of Commerce and Business Promotion		Dormant stakeholder, not sufficient capacity

Stakeholders	Mandate and Functions	Prospective role in the project
- Ministry of Infrastructure and Construction and Urban Planning - EAGB: Public Electricity and Water Company of Guinea Bissau (Empresa Publica de Electricidade e Aguas da Guine Bissau) -		to act beyond Bissau,
GROUP 2 **National Institute for Research and Applied Technology (INITA)**	INITA is the autonomous technical body of MEIRN in charge of (mission)	• Conduct of required project researches and developments (R&D) • Conduct of tailor-made learning programs on low carbon technologies • Assist the project in the creation and production of informative material about low carbon technologies
GROUP 2 **Institute of Biodiversity and Protected Areas (IBAP)**	Decree-Law No. 5A / 2011 - Framework Law on Protected Areas IBAP promotes sustainable management of biodiversity resources and conservation policies is in line with the agreements Guinea-Bissau. It is responsible agent for the management and monitoring of the Protected Areas, such as Cufada Lake and Cantanhez Forrest where project sites are situated.	• Identification of forest communities and resources in the project pilot sites within Protected Areas • Organization and conduct of awareness raising campaigns • Assist with the management of RE solutions at the community level in the Protected Areas • Ensure good buy-in from direct beneficiaries of the project
GROUP 2 **National Institute of Studies and Research** (INEP), which incorporates the ***Center for Environmental Studies and Relevant Technology (CEATA)***	INEP is mandated to support national development strategies and has a long history in environmental studies and technologies as well as in participatory work with local communities and communication	• Communication Strategy and development of material (flyer etc.) for RE applications and improved cooking stoves, charcoal • Further environmental and non-technical studies for project implementation • Drafting of national policies/strategies of RE, off grid and low carbon biotechnologies
GROUP 2 - National Institute of Statistics (INE), data energy etc.	Research, provision, compilation and interpretation of energy related data bank Facilitate economic, political and social development through meaningful statistical evidence	• Provision and interpretation of RE energy data, aggregated per sex
GROUP 1	Decree-Law No. 5/2011 approving	Ensure the integration of proposed low-

Stakeholders	Mandate and Functions	Prospective role in the project
Ministry of Agriculture, Forestry and Livestock **Direction General of Forests and Fauna (DGFF)**	the New Forestry Law. Mandated to control the use of forest resources, forest management and monitoring MORE	carbon bioenergy technologies related policies in the national policy and institutional framework prescribed in the Domestic Energy Action Plan • Identification of forest communities and resources in the project pilot sites outside the Protected Areas • Organization and conduct of awareness raising campaigns • Ensure the inclusiveness and good buy-in from farmers, charcoal and fuelwood producers
GROUP 3 **Private sector: mini-grid operators, installers of RE systems**	National/international enterprises in the RE sector, such as SDD/FRES, Impar, Dura Energy, Prosolia, PP Energy, ELMI SARL Private installers and mini-grid operators with interest in public-private partnerships and	• Provide equity investment for pilot projects • Technology needs assessment for PV supply chain • Design of O&M models
GROUP 3 **Local and international finance institutions**	National Commercial Banks Regional Development Banks Small grant funds (GEF) Manifested interest in cooperation for RE solutions, public-private partnerships	• Provision of loan financing models for pilot projects • Loans & funds for pilot communities
GROUP 3 **2 Chambers of Commerce**	- Commerce Industry and Services, - Chamber of Commerce, small scale farmers	• Stimulation of green business, women training and entrepreneurship in the RE sector in GB; • Integration of energy component in private commercial investment plans etc., • Develop public-private partnerships
GROUP 3 **National and International NGOs** **AD (Action for Development)** **ADPP (Development from People to People)** **TESE (Association for Development)** **GDVER (Gestão Durável Verde e Energias Renováveis)**	**AD** NGO is dedicated to the development of key actions such as food security, creation and support of community radios, art training and other professions as well as environmental actions to protect the forests of Guinea-Bissau and conducting bio-ecological studies in the wetlands of Cantanhez	Provision of services as sustainable management of resources and local development actions, Community based management of RE off grid solutions, Management & Finance Models Project design & implementation of off-grid hybrid PV/Diesel plant solutions in GB Sustainable energy education programs in Basic Schools Training in electricity & RE PV solutions of local communities;

Stakeholders	Mandate and Functions	Prospective role in the project
	ADPP Leading NGO in sustainable energy solutions and vocational training, sustainable integrative rural development **TESE** Leading NGO in sustainable energy solutions &community management **GDVER** Renewable energy project design and implementation of Biogas plant for sustainable development	Cleaner cooking – cook stove production and marketing Organization and conduct of awareness-raising campaigns for RE and low carbon biotechnologies Ensure strong support and buy-in from direct beneficiaries of the project
Group 4 **AMAE and other specific women organizations**	NGO works for support of economic activities of women nationwide; has about 5600 members/largest platform of this kind in Bissau	Promotion of women entrepreneurship in sustainable development and RE, training and support for start-ups
GROUP 4 **Local communities and their associations** **Pitche (Com. 1)** **TcheTche (Com. 2)** **Duta Djara (Com. 3)** **Lamane/Cufada R. (Com. 4)** **Cantanhez /Cacine (Com. 5)** **Cantanhez/ Iemberem (Com. 6)** **Cacheu (Com. 7)**	Impoverished forest dependent populations with manifested interest in RE energy supply and low carbon biotechnologies: Women Groups, women community committee Fishermen Association, Association of Inhabitants, Youth Association, Association of Honey Producers Traditional Authorities (regulo, village chief) Local State Authorities (Admistrator & staff, Province Governor, Fishing & Port Authorities)	Engagement in women and community management of RE solutions, including multifunctional platforms, cooling systems, water pumps, schools and health posts; Participation in Irrigation projects of women horticulture/gardens Commitment to development & participation in Renewable Energy driven business solutions for rural population Participation in functional Alphabetization for women Local management of Domestic Electricity supply
GROUP 4 **Farmers, charcoal and fuelwood producers and other cooking value-chain stakeholders**	Positive and pro-active engagement for improvement of the sector/ charcoal industry + reduction of firewood/biomass through low carbon biotechnologies; Lead by women association/ foster women entrepreneurship	• Ensure initiative sustainability • Participate in the design of the financial mechanisms • Apply and further disseminate alternative cook stoves and charcoal kilns • Awareness campaigns for cleaner energy • Increase income of community participants

Stakeholders	Mandate and Functions	Prospective role in the project
UNDP Guinea-Bissau	GEF Implementation Partner	• Project oversight • Financial administration and technical support • Monitoring and Evaluation • Overall Support • Partnership with UN-system

4.5 Gender equality and empowering women:

During the PPG, a gender analysis for selected forest dependent rural communities in the four (4) project areas (Cacheu, Gabu, Cufada Lake, Cantanhez Natural Park) and a detailed Gender Mainstreaming Plan (included as Annex G) were developed to ensure gender mainstreaming in the project.

The project will have specific gender goal indicators which include the collection of gender-disaggregated data and a strong monitoring and evaluation mechanism to operate and advance gender mainstreaming and social equity (based on GEF Core Gender Indicators, see GEF Gender Equality Action Plan, 2015).

The challenges of gender mainstreaming

According to the project objectives and proposed actions, it is categorized as gender-responsive (gender marker 2): the envisaged results address different energy needs and priorities of men and women, different - gendered socio-economic visions on the path to clean energy as well as equitable distribution of benefits, resources, status and rights. The project does not directly address root causes of inequalities in women`s life in Guinea-Bissau (qualifying for gender marker 3).

The majority of the beneficiaries of better access to energy needs in rural areas under this project are women and children. Providing energy access to these most often-poor households adds value to agricultural production and to micro, small and medium enterprises. It generates high positive impacts on women as consumers of electricity and managers of RE community systems. While electrification will benefit both women and men by enhancing their engagement in more productive activities, gender gains are derived mainly from reducing the workload of women and girls related to biomass use.

The project creates a strong link between the forest and basic energy needs: by promoting the use of sustainable charcoal and firewood the project will not only reduce the pressure on biomass resources and decrease GHG emissions but also relief women from the burden of wood collection for cooking and the traditional method of fish smoking (improved and / or replaced through cooling). Fuelwood use for domestic purposes is synonymous with women in Guinea-Bissau. As rural women and girls are primarily responsible for most domestic work, access to energy will make a significant difference in their quality of life, including their health.

Biomass represents up to 95 % of the total energy consumed for cooking in Guinea-Bissau. Poor women work in the informal economic sector (often in food production) that relies heavily on biomass as main source of energy.

The project will directly impact women. It is estimated that the time spent for wood collection varies between 2 to 3 hours per woman per day in the country. With adequate management of firewood and improved cooking stoves, this can be reduced to only 2 or 3[8] hours per week. Pollution caused by the use of traditional cooking stoves is a very serious health problem for women and girls. In addition, the number of female-headed households is increasing throughout Guinea-Bissau, and these tend to be among the poorest and most vulnerable households in rural communities.

Main gender gaps in social and institutional matters

Inclusive participation in Guinea-Bissau remains an ongoing challenge. Gender inequality persists, as women are less educated and generally have a lower socio-economic status, with inadequate access to basic social and health services and reduced economic opportunities compared to men.

Structural inequalities between men and women in Guinea-Bissau are dictated by cultural and economic context; exclusion and discrimination of women are based on the logics of patriarchal power, reflected in phenomena such as female mutilation, domestic violence, early marriage, limited access to resources, especially (agricultural) land, weak institutional representation (only 15% women in parliament, max. 30% women in ministries), inequalities in the judicial and economic fields as well as in education, especially in rural areas.

Although equal in terms of the country`s constitution (Art. 24 and 25), gender balance in society is far from being a reality in Guinea-Bissau. Women are not decision-makers when it comes to access to and management of household goods, even though their contribution to family income is generally growing. Women land ownership rights are very much limited, the inheritance laws are discriminating, and women have nearly no access to loans or other form of financial support. They overwhelmingly live according to traditional gender-specific roles, where a man decides for the family and women are underrepresented in higher levels of value chains in agriculture. Moreover, also high fertility rates, malnutrition, high maternal mortality and high illiteracy hinder women´s equal share and equity in society.

Poverty in Guinea-Bissau is feminized and energy scarcity is closely linked to poverty. Women, particularly in rural/forest areas addressed by this project, make up the majority of the poor. The efficiency of their work is very low in light of the lack of basic sanitary infrastructure, lack of water, lack of firewood, lack of electricity etc.

Awareness on gender issues was raised systematically, for the first time, in DENARP II (2011) – the second National Poverty Reduction Strategy of Guinea-Bissau, which links structural gender inequalities to the economic development of the country and applies gender analysis and gender indicators to explain the situation of Bissau-Guinean women. The PNIEG (2012), the “National Policy for Gender Equity and Equality” follows this approach. New laws against Female Mutilation (2011) and Domestic Violence (2014) are recent achievements in the direction of increased women’s empowerment and gender equality.

Women and the energy sector

The abovementioned barriers constrain the full and equal participation of women in Guinea-Bissau in social development and environmental sustainability. Therefore, their empowerment and efforts towards gender equality in all sectors is crucial to ensure the effectiveness of development activities

[8] ECREEE, Situation Analysis, 2015

involving the use of energy. Gender mainstreaming in RE policies and programming can increase the efficiency of energy policies and free women from time consuming household obligations.

So far the environment and renewable energy policies and laws of the country do not address gender issues appropriately: they foresee only a few gender specific programs and do not analyze and consider gender differentiated needs and priorities: Instead they refer to gender mostly under the notion of vulnerability; the new "National Action Plan on Renewable Energy until 2030" (PANER, 2017) makes some gender references, but does not contain a specific analysis of women's energy needs and specific expected results of the provision of energy for women.

All in all, gender is hidden and nearly completely invisible in these plans. *Gender equality and gender gaps are so far poorly addressed in the relevant Guinea-Bissau National Energy Policy documents: gender differentiated energy needs and applications; the impact of new RE energy use on women; the question of resource ownership and female participation in the energy sector as technicians, employees and women entrepreneurs (Detail on this invisibility in Gender Mainstreaming Action Plan, Annex G).*

Nevertheless, in the energy sector some efforts have been made to implement measures under the regional framework "ECOWAS Policy for Gender Mainstreaming in Energy Access", as well as at the national level (see Gender Action Plan). Women play a pivotal role in energy production, distribution and utilization, especially in poor communities. Thus, the lack of recognition of their role due to wide spread and dominant gender stereotypes can undermine the effectiveness and sustainability of much-needed energy projects and policies.

Women may access mini grid electricity in lower numbers than men; likely because women are overrepresented in poor households and marginalized communities and may not be able to afford connection fees. Whether electrification programs succeed in incorporating gender dimension depends partly on the extent to which they understand and account for different energy needs of women and men (ECREEE, p. 33). There is the risk that energy supply and electrification can increase social inequalities, particularly for women, giving access to a broad range of opportunities to those who can afford to connect and leaving the others still further behind.

Reducing the barriers for women to access and use RE electricity, within the household or outside, will depend on two key factors:

- the provision of the essential complementary infrastructure and a supply with potential appliances;
- the ability of people to pay both for the supply and the end use equipment which rests on availability of credit, as well as of training and extension.

Lack of female participation in the energy ministry and other lead ministries (environment, agriculture) is a very serious barrier. As soon as women share ideas and project design, even if they don´t have the complete technical qualification, the agenda setting, and project selection will change to more accurately reflect the whole population.

Addressing the gender gap

The most urgent interventions aimed at addressing the abovementioned barriers with the objective to secure gender balanced access to RE clean energy and equal benefits for women and men through this project are:

- ***RE off grid energy plants supply energy primarily for female dominated domestic and productive activities*** in the rural areas of Guinea-Bissau at four (4) project sites (Gabu region, Cacheu region, Lake Cufada and Cantanhez Natural Park): multifunctional platform, cooling and storage of fish, fruits, meat, ice, water etc.; water supply for irrigation of horticulture gardens, local Public infrastructure (schools, hospital, administration);

- Development of ***business models that privilege community / women management of new RE energy*** infrastructures and improved cooking stoves etc. with local women organization at the forefront (bottom-up approach);

- ***Access to sustainable energy service liberates women***, because any of the involved technologies frees women from unnecessary work associated with their daily tasks and presents an opportunity for development, income-generating opportunities and education;

- ***Creation of new jobs and increase in women`s economic activities, including women training*** in electricity, solar system maintenance, electricity driven business that are stimulated through RE minigrid installation in rural communities; improved cooking stoves production and commercialization;

- Providing ***women with access to financing***; special fund for loans and micro-credits for women engaged in green business with clean energy and improved cooking stoves; training and management of small business;

- ***Business and technical advisory services*** to the power utility and other mini-grid plant developers: train local authorities and community groups on RE solutions, including the integration of gender-responsive activities (30%-50% women trainees)

- ***Promotion of 'bideira' model of self-reliance***, encouraging women to undertake economic activities higher-up in the value chain;

- ***Mainstream gender in all policies, programs and processes*** of renewable energy management and clean energy production of Guinea-Bissau. This means making women's and men's different concerns and experiences an integral dimension of the design, implementation, monitoring and evaluation of policies and programs to meet the specific needs and priorities of women and men;

- ***Enhance capacities*** for gender analysis and gender mainstreaming in the overall energy management: this requires capacity building of staff and the implementation of gender management systems at the institutional level to provide guidance on gender responsive leadership and decision making;

- Establishment of a ***gender focal point*** at the key ministries to promote inclusive participation for women; ***strong women participation in the decision-making*** processes at national and local level; ***promotion of women entrepreneurship*** in the RE sector and low carbon bioenergy technologies. The increase of women at the political and economic leadership level are a

condition *sine qua non* for successfully gender mainstreaming into the energy sector and renewable alternatives.

- ***Targeted awareness-raising and outreach campaigns to increase women's participation*** in the design and implementation of project activities, - minigrid, - finance mechanism and - low carbon technologies,

- ***Generate and disseminate knowledge*** and information on the differentiated impacts of energy poverty and access to modern RE energy and low carbon bioenergy technologies on women and men, girls and boys in Guinea-Bissau, including through sensitization campaigns;

- ***Promote partnership and cooperation among key actors*** (state- and non-state, national and international) on women energy needs and priorities through off grid clean energy solutions;

- ***The lack, or limited availability, and low accuracy/authenticity of disaggregated data by sex*** prevent an in-depth gender analysis. Therefore, it is a challenge and urgency to gather more sex-disaggregated data related to energy and consumption needs within the framework of this project.

4.6 South-South and Triangular Cooperation (SSTrC):

Guinea is a member of ECOWAS whose countries have set up a center for renewable energies and energy efficiency (ECREEE). Several initiatives have been developed in this context, including the consolidation of the SE4ALL action agenda, national renewable energy action plans and national energy efficiency action plans. All ECOWAS member countries are privileged partners in all fields. Thus, this project fits well with triangular cooperation in the sense that the objectives of the project are a subset of those of ECREEE.

Research and lessons learned could also include projects in the Portuguese-speaking world, including Brazil and Angola, where Guinea-Bissau could benefit from similar experiences in very remote areas (to be verified). Partnerships with technical institutes, universities, research centers, etc. of southern countries such as Brazil, Angola and even Senegal.

Technically qualified staff and expert groups on the issues addressed by the project from these countries will have many opportunities to exchange their experiences and knowledge.
UNDP has an important role to play as a knowledge broker and advocate for capacity building and as a partnership facilitator when developing countries work together to find solutions to their common development challenges. This UNDP-GEF project will support South-South and Triangular Cooperation (SSTrC) through cooperative modalities that will involve bilateral knowledge exchange on implementation procedures, technology transfer and opportunities for gender-based activities. Revenues associated with other small mini-grid projects currently implemented by UNDP in Senegal, Congo-Brazzaville, Congo-Kinshasa, Equatorial Guinea and Sao Tome and Principe. In addition, collaboration will be sought with other countries in Asia and Latin America and the Caribbean where similar projects have been / are being implemented by UNDP-GEF.

Also, the project will liaise with the new regional proposed GEF funded initiative on Africa Mini-grids program (AMP). The objective of the program is to increase access to energy by improving the financial

viability and promoting scaled-up commercial investment in renewable energy mini-grids. The program is led by UNDP, in partnership with RMI, AfDB and BOAD. AMP consists of a cohort of National Projects in 12 countries and a one Regional Project. Coordination will include analysis and presentation of lessons learned, organization of regular face-to-face and virtual networking, knowledge-sharing and outreach activities and events.

4.7 Sustainability and Scaling Up:

Participatory management described in the theory of change will make the project sustainable. Indeed, with the actors, the defined operating mechanisms adopted will remain evolutive according to the conditions. The electric energy will not be free, the costs will be determined in agreement with all the stakeholders. The minimum price per kWh is defined by the conditions of profitability is 130 FCFA / kWh (US $ 0.24 with US $ = 550 FCFA). All villages with a population of at least 50 or more will be impacted by the project.

For improved kilns and stoves, craftsmen will be trained and will be national producers and distributors. They will be supervised and covered by regulatory and even legislative devices if necessary. The sale price of improved cookstoves may vary and be between 6 500 and 15 000 FCFA (US $ 11.82 and 27.27) depending on capacity. All households in the study area will be able to use low carbon technologies no more than 5 years after the project.

Improving the social and economic conditions of the populations affected by the project will facilitate its reproduction. This replication can be done by national and / or foreign investors. The latter will be able to capture the funds dedicated to the environment, to sustainable development ... The extension of the grids will depend on the income-generating activities but also on the success of the improved kilns and stoves.

Sustainability

(a) Technical viability: From a technical point of view, the viability of hybrid power stations, either for main grid supply or for isolated mini-grids for rural electrification, has been demonstrated in several developing countries particularly in Africa (we can mention the case of Senegal). By addressing non-technical barriers to the development of mini-grids in Guinea Bissau, the project will help create a sustainable niche by strengthening the political, institutional, legal, regulatory and operational capacities of key national institutions, supporting technology development through a market-based approach, developing national capabilities and disseminating information. These efforts should ensure the long-term viability of mini-grids for rural electrification in the country.

b) Financial viability: From a financial point of view, the project will support the integration of labor and local industries in the mini-grids sector and improved cookstoves and kilns. This will be achieved by providing targeted support to households willing to engage in small income-generating activities using electricity and / or firewood or charcoal, building the capacity of technical staff, and also specialized workshops. Premises for the manufacture of ancillary equipment for mini-grids and the construction of kilns / stoves, installation, operation, maintenance and repair of systems. With the increase over time of renewable-based mini-grid installations as well as improved cookstoves / kilns, it is expected that these efforts will intensify with job creation opportunities with additional actors entering this domain into the future.

With regard to the financial support given to project promoters, the key to sustainability is the recognition by loan guarantee funds that low-carbon technologies and mini-grids are viable investments. Achieving this point signifies their involvement throughout the project, especially to support project developers for investment. The latter will be provided by the financial support of the pre-project studies while the first will be reinforced by the project management team. The Ministry in charge of guarantee funds may be part of its steering committee. Once the funds are raised, financial institutions can provide loans to project promoters and extend these loans to pre-project studies, especially if these studies can also be covered by the loan guarantee. At this point, subsidies may no longer be necessary.

But, in addition to integrating the guarantee funds, it is important to involve the private sector by making promoters aware of investment opportunities in mini-grids and low-carbon technologies, educating financial institutions about the particularities of investments in the sustainable development sector and strengthening the role of government and development partners as facilitators.

With regard to the promoters, the main professional associations of the sector will be regularly kept informed of the progress of the project. In addition, the success of the first seven mini-power plants will be presented using meeting platforms, newspapers, professional association newsletters, etc. A regular meeting will be organized to share experiences, lessons learned and challenges.

(c) Socio-economic sustainability: The project fully supports the human rights-based approach and will not have any negative impact on the enjoyment of human rights (civil, political, economic, environmental, social or cultural) key or potential stakeholders, the communities involved or the population as a whole.

The project will focus on providing modern and sustainable decentralized energy services to the rural population and, in the process, demonstrate the benefits that sustainable technology (renewable energy and improved stoves / stoves) can bring to improve livelihoods in rural areas. These relate to the social and economic benefits to the villages in terms of a healthier environment for the rural population, opportunities for income-generating activities and improved management of natural resources. Particular attention will be given to strengthening the role of women as actors in the energy sector rather than mere beneficiaries. Women entrepreneurs will be encouraged to manage facilities. Those engaged in the processing and packaging of agricultural products will be at the center of the promotion of electricity for productive purposes. In addition, on-the-job capacity building for mini-grids (Output 2.2) will focus on men and women. These combined activities will help reduce the gender gaps that traditionally exist in the energy sector.

In addition, the use of electricity for the provision of these services as well as low-carbon technologies instead of imported fossil fuels will reduce the country's GHG emissions and contribute to a safer environment for the rural population. In doing so, the development of electricity consumers' capacities will highlight the importance of best practices in energy management and the use of eco-energy devices such as the energy-saving of lamps / radios / televisions when they are not available, the use of LEDs for lighting and the use of efficient energy efficient appliances / motors, etc.

(d) Environmental sustainability: Guinea Bissau will use all of its strategies to address climate change to systematically integrate climate change considerations into the development of mini-grids and low-carbon technologies. This will facilitate decision-making on energy infrastructure and service delivery

options to account for the uncertainty associated with climate change predictions and to assess the climate resilience of different options. For example, decisions to invest in mini-grids should take into account weather forecasts. The project will ensure that the country's climate change portfolio agencies are actively involved in the project coordination mechanism to promote an integrated approach.

The project will have a direct positive effect on environmental sustainability, as the main objective of the project is to accelerate the use of renewable energy and low-carbon technologies for the overall good of the rural population. This will benefit both the country's economy and the global environment by reducing greenhouse gas emissions.

Scaling up

The potential for extension of the project in the country is very good as 60% of the country's population lives in rural areas without access to electricity or modernized energy services. This represented about 1.1 million people in Guinea Bissau in 2018 and represents about 142,000 households. The project will adopt a bottom-up approach in the overall policy / investment framework that should be developed to promote renewable energies focused on mini-grids for rural electrification. Technical assistance for barrier removal and Institutional strengthening to be provided under the project will facilitate this replicability, as it will create the institutional, political and technical conditions necessary to generate renewed investor interest in the development of additional projects in this area. In addition, the lessons learned will be of great value to neighbouring countries sharing a similar resource base, respective renewable energy resources for rural mini-grid rural electrification.

As noted above, 60% of the country's population lives in rural areas without access to electricity services. With regard to the annual per capita electricity consumption in the whole country, it is 29 kWh (PNER, 2015), well below the African average of 579 kWh and the world average of 2 777 kWh. On the other hand, the country has a potential of more than 31 MW of renewable energy resources (hydroelectricity and tidal power), without any development so far. For solar photovoltaic the potential is 5.5 kWh / m2 / day and whose development has boomed in the last five years. The wind potential with winds estimated between 2.5 and 5.3 m / s of speed remains as unexploitable today. (PNER, 2015) This situation therefore presents enormous potential for scaling up, using a strong economic model and capacity building for mini-grids and low-carbon technology provided to stakeholders at various levels, combined with a strong outreach / sensitization program, will encourage private sector participation in electricity generation and low-carbon technologies through improved networks and stoves / stoves to meet the needs of rural consumers according to the decentralized energy policy this will aim to provide "access to electricity services to all rural and urban residents at an affordable cost".

V. PROJECT MANAGEMENT

5.1. Cost efficiency and effectiveness:

Project management: The project will be operationalized using a project management unit (PMU). The main management roles of the PMU include:

- Lead the development of project design, including the preparation of terms of reference for consultants and subcontractors, identification and selection of national and international subcontractors / consultants, cost estimation, scheduling, procurement and establishment reports on project activities and budget.
- Support the activities of international / national experts, potential investors and subcontractors and provide general administrative / financial support to project activities.
- Grant Management.

Agreement on intellectual property rights and the use of the logo on project deliverables and disclosure of information: In order to grant GEF appropriate recognition for grants, the GEF logo will appear with the UNDP logo on all promotional material, other written material such as project-developed publications, and project material.. Any reference to publications on GEF-funded projects will also grant appropriate recognition to the GEF. The information will be disclosed in accordance with relevant policies, including the UNDP Disclosure Policy and the Public Participation Policy of the GEF..

5.2. Theory of change

Sustainable development is defined as: "Development that meets the needs of the present without compromising the ability of future generations to meet their own needs. (Bruntland Commission, 1987). This implies taking into account the concept of need, and more particularly the basic needs of the poor, who should be given the highest priority on the one hand; on the other hand, the idea of the limitations that the state of our techniques and our social organization impose on the capacity of the environment to meet current and future needs.

Sustainable development is essentially based on three pillars:

• The social pillar: social consequences of economic activity, problem of inequalities, working and living conditions

• The environmental pillar: Respect of biodiversity and ecosystems, reduction of polluting emissions, non-destruction of natural capital

• The economic pillar: economic development

Figure 5 presents the concept of sustainable development that will be supported by the theory of change.

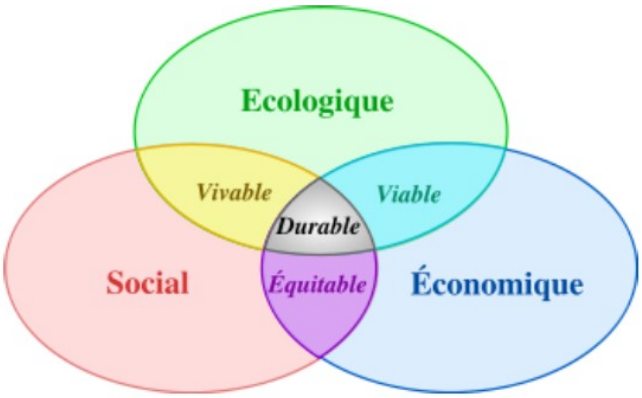


Figure 5 : Pillars of sustainable development

Guinea Bissau is a country with an ecological richness (73% of the territory covered by the forest) which constitutes a strong economic potential. Low social development ranks Guinea Bissau among the poorest countries in the world, with a Human Development Index (HDI) of 0.455.

The country has a range of renewable energy sources, with different development potential. Biomass, hydroelectricity, solar and wind are available.

Biomass: Guinea Bissau has forest resources (estimated at about 2 million hectares) and agricultural residues (rice husks, cotton stalks, cashew nuts, sugar cane residues, etc.). The amount of biomass used is estimated at about 738 000 tonnes per year. But biomass is not currently exploited in a sustainable or renewable way.

Hydropower: The hydroelectric potential of Guinea-Bissau is estimated at 33 MW. This potential would be achieved with the implementation of two hydroelectric dams, Cusselinta (13 MW) and Saltinho (14 MW), and several small / micro hydro systems, according to studies conducted in the 1980s.

Solar: Solar energy is the most abundant source of renewable energy in the country. Annual irradiation ranges from 4.79 kWh / m2 / day to 6.96 kWh / m2 / day, with an annual average of 5.87 kWh / m2 / day. So far, the solar resource has been used to a limited extent for the production of electricity with photovoltaic (PV) systems in rural households, schools, offices, hospitals and health centers; for solar pumping of water; for solar telecommunication systems; and for specific applications of solar water heater. But these were essentially donor-driven and unsustainable initiatives.

Wind: There are no appropriate studies on the wind potential of Guinea-Bissau. Roughly, wind resources are considered to be average, wind speed (measured at 50 meters) being greater than 4 m / s. Wind energy can be exploited throughout the year on the mainland coast or between the islands and in the Boe region (east), where the altitude exceeds 200 m. There is no current exploitation of the wind resource.

Guinea-Bissau could thus be described as an "energy paradox" because of its considerable resources and the rate of electrification and access to other forms of modern energy in the country. Indeed, despite the enormous quantity and rich diversity of renewable energy sources, Guinea-Bissau faces innumerable obstacles to adequate access to modern energy services. The production and distribution grid is dilapidated, the cost of electricity, despite a subsidy of about 100 FCFA (US $ 0.18) per kilowatt, is among the highest, the overall rate of electrification is one of the lowest in Africa (11.5%) and less than 1% in rural areas. In addition, the politico-institutional and financial environment is so weak that energy supply has fallen by 80% within 10 years (2003-2013). The difference between supply and demand is enormous (supply varies between 2 MW and 8 MW and demand is 30 MW). In addition to the significance of the significant energy deficit, the rate of deforestation is worrisome. In fact, 1% of the forest sector has been deforested every year since 2007. Every year, 738,000 tons of firewood and

charcoal are harvested from the forests of Guinea-Bissau. Bioenergy production technologies are very inefficient. Their efficiency varies from 5% to 7% for the three stone stoves, up to 33% for kilns and 12% to 15% for metal kilns.

The RE supply model designed by the project not only impacts most positively on women as beneficiaries and consumers of energy but puts the productive economic activities of women as centrepiece of power supply design and participatory energy management structures (business model).

The gender responsive project design is based on three core elements with a history of evidenced successful implementation.

WOMEN AS AGENTS OF CHANGE

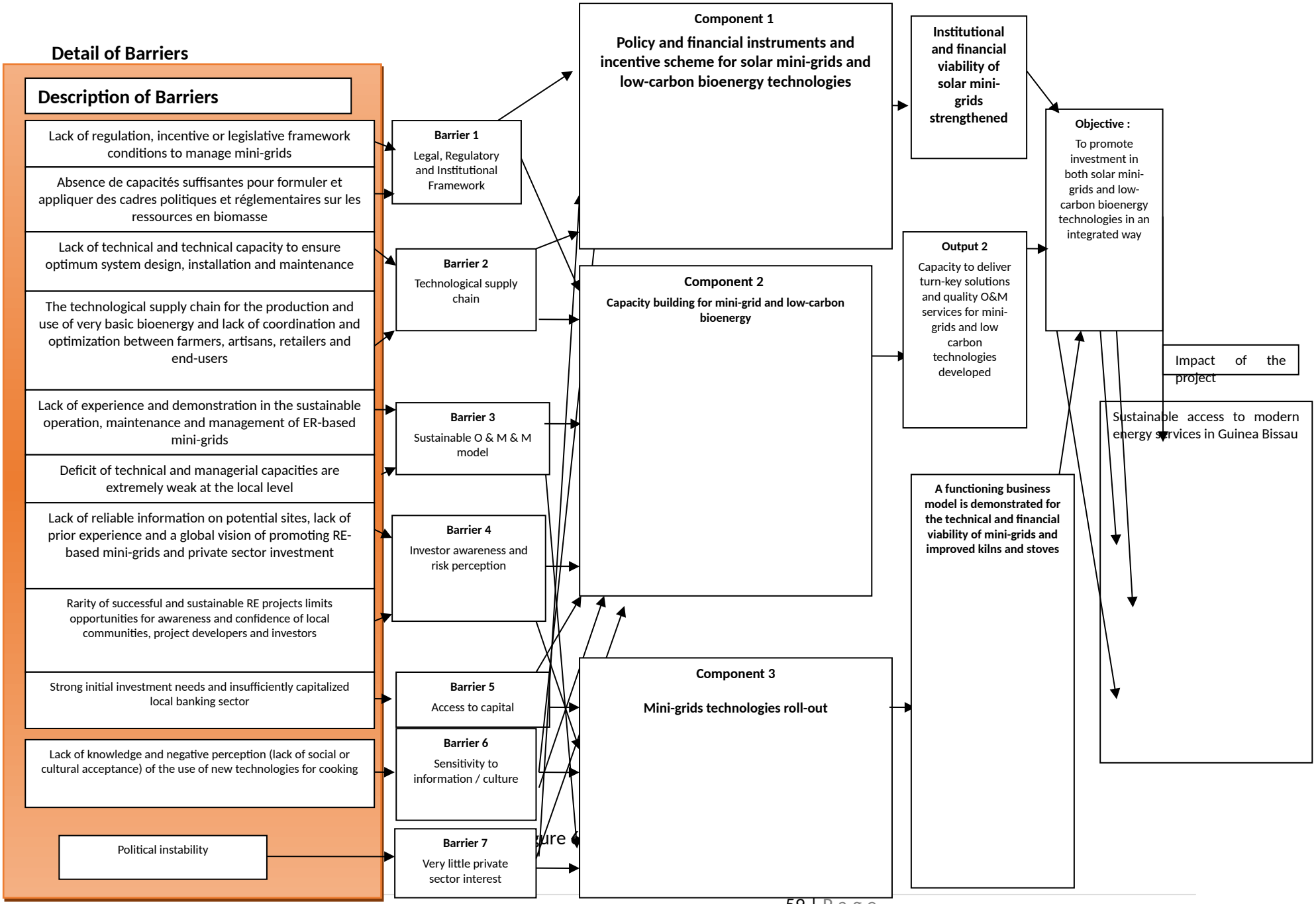
1. Energy installation focused on agro-processing machines (existing or partly existing in the visited communities) which enable women farmers and women led households to increase and diversify their income
2. Provision of energy centred on cooling and storage of fish as well as other products of seasonal character (different types of fruits and vegetables, cold water, ice, meat, chicken). As women traditionally commercialize fish, the management of cooling and storage systems is with local women groups.
3. Application of low carbon bioenergy technologies by and for women who are active agents in biomass management: they are the primary household cooks and they participate in many biomass-intensive processing sectors (on the subsistence and income generating level). As such, they will play a key role in shifting the country onto a cleaner development pathway.

The objectives of this project are the adoption and promotion of low-carbon energy technologies for communities dependent on protected areas. This will take into account renewable energy mini-grids, improved kilns and stoves. The achievement of these objectives remains conditional on the satisfaction of the following assumptions:

- Hypothesis 1: If the strengthening of the legislative and regulatory framework is effective,
- Hypothesis 2: If the involvement of stakeholders (stakeholders) is real,
- Hypothesis 3: If the capacities and skills of human resources are strengthened
- Hypothesis 4: If the necessary means (financial and technical) are implemented

Therefore, living populations in protected areas will have a sustainable source of electricity and will sustainably exploit forest resources. They will contribute to greenhouse gas reductions.

The diagram in Figure 6 presents the theory of project change.


Detail of Barriers
Description of Barriers
Lack of regulation, incentive or legislative framework conditions to manage mini-grids
Absence de capacités suffisantes pour formuler et appliquer des cadres politiques et réglementaires sur les ressources en biomasse
Lack of technical and technical capacity to ensure optimum system design, installation and maintenance
The technological supply chain for the production and use of very basic bioenergy and lack of coordination and optimization between farmers, artisans, retailers and end-users
Lack of experience and demonstration in the sustainable operation, maintenance and management of ER-based mini-grids
Deficit of technical and managerial capacities are extremely weak at the local level
Lack of reliable information on potential sites, lack of prior experience and a global vision of promoting RE-based mini-grids and private sector investment
Rarity of successful and sustainable RE projects limits opportunities for awareness and confidence of local communities, project developers and investors
Strong initial investment needs and insufficiently capitalized local banking sector
Lack of knowledge and negative perception (lack of social or cultural acceptance) of the use of new technologies for cooking
Political instability
Barrier 1
Legal, Regulatory and Institutional Framework
Barrier 2
Technological supply chain
Barrier 3
Sustainable O & M & M model
Barrier 4
Investor awareness and risk perception
Barrier 5
Access to capital
Barrier 6
Sensitivity to information / culture
Barrier 7
Very little private sector interest
Component 1
Policy and financial instruments and incentive scheme for solar mini-grids and low-carbon bioenergy technologies
Component 2
Capacity building for mini-grid and low-carbon bioenergy
Component 3
Mini-grids technologies roll-out
Institutional and financial viability of solar mini-grids strengthened
Output 2
Capacity to deliver turn-key solutions and quality O&M services for mini-grids and low carbon technologies developed
A functioning business model is demonstrated for the technical and financial viability of mini-grids and improved kilns and stoves
Objective :
To promote investment in both solar mini-grids and low-carbon bioenergy technologies in an integrated way
Impact of the project
Sustainable access to modern energy services in Guinea Bissau

ure 6

Hypotheses are assumptions that fit into the "if- then" logic of the results framework: they reflect the conditions that must be met for UNDP activities to achieve the expected results.

Risks are developments or events, often external and difficult to control, that influence the implementation of objectives. To the extent possible, they should be mitigated through prevention, advocacy and political commitment. If risks arise, program adjustment mechanisms should be used.

3.2.1 Assumptions

The success of the low-carbon energy services access project is essentially based on four assumptions

- **Hypothesis 1: If the strengthening of the legislative and regulatory framework is effective**

Guinea Bissau is a member country of ECOWAS and UEMOA. In this context, several mechanisms and laws are being developed to promote renewable energy and energy efficiency in the sub-region. This country, like other ECOWAS members, has not adapted these mechanisms. The following factors will provide a legislative and regulatory framework conducive to the development of renewable energy and energy efficiency:

1. Support the government to establish a clear rural electrification policy that encourages low-cost and integrated planning for all options
2. Assist the government in developing a comprehensive and fully integrated electrification plan with low-cost planning to determine where expansion is the most efficient and sustainable approach to increasing energy access in relation to off-grid sector development - mini grids and autonomous systems powered by renewable resources
3. Assist the government to develop a comprehensive, low-cost and integrated plan for all rural electrification options
4. Assist the government in establishing the medium- and long-term rural electrification strategy in the country by developing and implementing a rural electrification master plan
5. Assist the government in creating a lead agency / rural electrification agency with a clear mandate to coordinate activities with the private sector, the donor community and the national and local levels to accelerate off-grid market growth to achieve energy access objectives
6. Assist the government in creating public-private partnership systems to share the high costs of project development and market entry, particularly with developers in remote areas
7. Assist the government in developing procurement and public finance laws that will facilitate stand-alone renewable energy system investments in public facilities (schools, health care facilities, etc.)

- **Hypothesis 2: If the involvement of actors (stakeholders) is real,**

Stakeholders are important to the success of any project. In Guinea Bissau, several actors are involved in the development of forest areas. The most successful projects are those that are developed and implemented with the population. Among these cases, we can mention the projects developed by IBAP in Lamane and Duta Djara or the electrification of Bambadinka with the NGO TESE. On the basis of these observations, the following points will help to ensure better involvement of the various stakeholders

1. Assist the government in developing a new flexible legal framework and creating appropriate incentives for private sector participation in off-grid market development
2. Support the government to improve the planning framework to encourage private participation in mini-grid and autonomous solar system options
3. Assist the government in developing clear licensing and concession permit procedures
4. Assist the government in developing appropriate information sharing and dissemination systems for project promoters and key stakeholders, including the creation of a single window for national authorisations and the rapid issuance of local permits.
5. Assist government, trade associations or civil society organizations in developing bilateral energy services and bilateral contracts
6. Support the development and implementation of programs to educate consumers, retailers and distributors on the benefits of quality certified solar products (compared to "non-standard" products)
7. Assist in the development of strategies to encourage the inclusive participation of men and women

- **Hypothesis 3: If human resources capacities and skills are strengthened**

All investment projects are intended for human resources, during and at the end of the process. However, their involvement requires a certain elementary expertise. Several projects in Africa are not successful due to a lack of technical expertise for monitoring. In Guinea Bissau, all multifunctional platforms are at a standstill due to a lack of maintenance for the most part. The following aspects will be taken into account to strengthen the capacities and skills of human resources to support this project.

8. Help the government to strengthen its technical and vocational training centres
9. Support the establishment of technical certification and vocational training programs through government, the private sector and/or universities for the installation and maintenance of autonomous solar systems.
10. Assist the government in establishing international quality standards for off-grid autonomous solar products, including minimum technical standards (IEC technical specifications), warranties, availability requirements and cost guidelines for after-sales services / O&M, etc.
11. Assist the government in integrating standards with appropriate supervisory agencies to ensure that quality assurance procedures are in place
12. Support the development of a database on best practices / information sharing services to ensure the transfer of expertise from international, local and regional initiatives
13. Assist the private sector and / or civil society organizations to ensure the implementation of environmental / social standards
14. Support the establishment of a framework for the repair and recycling of off-grid solar systems and equipment
15. Support government, trade associations and civil society organizations to develop and implement awareness, marketing and consumer education programs on the benefits of off-grid solar products and the existence of related national programs.

- **Hypothesis 4: If the necessary means (financial and technical) are implemented**
-

Most of the projects fail due to lack of technical and/or financial resources. In Africa, if these are available, they are not adapted and/or poorly managed. The success of this project depends on the choice of technical means and the management of financial means. The development of the following aspects will help to duplicate the pilot systems that will be implemented as part of the project.

16. Assist the government in establishing a legal framework that allows companies or public authorities to prosecute those caught distributing counterfeit or low-quality products that do not comply with the promulgated standards
17. Assist the government in analyzing areas where fossil fuel subsidies are an obstacle to the development of sustainable solutions
18. Assist the government in developing appropriate VAT and pricing policies covering the entire supply chain of off-grid / stand-alone solar products (including batteries, inverters or other system components) that would provide the necessary support to the sector.
19. Assist the government in establishing an ad hoc working group to mitigate potential difficulties in customs clearance and import logistics
20. Assist the government in setting up appropriate subsidy programs that require private funding
21. Inform regulators, governmental and non-governmental actors of the different pricing systems proposed by suppliers of autonomous solar systems to improve understanding and avoid unnecessary interventions.
22. Support public and off-grid companies to collaborate specifically in the development of pricing systems for the productive use market segment
23. Assist the government in developing appropriate procedures and guidelines to protect itself from blocked investments from competition between all rural grid and off-grid electrification approaches

4.3. Risks

In accordance with UNDP standard requirements, the project manager will monitor risks on a quarterly basis and report on the status of risks to the UNDP Country Office. The UNDP Country Office will record progress in the UNDP ATLAS risk register. Risks will be considered critical when the impact and probability are high (i.e. when the impact is assessed at 4 and the probability at 3 or more). Management responses to critical risks will also be reported to the GEF in the annual evaluation report.

The following table summarizes the various risks associated with the project.

Table 4: Risk table

Description	Type	Probability and Impact	Mitigation Action	Status
Instability Guinea-Bissau faces regular political instability. Several coups have taken place in the past, the last one in 2012. A new elected government has been in power since 2014. Elections will be held on November 18, 2018. Before 2019, a new legal government should be put in place to allow us to reassess the issue of instability. If further political instability occurs, it will certainly have a negative impact on the overall investment climate and delay the implementation of the project.	Political	P=4 I=4	The project will work as much as possible with decentralized authorities in provinces and rural areas. The political will to support this project in these regions is strong. The impact of political instability at the national level is most evident in the capital, Bissau. The project will also constitute a broad coalition of partners and stakeholders whose interest in rural development is likely to be maintained, even in the event of regime change. They include local businesses and communities, NGOs and international development agencies…	No change in the situation
Politics Timid support for a framework to encourage the private sector to invest in mini-grids for rural electrification.	Operational	P=2 I=4	There is a possibility that the government may not intervene early enough in a policy framework that will encourage the private sector to invest in mini-grids for rural electrification; for example, there is no rural electrification master plan. If this were to happen, the implementation of the project would be hindered. However, the Government is strongly motivated to provide access to modern energy services to rural populations using traditional forms of energy, to improve their quality of life and for income-generating activities, and is	New

Description	Type	Probability and Impact	Mitigation Action	Status
			motivated by its plans (PANER, PANEE, SE4ALL...) to achieve the Sustainable Development Goals. In addition, the project's interventions under Component 1 will help to mitigate this risk.	
Technology Insufficient quality of PV equipment leading to • the first failure of mini-grid, pumping and storage systems • consumer doubts about renewable energy.	Operational	P=2 I=2	There is a possibility that the government may not intervene early enough in a policy framework that will encourage the private sector to invest in mini-grids for rural electrification; for example, there is no rural electrification master plan. If this were to happen, the implementation of the project would be hindered. However, the Government is strongly motivated to provide access to modern energy services to rural populations using traditional forms of energy, to improve their quality of life and for income-generating activities, and is motivated by its plans (PANER, PANEE, SE4ALL...) to achieve the Sustainable Development Goals. In addition, the project's interventions under Component 1 will help to mitigate this risk. New Technology Insufficient quality of PV equipment leading to - the first failure of mini-grid, pumping and storage systems - consumer doubts about renewable energy. Given the low literacy rate and lack of technical capacity among rural communities, maintenance deliveries represent a significant risk for mini-grid system operations. Minor repairs must be done by local-qualified personnel to prevent equipment from idling for long periods of time. Spare parts must be standard among locations, locally manufactured if possible, easily available for transport and installed at minimal	No change

Description	Type	Probability and Impact	Mitigation Action	Status
			cost. Technical and operational capacity building among rural communities will be critical to mitigate these technical risks. This will be done by providing basic technical training in rural areas and by sponsoring local institutions that take on maintenance tasks.	
Financial Widespread poverty and lack of sustainable sources of income, leading to low capacity to pay for modern energy services. Households may not be able to purchase improved stoves; and improved stoves may not be affordable for operational charcoal producers In addition, there is a risk on financial sustainability for investment in min-grids and low-carbon bio-energy technology (for example, suspension of credit financing and failure of loan guarantee funds).	Operational	P=2 I=3	The project will conduct assessments of the capacity and willingness to pay of end-users. In addition, the combination of the electricity utility business model and the private sector business model through public-private partnerships (PPPs) will reduce financial risk on both sides (public and private sector). The role of microfinance (adapted to the market of low-income people) and loan guarantees (financial risk reduction) will also help to reduce risk… In regard to the risk of failure with credit financing and loan guarantee funds, both mini-grids and low carbon technology development are fairly well-known among lending institutions throughout the world; hence, securing loans in the international finance market for investment in this area does not pose much of a problem. Therefore, the project will work to establish a combination between local and international markets (for example bringing foreign private sector companies) in order to have a balance between local and international credit financing options and opportunities.	No change
Market	Operational	P=3 I=3	The introduction of financially and socially sustainable tariffs for mini-grids based on renewable energy will be a fundamental instrument of the proposed package, specifically aimed at addressing this risk by levelling the playing field with regard to available alternatives.	No change

Description	Type	Probability and Impact	Mitigation Action	Status
In Guinea-Bissau, RE systems will have to compete with locally available, subsidized alternatives to diesel. Without additional incentives, sustainable mini-grid companies can remain uncompetitive				
Social There will be limited social and/or cultural acceptance (this applies in particular to improved stoves, as people will have to adjust their behaviours, which is notoriously difficult to achieve). Although the project plans to address this directly, some aspects may remain beyond the project's control.	Operational	P=2 I=3	The PPG phase will develop a detailed analysis of the socio-economic aspects of the use of improved stoves. However, if a particular community or village does not wish to change its cooking habits, the project will focus on the benefits of improved stoves and provide examples of communities / villages that have adopted them.	No change

P = Probabilité sur une échelle de 1 (faible) à 5 (élevée). **I** = Impact sur une échelle de 1 (faible) à 5 (élevée

4.3 Knowledge management

The project results will be collected, documented and disseminated in an appropriate manner within and beyond the project intervention through existing networks and information sharing forums. A tailored audience, specific knowledge products and channels adapted to the target groups will be defined to ensure effective dissemination of best practices and lessons learned from the project.

The country will participate in relevant platforms and networks, such as the Renewable Energy Trade Platform in the West and African sub-region, providing and receiving inputs. The project proposal will therefore define how information on national projects will be shared and updated on these platforms. Sharing lessons learned and experiences through the platform will ensure the alignment of this proposed project with other national, regional and global transparency initiatives.

To promote learning, Component 2 training activities will be complemented by peer-to-peer visits and participation of relevant government staff in international conferences, workshops and meetings. Importantly, the exchange of lessons learned will be a two-way and participative process.

The project will follow seven elements that are recommended in a knowledge management approach as best practices: 1) Overview of existing lessons and best practice that inform project concept; 2) Plans to learn from relevant projects, programs, initiatives & evaluations; 3) Proposed processes to capture, assess and document info, lessons, best practice & expertise generated during implementation; 4) Proposed tools and methods for knowledge exchange, learning & collaboration; 5) Proposed knowledge outputs to be produced and shared with stakeholders; 6) Discussion on how knowledge and learning will contribute to overall project/program impact and sustainability and 7) Plans for strategic communications.

1) Overview of existing lessons and best practice that inform project concept.

The existing lessons and best practices that served for drafting the proposal are drawn from the ongoing solar mini grid projects in Guinea Bissau (GB) but also form the remarkable off grid market development in other countries in Africa.

Overall the energy access in Guinea Bissau is insufficiently developed due to the weak ecosystem: policies and enabling instruments, a narrow private window and investment in this nascent market, limited technical knowledge, lack of sustainable O&M practices and lack of ownership the beneficiaries.

On the opposite, Africa has become the leading region for energy access development with countries such as Kenya, Uganda, and Senegal paving the way with ground breaking market penetration strategies and innovative financial systems to provide clean and affordable energy to an increasing rural and peri-urban population. Some of these countries have reached a level of market maturity whereby donor money and government intervention are becoming quite redundant as the private sector has been enabled to deploy successful market penetration technologies and strategies. Some of the experience gathered from these countries can successfully be imported in GB.

The underlined proposal integrates most of the best practices gathered in Africa. Those are but not limited to: (1) designing and implementing genuine policies and regulations to remove barriers for the private sector; (2) involving a large variety of actors to enable all the technical, financial, social, operational and organisational facets to be in place; (3) derisking investment in the sector through incentives; (4) technology transfer, development of local skills and abilities in all manners; (5) strong involvement of local communities, local champions, etc to create ownership.

2) Plans to learn from relevant projects, programs, initiatives & evaluations;

   a. Through Component 2, the project (with coordination with the Government), can organise a study tour in Senegal for instance to get acquainted with the success factors underlying the booming off-grid market, specially getting relevant information from the role of the Government to trigger the market growth.
   b. It is equally required to assess thoroughly why the mini-grid market is not taking off in GB by determining also the missing ingredients in the sector.
      - What has not been done?
      - What is not in place?
      - What are the shared responsibilities of the Government institutions, the private sector and the population.

3) Proposed processes to capture, assess and document info, lessons, best practice & expertise generated during implementation;

   a. Put into place processes for regular information collection and reviews to detect trends, common roots barriers and successful practices;
   b. Provide feedback to policy and decision makers to enable the Improvement of the ecosystem.
   c. Regular projects reviews and evaluation by independent actors

4) Proposed tools and methods for knowledge exchange, learning & collaboration;

   a. Creation of a national team from different public and private institutions including representatives of the beneficiary that will oversee and monitor the different projects and programs in the sector and provide feedback and recommendation to decision makers.
   b. Design standard questionnaires and fiches to collect all relevant information on project development, implementation, operation in order to monitor the performance;
   c. Train all actors on the contents of the questionnaires and fiches
   d. Regular follow up to collect the requested information

   In terms of training and overall capacity development, the following activities are proposed:

   a. Training of policy makers and government institutions to evaluate the needs (policy gap) and formulate appropriate policies and regulation for unlocking the off-grid market
   b. Training program to develop / upgrade the technical skills for off-grid project preparation, implementation, O&M and monitoring;
   c. Training of local institution on monitoring and evaluation of off-grid projects

d. Assisting local education facilities to develop curriculum related to renewable energy technologies at different level: master, engineer, technician and skilled labor
e. Positioning 2 to 3 Technical assistants specialized in off-grid regulation, tariff setting and technical norms.
f. Developing expertise for systematic best practices information collection, processing and dissemination.

5) Proposed knowledge outputs to be produced and shared with stakeholders.
   a. Guidelines for communities' mobilizations and involvements; customer relations and conflict resolution procedures in case of illegal connections or theft, engagement of productive end-users, etc.
   b. Guidelines and recommendations for technical oversight over plant operations and responsibility for repairing faulty equipment;
   c. Practical guide to accounting and administrative and financial management of off-grid systems, billing and payment collection system;

6) Discussion on how knowledge and learning will contribute to overall project/program impact and sustainability
   a. Promoting quality with the view of gaining the confidence of the beneficiaries and users. Discussions will be initiated with relevant government institutions to adopt specifics norms and standards to safeguard the quality of the equipment and processes through certification and labelling. The custom office and relevant institutions need to be included in the discussion to prevent fraud and entrance of inappropriate equipment in the country.
   b. Used batteries and electronic waste are a real menace and can reduce strongly the sustainability of mini grid program. Discussion with relevant actors on waste collection and management need to be held in the early stage of the projects so as to find solutions for systematic waste control.
   c. Appropriate tariff setting is crucial to guarantee enough returns to investors while making the electricity services affordable to the population and small businesses. Several ministries and public institutions, private sector and consumers representatives should be involved in these discussions.

7) Plans for strategic communications.

The plan for strategic communication will seek to (i) establish trust between actors, partners and the community; (ii) manage expectations so that beneficiaries and partners are aware about the whereabout of project to forge ownership; (iii) encourage participation and collaboration to in meaningful dialogue about the programs. Beside traditional medium as a project website and regular publications the following can be undertaken:

   a. Organise periodic regional and national stakeholder dialogue workshops on the projects expected outputs and challenges,
   b. Document and disseminate lessons learned and best practices to relevant regional and national partners.
   c. Access global best practices share them with other local and international stakeholders and ensure incorporation into the Project planning stages.

d. Ensure programming is informed by Project audit and evaluation findings, formulating a detailed action plan for implementation of audit/evaluation recommendations.

4. 4. Innovativeness (innovative nature of the project, how the project is innovative compared to the projects written in the reference scenario)

Guinea Bissau has low access to electricity. This rate is estimated at 1% in rural areas. Wood energy is the most widely used source of energy and causes unprecedented deforestation in the country (7380,000 tonnes of wood and coal per year).

The project has several distinctive characteristics, which makes it highly innovative in the context of Guinea-Bissau. First, it creates a strong link between forest and basic energy needs: by promoting the use of charcoal and sustainable fuelwood, the project will not only reduce the pressure on biomass resources, but also promote solar energy. The project will also test a combination of three business models (utility business model, PPP business model, and the private sector business model), exploring the advantages of these models to support decentralized mini-grids. It will also focus on identifying and supporting private-sector-led RE projects (as opposed to the traditional public/donor approach), thereby maximizing long-term financial and operational sustainability. Finally, unlike the traditional approach of providing easily available turnkey solutions for rural electrification, the project will work with the entire national value chain for photovoltaics and bioenergy, starting with design, construction and commissioning, maintenance and management.

4.5. Potential for scaling up and replication

Improving the social and economic conditions of the populations affected by the project will facilitate its reproduction. This replication may be carried out by national and/or foreign investors. The latter will be able to collect funds dedicated to the environment, sustainable development... The extension of the networks will depend on income-generating activities but also on the success of improved kilns and stoves.

4.6. Sustainability

The participatory management described in the theory of change will make the project sustainable. Indeed, with the actors, the defined operating mechanisms adopted will remain scalable according to the conditions. Electricity will not be free, the costs will be determined in common agreement with all stakeholders. The minimum price per kWh, on the other hand, will be defined by the profitability conditions.
For improved kilns and stoves, artisans will be trained and will be national producers and distributors. They will be supervised and covered by regulatory and even legislative measures if necessary.

VI. PROJECT RESULTS FRAMEWORK

This project will contribute to the following Sustainable Development Goal (s): Goal 7: Ensure access to affordable, reliable, sustainable and modern energy for all; **Goal 13**: Take urgent action to combat climate change and its impacts; and **Goal 5:** Achieve gender equality and empower all women and girls.
This project will contribute to the following country outcome included in the UNDAF/Country Programme Document: Outcome 2.2 outcome 4: "National institutions implement appropriate gender-sensitive policies and strategies to promote sustainable economic development" and outcome 5: "the most vulnerable populations in the poorest regions have greater access to economic opportunities".
This project will be linked to the following output of the UNDP Strategic Plan 2018-2021: Development Setting B: Accelerating structural transformations for Sustainable Development. Signature solution 4: Promote nature - based solutions for a sustainable planet

Table 10: Logical framework of the project

Objective	Indicator / Sub-Indicator	Reference	Targets Mi -	Project Targets End of Project	Sources of Verification	Risks and Assumptions
Promote better access to modern energy services through sustainable mini-grids and low-carbon bioenergy technologies among forest-dependent communities in Guinea-Bissau	Emissions reduction (in tCO2 over a lifetime project of 25 years). Investment in mini-grids based on renewable energy. Installed capacity (MW) by mini-grids. Use of improved kilns and stoves Number of jobs created. Number of recipients among households and businesses in rural areas.	The country's GHG emissions were 378,081 million tonnes in 2011. The current contribution of rural energy services is negligible. No investment in the provision of rural energy services through the production of electricity from mini-grids based on RE.	A capacity of 0.800 MWp of solar PV has been installed, resulting in an investment of $ 5 million. Cumulative reduction of 876 tons of CO2 / year. Total of 50 jobs created. 500 beneficiary	A total capacity of 1,193 MWp installation of solar PV has enabled an investment of nearly $ 14 million. Reduction of 93,900 tons of CO2 over the 25-year lifetime of mini grids, improved kilns and stoves. Total of 200 jobs created. More than 1,500 beneficiary households and 50 small commercial / industrial enterprises in rural areas.	Annual project reports, GHG monitoring and verification reports. Mid-term review of the project and final evaluation reports.	Ongoing engagement of project partners, including government agencies and investors / developers.

Objective	Indicator / Sub-Indicator	Reference	Targets Mi -	Project Targets End of Project	Sources of Verification	Risks and Assumptions
			households and 25 small commercial / industrial enterprises in rural areas			
Component 1: Policy and financial instruments and incentive scheme for solar mini-grids and low-carbon bioenergy technologies						
Outcome 1: Institutional and financial viability of solar mini-grids strengthened	*Policies and strategies for the development of approved and operational mini grids.*	ECOWAS Texts (ECREEE)	Policies in place and approved	Policies operational	Project documentation	Commitment of government entities
Output 1.1: Policy package to operate Solar mini-grids developed	*Availability of a set of policies for the development of mini-grids.*	ECOWAS Texts (ECREEE)	Policies in place and approved	Policies operational	Project documentation	Cooperation and interest of government entities.
Output 1.2: Financial incentives and market mechanisms to secure supply and stimulate demand for energy-efficient stoves and firewood/charcoal developed	*Existence of a financial instrument to support the development of mini-grids.*		Policies in place and approved	Policies operational	Project documentation	Cooperation and interest of government entities.
Output 1.3: Developed gender-sensitive capacity development and modules for the	*Existence of incentive schemes for women producers and users of improved stoves*	Not available at the moment.	Completed within 9 months of project launch.	Completed within 09 months of the end of the project.	Project documentation.	Stakeholder cooperation and NGOs

Objective	Indicator / Sub-Indicator	Reference	Targets Mi -	Project Targets End of Project	Sources of Verification	Risks and Assumptions
production and utilization of improved kilns and cookstoves	*and stoves.*					
Output 1.4: Alternative sustainable solutions to wood energy explored and assessed	*Use of biogas and effective electric stoves*		Start of use of firewood alternatives within 24 months of project launch	Completed within 6 months of the end of the project	Project documentation. Mid-term report	Continued interest of the private sector
Component 2: Capacity building for mini-grid and low-carbon bioenergy						
Outcome 2: Capacity to deliver turn-key solutions and quality O&M services for mini-grids and low carbon technologies developed	Completion of capacity building activities of stakeholders.	Not available at the moment.	Existence of turnkey solutions	N/A (completed by Mid-term project lifetime)	Project documentation.	Cooperation of all stakeholders.
Output 2.1: Tailored capacity building programme delivered to public sector and civil society	Number of persons/institutions trained.	PERACOD guide and ECREEE document	100	200	Published documents.	Commitment of different government institutions and NGOs.
Output 2.2: Various support delivered to and established for the private sector	*Number of private sector initiative*	Not available at the moment.	2	7	Project reports.	Ongoing commitment of project developers.
Output 2.3: On-the-job capacity building programme for 100 solar mini-grid installers delivered, including on	Number of persons/institutions trained.	Not available at the moment.	40	100	Evidence of a fully operational advisory services unit. Project reports.	Ongoing commitment of different government institutions and project developers.

Objective	Indicator / Sub-Indicator	Reference	Targets Mi -	Project Targets End of Project	Sources of Verification	Risks and Assumptions
materials, plant design, combination, construction, O&M						
Output 2.4: Capacity building for the fuelwood value chain delivered	Number of persons/institutions trained.	Not available at the moment	60 craftsmen and other farmers trained.	300 craftsmen and other farmers trained.	Project documentation	Designation of staff by government departments / other institutions concerned.
Component 3: Mini-grids and low-carbon bioenergy technologies roll-out						
Outcome 3: A functioning business model is demonstrated for the technical and financial viability of mini-grids	Business model defined, demonstrated and ready to be widely used.	No such model available now.	1 business model is applied	1 business model is confirmed and demonstrated	Project reports.	Government entities and the private sector willing to cooperate.
Output 3.1: 7 pilot sites totaling 1.2 MW for mini-grids identified and assessed, and institutional / investment model defined.	*Existence of full feasibility studies and business plans or pre-feasibility studies for the 7 identified sites.* Installed Capacity	No feasibility study / business plan available at the moment.	3 sites defined 0.8 kWp	7 sites defined 1.2 kWp	Project reports.	Continued interest of government and private sector.
Output 3.2: Productive use promoted to increase electricity demand in the targeted sites	*Existence of productive equipments usage.*		20	40	Contracts confirming the establishment of partnerships.	Continued interest of government entities and private investors.
Output 3.3: 5,000 improved cookstoves commercialized and 50 improved kilns	*Deployment plan for kilns and improved stoves defined and*	No plans of this type available.	1,000 stoves and 10 kilns disseminated	5,000 stoves and 50 kilns disseminated	Project reports.	Designation of staff by government departments / other

Objective	Indicator / Sub-Indicator	Reference	Targets Mi -	Project Targets End of Project	Sources of Verification	Risks and Assumptions
disseminated	*approved*					institutions concerned.

VII. MONITORING AND EVALUATION (M&E) PLAN

Project-level monitoring and evaluation will be undertaken in compliance with UNDP requirements as outlined in the UNDP POPP (including guidance on GEF project revisions) and UNDP Evaluation Policy **The UNDP Country Office is responsible for ensuring full compliance with all UNDP project M&E requirements including project monitoring, UNDP quality assurance requirements, quarterly risk management, and evaluation requirements.**

Additional mandatory GEF-specific M&E requirements will be undertaken in accordance with the GEF Monitoring Policy and the GEF Evaluation Policy and other relevant GEF policies[9]. The M&E plan and budget included below will guide the GEF-specific M&E activities to be undertaken by this project.

In addition to these mandatory UNDP and GEF M&E requirements, other M&E activities deemed necessary to support project-level adaptive management will be agreed – including during the Project Inception Workshop - and will be detailed in the Inception Report.

Minimum project monitoring and reporting requirements as required by the GEF:

Inception Workshop and Report: A project inception workshop will be held within 2 months from the First disbursement date, with the aim to:

a. Familiarize key stakeholders with the detailed project strategy and discuss any changes that may have taken place in the overall context since the project idea was initially conceptualized that may influence its strategy and implementation.
b. Discuss the roles and responsibilities of the project team, including reporting lines, stakeholder engagement strategies and conflict resolution mechanisms.
c. Review the results framework and monitoring plan.
d. Discuss reporting, monitoring and evaluation roles and responsibilities and finalize the M&E budget; identify national/regional institutes to be involved in project-level M&E; discuss the role of the GEF OFP and other stakeholders in project-level M&E.
e. Update and review responsibilities for monitoring project strategies, including the risk log; SESP report, Social and Environmental Management Framework (where relevant) and other safeguard requirements; project grievance mechanisms; gender strategy; knowledge management strategy, and other relevant management strategies.
f. Review financial reporting procedures and budget monitoring and other mandatory requirements and agree on the arrangements for the annual audit.
g. Plan and schedule Project Board meetings and finalize the first-year annual work plan. Finalize the TOR of the Project Board.
h. Formally launch the Project.

GEF Project Implementation Report (PIR):

The annual GEF PIR covering the reporting period July (previous year) to June (current year) will be completed for each year of project implementation. UNDP will undertake quality assurance of the PIR before submission to the GEF. The PIR submitted to the GEF will be shared with the Project Board. UNDP will conduct a quality review of the PIR, and this quality review and feedback will be used to inform the preparation of the subsequent annual PIR.

GEF and/or LDCF/SCCF Core Indicators:

The GEF and/or LDCF/SCCF Core indicators included as Annex 15 will be used to monitor global environmental benefits and will be updated for reporting to the GEF prior to MTR and TE. Note that the project team is

[9] See https://www.thegef.org/projects-operations/policies-guidelines

responsible for updating the core indicators status. The updated monitoring data must be shared with MTR/TE consultants prior to required evaluation missions, so these can be used for subsequent groundtruthing. The methodologies to be used in data collection have been defined by the GEF and are available on the GEF. If relevant to the project: The required Protected Area Management Effectiveness Tracking Tool (METTs) must be prepared and the scores included in the GEF Core Indicators.

Independent Mid-term Review (MTR):

An independent mid-term review (MTR) will be completed by the mid-point of the project. The terms of reference, the MTR process and the final MTR report will follow the standard templates and MTR guidance for UNDP-supported GEF-financed projects available on the UNDP Evaluation Resource Center. The MTR must be submitted to the GEF by the mid-point of the project but no later than 48 months after CEO Endorsement. *Mid-point is determined by taking the expected ProDoc signature date plus number of months duration and finding the middle date. (For example, if a project has a 5-year duration, the MTR must be submitted 2.5 years after ProDoc signature.)* To meet the submission deadline, final MTR reports must be completed and submitted to BPPS NCE team no later than 2 months in advance of the submission deadline to allow sufficient time for internal review/clearance that is required prior to submission.

Provisions must be taken to complete and submit the MTR within the submission deadline. Therefore, the MTR process must start no later than 8 months before the expected date of submission of the MTR.

The MTR will be 'independent, impartial and rigorous'. The evaluator(s) that UNDP will hire to undertake the assignment will be independent from organizations that were involved in designing, executing or advising on the project to be reviewed. Equally, the evaluators should not be in a position where there may be the possibility of future contracts regarding the project under review.

The GEF Operational Focal Point and other stakeholders will be actively involved and consulted during the MTR process. Additional quality assurance support is available from BPPS/NCE.

The final MTR report will be publicly available in English and will be posted on the UNDP ERC by the MTR submission date included on cover page of this project document. A management response to MTR recommendations will be posted in the ERC within six weeks of the MTR report's submission to the GEF.

Terminal Evaluation (TE):

An independent terminal evaluation (TE) will take place upon completion of all major project outputs and activities. The terms of reference, the evaluation process and the final TE report will follow the standard templates and TE guidance for UNDP-supported GEF-financed projects available on the UNDP Evaluation Resource Center. TE must be submitted to the GEF no later than 6 months after the Completion Date. This is a hard deadline that, if not met, can only be extended through a formal extension request. To meet the submission deadline, final TE reports must be completed and submitted to BPPS NCE team no later than 2 months in advance of the deadline to allow sufficient time for internal review/clearance that is required prior to submission.

Provisions must be taken to complete and submit the TE within the submission deadline. Therefore, TE must start no later than 8 months before the expected date of submission of the TE (or 11 months prior to the estimated operational closure date).

The evaluation will be 'independent, impartial and rigorous'. The evaluator(s) that UNDP will hire to undertake the assignment will be independent from organizations that were involved in designing, executing or advising on the project to be evaluated. Equally, the evaluators should not be in a position where there may be the possibility of future contracts regarding the project being evaluated.

The GEF Operational Focal Point and other stakeholders will be actively involved and consulted during the terminal evaluation process. Additional quality assurance support is available from BPPS NCE.

The final TE report will be publicly available in English and posted on the UNDPHY ERC by the TE submission date included on cover page of this project document. A management response to the TE recommendations will be posted to the ERC within six weeks of the TE report submission to the GEF.

Per the GEF Terminal Evaluation requirements, for cancelled full-sized projects, Terminal Evaluations are required if the GEF grant expenditure exceeds more than US$ 2 million.

Final Report:

The project's final GEF PIR along with the terminal evaluation (TE) report and corresponding management response will serve as the final project report package. The final project report package shall be discussed with the Project Board during an end-of-project review meeting to discuss lesson learned and opportunities for scaling up.

In accordance with UNDP's programming policies and procedures, the project will be monitored through the following monitoring and evaluation plans.

Mandatory GEF M&E Requirements and M&E Budget:

Table 5 : budget de monitoring et évaluation

GEF M&E requirements	Primary responsibility	Indicative costs to be charged to the Project Budget[10] (US$)		Time frame
		GEF grant	Co-financing	
Inception Workshop	UNDP Country Office	11,000		Within two months of project document signature
Inception Report	Project Manager	None	None	Within two weeks of inception workshop
Standard UNDP monitoring and reporting requirements as outlined in the UNDP POPP	UNDP Country Office	None	None	Quarterly, annually
Risk management	Project Manager Country Office	None	None	Quarterly, annually
Monitoring of indicators in project results framework	Project Manager	19,000	*13,000*	Annually before PIR Per year: USD 4,000
GEF Project Implementation Report (PIR)	Project Manager and UNDP Country Office and UNDP-GEF team	None	None	Annually
Lessons learned and knowledge generation	Project Manager	None	*3,000*	Annually
Monitoring of environmental and social risks, and corresponding management plans as relevant	Project Manager UNDP Country Office	*None*	*3,000*	On-going
Stakeholder Engagement Plan	Project Manager UNDP Country Office			On-going
Gender Action Plan	Project Manager UNDP Country			On-going

[10] Excluding project team staff time and UNDP staff time and travel expenses.

GEF M&E requirements	Primary responsibility	Indicative costs to be charged to the Project Budget (US$)		Time frame
		GEF grant	Co-financing	
	Office UNDP GEF team			
Addressing environmental and social grievances	Project Manager UNDP Country Office			On-going
Project Board meetings	Project Board UNDP Country Office Project Manager	*None*	*3,000*	At minimum annually
Supervision missions	UNDP Country Office	None[11]	6,000	Annually
Oversight missions	UNDP-GEF team	NoneError: Reference source not found	6,000	Troubleshooting as needed
GEF Secretariat learning missions/site visits	UNDP Country Office and Project Manager and UNDP-GEF team	None	None	To be determined.
Independent Mid-term Review (MTR) and management response	UNDP Country Office and Project team and UNDP-GEF team	30,000	6,000	Between 2nd and 3rd PIR.
Independent Terminal Evaluation (TE) included in UNDP evaluation plan, and management response	UNDP Country Office and Project team and UNDP-GEF team	50,000	10,000	At least three months before operational closure
Translation of MTR and TE reports into English	UNDP Country Office	None	5,000	As required. GEF will only accept reports in English.
TOTAL indicative COST Excluding project team staff time, and UNDP staff and travel expenses		*110,000*	*55,000*	

[11] The costs of UNDP Country Office and UNDP-GEF Unit's participation and time are charged to the GEF Agency Fee.

VIII. GOVERNANCE AND MANAGEMENT ARRANGEMENTS

Section 1: General roles and responsibilities in the projects' governance mechanism

Implementing Partner: The Implementing Partner for this project is Ministry of Environment and Biodiversity. The Implementing Partner is the entity to which the UNDP Administrator has entrusted the implementation of UNDP assistance specified in this signed project document along with the assumption of full responsibility and accountability for the effective use of UNDP resources and the delivery of outputs, as set forth in this document.

The Implementing Partner is responsible for executing this project. Specific tasks include:

- Project planning, coordination, management, monitoring, evaluation and reporting. This includes providing all required information and data necessary for timely, comprehensive and evidence-based project reporting, including results and financial data, as necessary. The Implementing Partner will strive to ensure project-level M&E is undertaken by national institutes and is aligned with national systems so that the data used and generated by the project supports national systems.
- Risk management as outlined in this Project Document;
- Procurement of goods and services, including human resources;
- Financial management, including overseeing financial expenditures against project budgets;
- Approving and signing the multiyear workplan;
- Approving and signing the combined delivery report at the end of the year; and,
- Signing the financial report or the funding authorization and certificate of expenditures.

UNDP: UNDP is accountable to the GEF for the implementation of this project. This includes overseeing project execution undertaken by the Implementing Partner to ensure that the project is being carried out in accordance with UNDP and GEF policies and procedures and the standards and provisions outlined in the Delegation of Authority (DOA) letter for this project. **The UNDP GEF Executive Coordinator, in consultation with UNDP Bureaus and the Implementing Partner, retains the right to revoke the project DOA, suspend or cancel this GEF project.** UNDP is responsible for the Project Assurance function in the project governance structure and presents to the Project Board and attends Project Board meetings as a non-voting member.

Section 2: The Project Governance Stricture

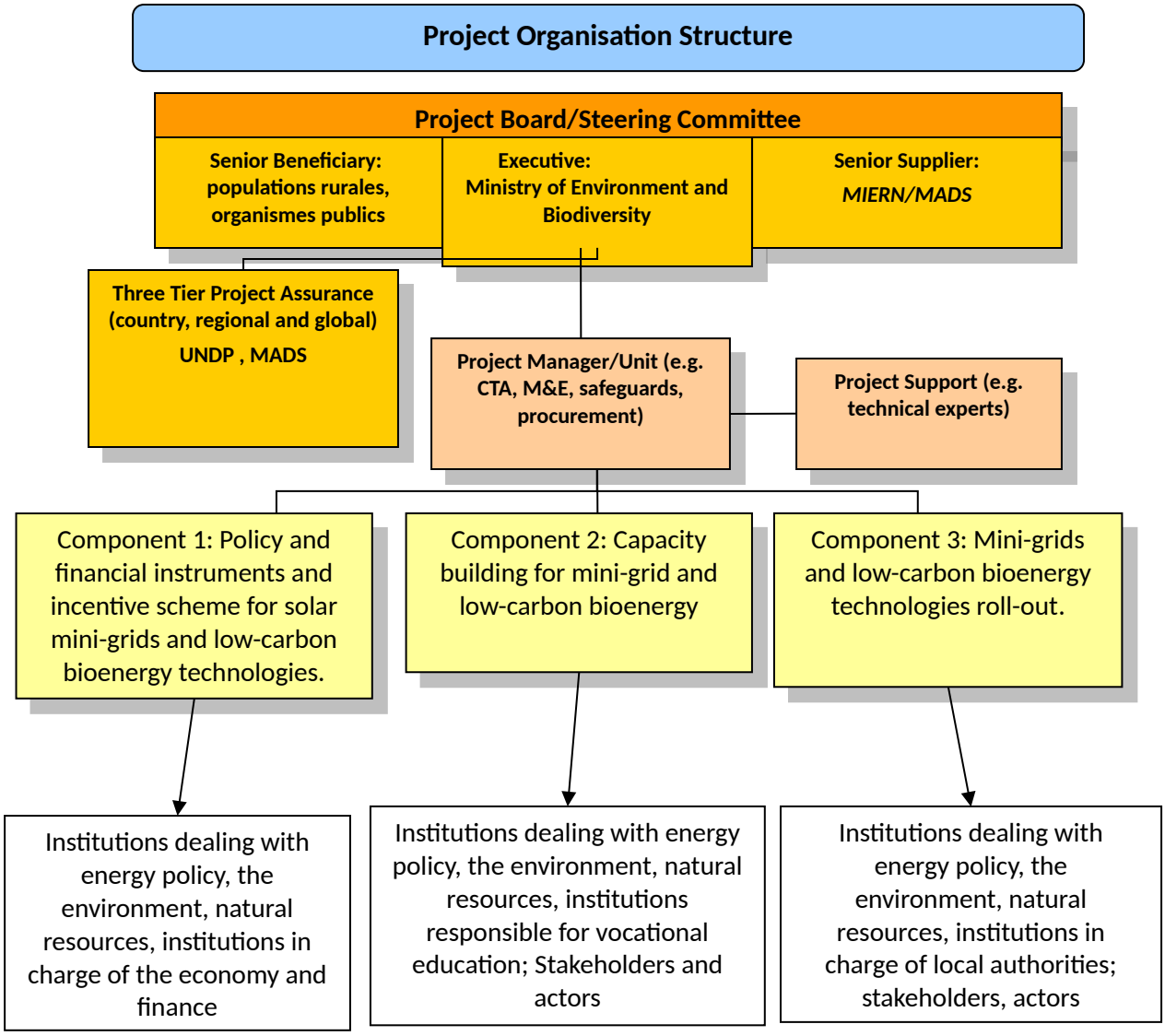


Figure 7 : Organization chart of the project

The UNDP Resident Representative assumes full responsibility and accountability for oversight and quality assurance of this Project and ensures its timely implementation in compliance with the GEF-specific requirements and UNDP's Programme and Operations Policies and Procedures (POPP), its Financial Regulations and Rules and Internal Control Framework. A representative of the UNDP Country Office will assume the assurance role and will present assurance findings to the Project Board, and therefore attends Project Board meetings as a non-voting member.

Section 3: Segregation of duties and firewalls vis-à-vis UNDP representation on the project board:

As noted in the Minimum Fiduciary Standards for GEF Partner Agencies, in cases where a GEF Partner Agency (i.e. UNDP) carries out both implementation oversight and execution of a project, the GEF Partner Agency (i.e. UNDP) must separate its project implementation oversight and execution duties, and describe in the relevant project document a: 1) Satisfactory institutional arrangement for the separation of implementation oversight and

executing functions in different departments of the GEF Partner Agency; and 2) Clear lines of responsibility, reporting and accountability within the GEF Partner Agency between the project implementation oversight and execution functions.
In this case, UNDP is only performing an implementation oversight role in the project vis-à-vis our role in the project board and in the project assurance function and therefore a full separation of project implementation oversight and execution duties has been assured.

a) **Project Board:** All UNDP projects must be governed by a multi-stakeholder board or committee established to review performance based on monitoring and evaluation, and implementation issues to ensure quality delivery of results. The Project Board (also called the Project Steering Committee) is the most senior, dedicated oversight body for a project.

The two main (mandatory) roles of the project board are as follows:

1) **High-level oversight of the execution of the project by the Implementing Partner** (as explained in the "Provide Oversight" section of the POPP). This is the primary function of the project board and includes annual (and as-needed) assessments of any major risks to the project, and decisions/agreements on any management actions or remedial measures to address them effectively. The Project Board reviews evidence of project performance based on monitoring, evaluation and reporting, including progress reports, evaluations, risk logs and the combined delivery report. The Project Board is responsible for taking corrective action as needed to ensure the project achieves the desired results.
2) **Approval of strategic project execution decisions of the Implementing Partner** with a view to assess and manage risks, monitor and ensure the overall achievement of projected results and impacts and ensure long term sustainability of project execution decisions of the Implementing Partner (as explained in the "Manage Change" section of the POPP).

Requirements to serve on the Project Board:

- ✓ Agree to the Terms of Reference of the Board and the rules on protocols, quorum and minuting.
- ✓ Meet annually; at least once.
- ✓ Disclose any conflict of interest in performing the functions of a Project Board member and take all measures to avoid any real or perceived conflicts of interest. This disclosure must be documented and kept on record by UNDP.
- ✓ Discharge the functions of the Project Board in accordance with UNDP policies and procedures.
- ✓ Ensure highest levels of transparency and ensure Project Board meeting minutes are recorded and shared with project stakeholders.

Responsibilities of the Project Board:

- ✓ Consensus decision making:
  - The project board provides overall guidance and direction to the project, ensuring it remains within any specified constraints, and providing overall oversight of the project implementation.
  - Review project performance based on monitoring, evaluation and reporting, including progress reports, risk logs and the combined delivery report;
  - The project board is responsible for making management decisions by consensus.
  - In order to ensure UNDP's ultimate accountability, Project Board decisions should be made in accordance with standards that shall ensure management for development

results, best value money, fairness, integrity, transparency and effective international competition.
    - In case consensus cannot be reached within the Board, the UNDP representative on the board will mediate to find consensus and, if this cannot be found, will take the final decision to ensure project implementation is not unduly delayed.
- ✓ Oversee project execution:
    - Agree on project manager's tolerances as required, within the parameters outlined in the project document, and provide direction and advice for exceptional situations when the project manager's tolerances are exceeded.
    - Appraise annual work plans prepared by the Implementing Partner for the Project; review combined delivery reports prior to certification by the implementing partner.
    - Address any high-level project issues as raised by the project manager and project assurance;
    - Advise on major and minor amendments to the project within the parameters set by UNDP and the donor and refer such proposed major and minor amendments to the UNDP BPPS Nature, Climate and Energy Executive Coordinator (and the GEF, as required by GEF policies);
    - Provide high-level direction and recommendations to the project management unit to ensure that the agreed deliverables are produced satisfactorily and according to plans.
    - Track and monitor co-financed activities and realisation of co-financing amounts of this project.
    - Approve the Inception Report, GEF annual project implementation reports, mid-term review and terminal evaluation reports.
    - Ensure commitment of human resources to support project implementation, arbitrating any issues within the project.
- ✓ Risk Management:
    - Provide guidance on evolving or materialized project risks and agree on possible mitigation and management actions to address specific risks.
    - Review and update the project risk register and associated management plans based on the information prepared by the Implementing Partner. This includes risks related that can be directly managed by this project, as well as contextual risks that may affect project delivery or continued UNDP compliance and reputation but are outside of the control of the project. For example, social and environmental risks associated with co-financed activities or activities taking place in the project's area of influence that have implications for the project.
    - Address project-level grievances.
- ✓ Coordination:
    - Ensure coordination between various donor and government-funded projects and programmes.
    - Ensure coordination with various government agencies and their participation in project activities.

The composition of the Project Board must include the following roles:

1. **Project Executive:** Is an individual who represents ownership of the project and chairs the Project Board. The Executive is normally the national counterpart for nationally implemented projects. The Project Executive is the National Institute of Meteorology

2. **Beneficiary Representative(s):** Individuals or groups representing the interests of those who will ultimately benefit from the project. Their primary function within the board is to ensure the realization of project results from the perspective of project beneficiaries. Often civil society representative(s) can fulfil this role. The Beneficiary representative (s) are: the National Institute of Meteorology, the General Directorate for Water Resources

3. **Development Partner(s):** Individuals or groups representing the interests of the parties concerned that provide funding and/or technical expertise to the project. The Development Partner(s) are: ECOWAS, EU, AfDB and IFAD through PADES project, World Bank, Adaptation Fund / BOAD, IUCN.

b) **Project Assurance**: UNDP performs the quality assurance role and supports the Project Board and Project Management Unit by carrying out objective and independent project oversight and monitoring functions. This role ensures appropriate project management milestones are managed and completed. The Project Board cannot delegate any of its quality assurance responsibilities to the Project Manager. UNDP provides a three – tier oversight services involving the UNDP Country Offices and UNDP at regional and headquarters levels. Project assurance is totally independent of the Project Management function.

A designated representative of UNDP playing the project assurance role is expected to attend all board meetings and support board processes as a non-voting representative. It should be noted that while in certain cases UNDP's project assurance role across the project may encompass activities happening at several levels (e.g. global, regional), at least one UNDP representative playing that function must, as part of their duties, specifically attend board meeting and provide board members with the required documentation required to perform their duties.

c) **Project Management – Execution of the Project:** The Project Manager (PM) (also called project coordinator) is the senior most representative of the Project Management Unit (PMU) and is responsible for the overall day-to-day management of the project on behalf of the Implementing Partner, including the mobilization of all project inputs, supervision over project staff, responsible parties, consultants and sub-contractors. The project manager typically presents key deliverables and documents to the board for their review and approval, including progress reports, annual work plans, adjustments to tolerance levels and risk registers

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IX. FINANCIAL PLANNING AND MANAGEMENT

The total cost of the project is $ 17,412,702. This is financed through a GEF grant of USD 2,912,702, in USD 500,000 cash co-financing to be administered by UNDP and USD 14,000,000 in parallel co-financing. UNDP, as the GEF Implementing Agency, is responsible for the execution of the GEF resources and the cash co-financing transferred to UNDP bank account only.

Co-financing: The actual realization of project co-financing amounts will be monitored by the UNDP Country Office and the PMU on an annual basis in the GEF PIF and will be reported to the GEF during the *mid-term review* and terminal evaluation process as follows:

Co-financing source	**Co-financing type**	**Name of Co-financier**	**Investment Mobilized**	**Co-financing amount**
Government	*In cash/in Kind*	*Ministry of Energy*	*Investment mobilized*	*10,000,000*
GEF	Cash	UNDP	*Investment Mobilized*	500,000
Privat Secutor	Equity	Rural Energy Services SSD/FRES	*Investment Mobilized*	4,000,000
Total				14,500,000

Budget Revision and Tolerance: As per UNDP POPP, the project board may agree with the project manager on a tolerance level for each detailed plan under the overall multi-year workplan. The agreed tolerance should be written in the project document or approved project board meeting minutes. It should normally not exceed 10 percent of the agreed annual budget at the activity level, but within the overall approved multi-year workplan at the activity level. Within the agreed tolerances, the project manager can operate without intervention from the project board. Restrictions apply as follows:

Should the following deviations occur, the Project Manager/IP through UNDP Country Office will seek the approval of the BPPS/NCE-VF team to ensure accurate reporting to the GEF. It is **strongly encouraged** to maintain the expenditures within the approved budget at the budgetary account and at the component level:

a) Budget reallocations must prove that the suggested changes in the budget will not lead to material changes in the results to be achieved by the project. A strong justification is required and will be approved on an exceptional basis. Budget re-allocations among the components (including PMC) of the approved Total Budget and Work Plans (TBWP) that represent a value greater than 10% of the total GEF grant.
b) Introduction of new outputs/activities (i.e. budget items) that were not part of the agreed project document and TBWP that represent a value greater than 5% of the total GEF grant. The new budget items must be eligible as per the GEF and UNDP policies.
c) Project management cost (PMC): budget under PMC component is capped and cannot be increased.

UNDP is not in a position to increase the total budget above the amount approved by the donor, therefore any over-expenditure would have to be absorbed from non-GEF resources by the Implementing Partner (GEF Executing Entity)

Project extensions: The UNDP-BPPS-NCE team Executive Coordinator must approve all requests for extension of the Project Completion Date and for other milestone extensions with hard deadlines. All extensions impose additional time and cost burdens at all levels and the GEF project budget cannot be increased beyond its originally approved amount. A single extension may be granted on an exceptional basis and subject to the conditions and maximum durations set out in the UNDP POPP. The project management costs during the extension period must remain within the originally approved amount, and any increase in PMC costs shall be covered by non-GEF resources; the additional UNDP oversight costs during the extension period must be covered by non-GEF resources, in accordance with UNDP's policy as set out in UNDP POPP.

For any extension request, UNDP CO and IP will consult and jointly present a clear plan indicating how and from which specific sources the additional oversight costs that will be incurred by UNDP will be covered during the extended period. The BPPS-NCE Executive Coordinator will consult the Regional Bureaux (RBX) and may reject the extension request if no (external co-financing by the IP or internal UNDP CO resources) can be identified.

All extension requests, along with all supporting documentation, shall be submitted by the IP to the UNDP CO in line with the requirements and within the deadlines set out in the UNDP SOPs and policies in UNDP POPP.
Audit: The project will be audited as per UNDP Financial Regulations and Rules and applicable audit policies. Audit cycle and process must be discussed during the Inception workshop. If the Implementing Partner is an UN Agency, the project will be audited according to that Agencies applicable audit policies.

Transfer or disposal of assets: In consultation with the Implementing Partner and other parties of the project, UNDP is responsible for deciding on the transfer or other disposal of assets. Transfer or disposal of assets is recommended to be reviewed and endorsed by the project board following UNDP rules and regulations. Assets may be transferred to the government for project activities managed by a national institution at any time during the life of a project, however **must be done before the operational closure date**. In all cases of transfer, a transfer document must be prepared and kept on file[12]. The transfer should be done before Project Management Unit complete their assignments.

Completion Date: The project completion date is the date of Project Document Signature plus project duration. This date can only be extended through a formal extension request. Prior to completion date, all UNDP-financed inputs must be provided and related activities for the Project completed. No activities, except for the final clearance of the Terminal Evaluation Report and the corresponding management response and the end-of-project review Project Board Meeting should take place after the Completion Date.

Project Closure: Project closure will be conducted as per UNDP requirements outlined in the UNDP POPP. All costs incurred to close the project must be included in the project closure budget and reported as final project commitments presented to the Project Board during the final project review. The only costs a project may incur following the final project review are those included in the project closure budget.

- **Operational Closure**: **Operational closure must happen within 9 months from project completion date.** Prior to operational closure, the Terminal Evaluation must have been submitted and the corresponding TE management response and the end-of-project review Project Board meeting must have been completed. The Implementing Partner through a Project Board decision will notify the UNDP Country Office when operational closure has been completed. Before Operational Closure, the project must have completed the transfer or disposal of any equipment that is still the property of UNDP.
- **Financial Closure: Financial closure must happen within 6 months of operational closure or after the date of cancellation.** The project will be financially closed when the following conditions have been met:

[12] See https://popp.undp.org/_layouts/15/WopiFrame.aspx?sourcedoc=/UNDP_POPP_DOCUMENT_LIBRARY/Public/PPM_Project%20Management_Closing.docx&action=default.

a) the project is operationally completed or has been cancelled; b) the Implementing Partner has reported all financial transactions to UNDP; c) UNDP has closed the accounts for the project; d) UNDP and the Implementing Partner have certified a final Combined Delivery Report (which serves as final budget revision).

Between operational and financial closure, the implementing partner will identify and settle all financial obligations and prepare a final expenditure report. The UNDP Country Office will send the final signed closure documents including confirmation of final cumulative expenditure and unspent balance to BPPS/NCE for confirmation before the project will be financially closed in Quantum by the UNDP Country Office.

Cancellation and Suspension: All projects considering going through cancellation or suspension must follow UNDP and GEF requirements. Guidance can be found in the UNDP POPP (SOPs for management actions of Vertical Fund projects escalated to the Executive Coordinator and Guidance for GEF project revisions).

Refund to GEF: Should a refund of unspent funds to the GEF be necessary, this will be managed directly by the BPPS/NCE team Directorate in New York. No action is required by the UNDP Country Office on the actual refund from UNDP project to the GEF. Unspent project balance is not permitted to be transferred to any other projects.

Table 13: Total Budget Work Plan

X. TOTAL BUDGET AND WORK PLAN

Quantum Business Unit	UNDP-GNB		
Quantum Project ID:	00116217	Quantum Project Title:	Promoting access to energy-UNDP-GNB-00116217
Quantum Award ID: Quantum TAR Award ID	1185178 00130599.1	Quantum Award Title:	Promoting access to energy-UNDP-GNB-00116217
UNDP-GEF PIMS No.	5885		
Implementing Partner	Ministry of Environment and Biodiversity (MEB)		

GEF Result / Activity Atlas	Resp. / Implementing partner	ID Fund	Name of the donor	ATLAS Budget Code	Atlas Description of the budget	Amount Year 1 (USD)	Amount Year 2 (USD)	Amount Year 3 (USD)	Amount Year 4 (USD)	Amount Year 5 (USD)	Total amount (USD)	Notes
Component 1: Policy and financial instruments and incentive scheme for solar mini-grids and low-carbon bioenergy technologies	MEB	62000	GEF	71200	International Consultants	25,000.00	25,000.00	25,000.00	25,000.00	25,000.00	125,000.00	a
				71300	National Consultants	10,000.00	10,000.00	15,000.00	10,000.00	15,000.00	60,000.00	b
				71600	Travel	10,000.00	10,000.00	10,000.00	10,000.00	15,000.00	55,000.00	c
				72200	Equipment and furniture	15,000.00	10,000.00	5,000.00	-	-	30,000.00	d
				74200	AV & printing cost	4,000.00	4,000.00	4,000.00	4,000.00	4,000.00	20,000.00	e
				74500	Miscellaneous Expenses	1,000.00	1,000.00	1,000.00	1,000.00	1,000.00	5,000.00	f
				75700	Training, workshops and conferences	5,000.00	5,000.00	5,000.00	5,000.00	5,000.00	25,000.00	g
					Total Result 1	**70,000.00**	**65,000.00**	**65,000.00**	**55,000.00**	**65,000.00**	**320,000.00**	
Component 2: Capacity	MEB /UNDP	62000	GEF	71200	International Consultants	40,000.00	40,000.00	40,000.00	40,000.00	40,000.00	200,000.00	h

building for mini-grid and low-carbon bioenergy				71300	National Consultants	15,000.00	15,000.00	10,000.00	10,000.00	10,000.00	60,000.00	l
				71600	Travel	5,800.00	5,800.00	5,800.00	5,800.00	5,800.00	29,000.00	j
				72200	Equipment and furniture	40,000.00	42,500.00	-	-	-	82,500.00	k
				71200	International Consultants			30,000.00		50,000.00	80,000.00	l
				74500	Miscellaneous Expenses	500	500	500	500.00	500	2,500.00	m
					Total Result 2	**101,300.00**	**103,800.00**	**86,300.00**	**56,300.00**	**106,300.00**	**454,000.00**	
Component 3: Mini-grids and low-carbon bioenergy technologies roll-out	MEB/ UNDP	62000	GEF	71200	International Consultants	75,000.00	75,000.00	75,000.00	60,000.00	60,000.00	345,000.00	n
				71300	Local Consultants	28,000.00	28,000.00	28,000.00	28,000.00	28,000.00	140,000.00	o
				71600	Travel	5,000.00	5,000.00	5,000.00	5,000.00	5,000.00	25,000.00	p
				72100	Contractual Service Companies	-	317,500.00	317,500.00	317,500.00	317,500.00	1,270,000.00	q
				72200	Equipment and furniture	45,000.00	45,000.00	45,000.00	20,000.00	-	155,000.00	r
				75700	Training, workshops and conferences	12,000.00	7,000.00	9,000.00	7,000.00	15,000.00	50,000.00	s
				74500	Miscellaneous Expenses	3,000.00	3,000.00	3,000.00	3,000.00	3,000.00	15,000.00	t
					Total Result 3 (FEM only)	**168,000.00**	**480,500.00**	**482,500.00**	**440,500.00**	**428,500.00**	**2,000,000.00**	
		4000	UNDP	72100	Contractual Service Companies	-	75,000.00	75,000.00	75,000.00	75,000.00	300,000.00	u
					Total Result 3 (UNDP only)	**-**	**75,000.00**	**75,000.00**	**75,000.00**	**75,000.00**	**300,000.00**	
					Total Result3 (FEMF + UNDP)	**168,000.00**	**555,500.00**	**557,500.00**	**515,500.00**	**503,500.00**	**2,300,000.00**	

Project management	MEB/ UNDP	62000	GEF	71400	Contractual Service - Individual	18,502.00	18,550.00	18,550.00	18,550.00	18,550.00	92,702.00	v
		62000	GEF	71600	Travel	4,700.00	4,700.00	4,700.00	4,700.00	4,700.00	23,500.00	w
		62000	GEF	74100	Professional Services	3,000.00	3,000.00	3,000.00	3,000.00	3,000.00	15,000.00	x
		62000	GEF	72200	Supplies	1,500.00	1,500.00	1,500.00	1,500.00	1,500.00	7,500.00	y
					Total GEF management	**27,702.00**	**27,750.00**	**27,750.00**	**27,750.00**	**27,750.00**	**138,702.00**	
		4000	UNDP	71400	Contractual Service - Individual	40,000.00	40,000.00	40,000.00	40,000.00	40,000.00	200,000.00	z
					Total management of UNDP	**40,000.00**	**40,000.00**	**40,000.00**	**40,000.00**	**40,000.00**	**200,000.00**	
					Total management	**67,702.00**	**67,750.00**	**67,750.00**	**67,750.00**	**67,750.00**	**338,702.00**	
SUBTOTAL GEF						**367,002.00**	**677,050.00**	**661,550.00**	**579,550.00**	**627,550.00**	**2,912,702.00**	
SUB-TOTAL UNDP TRAC						**40,000.00**	**115,000.00**	**115,000.00**	**115,000.00**	**115,000.00**	**500,000.00**	

Budget notes	
a	Senior Technical Advisor and International Consultants for Mini Grid Policies and Strategies (unit rate - 500$, nr of days: 250).

b	Local consulting assistance to NR CTP and Int. Consultants for mini-grid policies and strategies (unit rate - 300$, nr of days: 200).
c	Inland travel on the project sites.
d	Project equipment and software.
e	Publication of policy and strategy documents, training materials, etc.
f	Miscellaneous costs.
g	Training, workshop, meetings related to policy design, enact and enforcement (5 sessions, $,5000 per unit)
h	Partial costs of NR CTP and International Consultants for capacity building. (unit rate - 500$, nr of days: 400)
i	Local advisory assistance to NR CTP and International Consultants for Capacity Building. (unit rate - 300$, nr of days: 200)
j	Inland travel on the project sites.
k	Equipment and software for data entry and processing.
l	Mid-Term Review and Final Evaluation International Consultant.
m	Miscellaneous costs.
n	Partial costs of NR CTP and International Consultants for the electrification of villages. (unit rate - 500$, nr of days: 690)
o	Local consultants to support NR CTP and International Consultants for the electrification of villages. (unit rate - 300$, nr of days: 415)
p	Inland travel on the project sites.
q	Financial support for preliminary design studies and the purchase of equipment or construction.
r	Hardware and software for the design and implementation / review of pre-feasibility / feasibility studies.
s	Training, workshop, meetings related to investment planning, stakeholder engagement, monitoring of environmental and social risks.
t	Miscellaneous costs.
u	Support for preliminary design studies and the purchase of equipment or construction
v	Project staff costs - This budget line is for covering the Project management unit cost (Project Manager, Finance assistant)
w	Inland travel on the project sites.
x	Audit fee – Annually as per UNDP audit Policy
y	Office supplies and stationery
z	Project staff costs- This budget line is for co-sharing the PMU cost and UNDP's project assistants cost

Table 14: Annual Breakdown of the Budget by Funder

	Amount ($)	Amount ($)	Amount ($)	Amount ($)	Amount ($)	Total ($)
	Year 1	Year 2	Year 3	Year 4	Year 5	
GEF	367,002	677,050	661,550	579,550	627,550	**2,912,702**
UNDP	40,000	115,000	115,000	115,000	115,000	**500,000**
National Government	2,000,000	2,000,000	2,000,000	2,000,000	2,000,000	**10,000,000**
Private sector	300,000	1,000,000	1,000,000	1,000,000	700,000	**4,000,000**
TOTAL	**2,707,402**	**3,787,450**	**3,775,950**	**3,690,950**	**3,448,950**	**17,412,702**

XI. LEGAL CONTEXT

Option a. Where the country has signed the **Standard Basic Assistance Agreement (SBAA)**

This project document shall be the instrument referred to as such in Article 1 of the Standard Basic Assistance Agreement between the Government of Guinea Bissau and UNDP, signed on 1975. All references in the SBAA to "Executing Agency" shall be deemed to refer to "Implementing Partner."

This project will be implemented by **Ministry of Environment and Biodiversity** in accordance with its financial regulations, rules, practices, and procedures only to the extent that they do not contravene the principles of the Financial Regulations and Rules of UNDP. Where the financial governance of an Implementing Partner does not provide the required guidance to ensure best value for money, fairness, integrity, transparency, and effective international competition, the financial governance of UNDP shall apply.

XII. RISK MANAGEMENT

Option a. Implementing Partner is a Government Entity (NIM)

1. Consistent with the Article III of the SBAA *[or the Supplemental Provisions to the Project Document]*, the responsibility for the safety and security of the Implementing Partner and its personnel and property, and of UNDP's property in the Implementing Partner's custody, rests with the Implementing Partner. To this end, the Implementing Partner shall:
   a) put in place an appropriate security plan and maintain the security plan, taking into account the security situation in the country where the project is being carried;
   b) assume all risks and liabilities related to the Implementing Partner's security, and the full implementation of the security plan.

2. UNDP reserves the right to verify whether such a plan is in place, and to suggest modifications to the plan when necessary. Failure to maintain and implement an appropriate security plan as required hereunder shall be deemed a breach of the Implementing Partner's obligations under this Project Document.

3. The Implementing Partner agrees to undertake all reasonable efforts to ensure that no UNDP funds received pursuant to the Project Document are used to provide support to individuals or entities associated with terrorism and that the recipients of any amounts provided by UNDP hereunder do not appear on the United Nations Security Council Consolidated Sanctions List, and that no UNDP funds received pursuant to the Project Document are used for money laundering activities. The United Nations Security Council Consolidated Sanctions List can be accessed via https://www.un.org/securitycouncil/content/un-sc-consolidated-list.

4. The Implementing Partner acknowledges and agrees that UNDP will not tolerate sexual harassment and sexual exploitation and abuse of anyone by the Implementing Partner, and each of its responsible parties, their respective sub-recipients and other entities involved in Project implementation, either as contractors or subcontractors and their personnel, and any individuals performing services for them under the Project Document.

   (a) In the implementation of the activities under this Project Document, the Implementing Partner, and each of its sub-parties referred to above, shall comply with the standards of conduct set forth in the Secretary General's Bulletin ST/SGB/2003/13 of 9 October 2003, concerning "Special measures for protection from sexual exploitation and sexual abuse" ("SEA").

   (b) Moreover, and without limitation to the application of other regulations, rules, policies and procedures bearing upon the performance of the activities under this Project Document, in the implementation of activities, the Implementing Partner, and each of its sub-parties referred to above, shall not engage in any form of sexual harassment ("SH"). SH is defined as any unwelcome conduct of a sexual nature that might reasonably be expected or be perceived to cause offense or humiliation, when such conduct interferes with work, is made a condition of employment or creates an intimidating, hostile or offensive work environment. SH may occur in the workplace or in

connection with work. While typically involving a pattern of conduct, SH may take the form of a single incident. In assessing the reasonableness of expectations or perceptions, the perspective of the person who is the target of the conduct shall be considered.

5. a) In the performance of the activities under this Project Document, the Implementing Partner shall (with respect to its own activities), and shall require from its sub-parties referred to in paragraph 4 (with respect to their activities) that they, have minimum standards and procedures in place, or a plan to develop and/or improve such standards and procedures in order to be able to take effective preventive and investigative action. These should include: policies on sexual harassment and sexual exploitation and abuse; policies on whistleblowing/protection against retaliation; and complaints, disciplinary and investigative mechanisms. In line with this, the Implementing Partner will and will require that such sub-parties will take all appropriate measures to:

   i. Prevent its employees, agents or any other persons engaged to perform any services under this Project Document, from engaging in SH or SEA;

   ii. Offer employees and associated personnel training on prevention and response to SH and SEA, where the Implementing Partner and its sub-parties referred to in paragraph 4 have not put in place its own training regarding the prevention of SH and SEA, the Implementing Partner and its sub-parties may use the training material available at UNDP;

   iii. Report and monitor allegations of SH and SEA of which the Implementing Partner and its sub-parties referred to in paragraph 4 have been informed or have otherwise become aware, and status thereof;

   iv. Refer victims/survivors of SH and SEA to safe and confidential victim assistance; and

   v. Promptly and confidentially, record and investigate any allegations credible enough to warrant an investigation of SH or SEA. The Implementing Partner shall advise UNDP of any such allegations received and investigations being conducted by itself or any of its sub-parties referred to in paragraph 4 with respect to their activities under the Project Document, and shall keep UNDP informed during the investigation by it or any of such sub-parties, to the extent that such notification (i) does not jeopardize the conduct of the investigation, including but not limited to the safety or security of persons, and/or (ii) is not in contravention of any laws applicable to it. Following the investigation, the Implementing Partner shall advise UNDP of any actions taken by it or any of the other entities further to the investigation.

   a) The Implementing Partner shall establish that it has complied with the foregoing, to the satisfaction of UNDP, when requested by UNDP or any party acting on its behalf to provide such confirmation. Failure of the Implementing Partner, and each of its sub-parties referred to in paragraph 4, to comply of the foregoing, as determined by UNDP, shall be considered grounds for suspension or termination of the Project.

6. Social and environmental sustainability will be enhanced through application of the UNDP Social and Environmental Standards (http://www.undp.org/ses) and related Accountability Mechanism (http://www.undp.org/secu-srm).

7. The Implementing Partner shall: (a) conduct project and programme-related activities in a manner consistent with the UNDP Social and Environmental Standards, (b) implement any management or mitigation plan prepared for the project or programme to comply with such standards, and (c) engage in a constructive and timely manner to address any concerns and complaints raised through the Accountability Mechanism. UNDP will seek to ensure that communities and other project stakeholders are informed of and have access to the Accountability Mechanism.

8. All signatories to the Project Document shall cooperate in good faith with any exercise to evaluate any programme or project-related commitments or compliance with the UNDP Social and Environmental Standards. This includes providing access to project sites, relevant personnel, information, and documentation.

9. The Implementing Partner will take appropriate steps to prevent misuse of funds, fraud or corruption, by its officials, consultants, responsible parties, subcontractors and sub-recipients in implementing the project or using UNDP funds.

10. In the implementation of the activities under this Project Document, UNDP places reasonable reliance upon the Implementing Partner for it to apply its laws, regulations and processes, and applicable international laws regarding anti money laundering and countering the financing of terrorism, to ensure consistency with the principles of then in force the UNDP Anti-Money Laundering and Countering the Financing of Terrorism Policy.

11. The Implementing Partner will ensure that its financial management, anti-corruption, anti-fraud and anti-money laundering and countering the financing of terrorism policies are in place and enforced for all funding received from or through UNDP.

12. The requirements of the following documents, then in force at the time of signature of the Project Document, apply to the Implementing Partner: (a) UNDP Policy on Fraud and other Corrupt Practices and (b) UNDP Office of Audit and Investigations Investigation Guidelines. The Implementing Partner agrees to the requirements of the above documents, which are an integral part of this Project Document and are available online at www.undp.org.

13. In the event that an investigation is required, UNDP has the obligation to conduct investigations relating to any aspect of UNDP projects and programmes in accordance with UNDP's regulations, rules, policies and procedures. The Implementing Partner shall provide its full cooperation, including making available personnel, relevant documentation, and granting access to the Implementing Partner's (and its consultants', responsible parties', subcontractors' and sub-recipients') premises, for such purposes at reasonable times and on reasonable conditions as may be required for the purpose of an investigation. Should there be a limitation in meeting this obligation, UNDP shall consult with the Implementing Partner to find a solution.

14. The signatories to this Project Document will promptly inform one another in case of any incidence of inappropriate use of funds, or credible allegation of fraud or corruption with due confidentiality.

    Where the Implementing Partner becomes aware that a UNDP project or activity, in whole or in part, is the focus of investigation for alleged fraud/corruption, the Implementing Partner will inform the UNDP Resident Representative/Head of Office, who will promptly inform UNDP's Office of Audit

and Investigations (OAI). The Implementing Partner shall provide regular updates to the head of UNDP in the country and OAI of the status of, and actions relating to, such investigation.

15. UNDP shall be entitled to a refund from the Implementing Partner of any funds provided that have been used inappropriately, including through fraud, corruption or other financial irregularity, or otherwise paid other than in accordance with the terms and conditions of the Project Document. Such amount may be deducted by UNDP from any payment due to the Implementing Partner under this or any other agreement. Recovery of such amount by UNDP shall not diminish or curtail the Implementing Partner's obligations under this Project Document.

    Where such funds have not been refunded to UNDP, the Implementing Partner agrees that donors to UNDP (including the Government) whose funding is the source, in whole or in part, of the funds for the activities under this Project Document, may seek recourse to the Implementing Partner for the recovery of any funds determined by UNDP to have been used inappropriately, including through fraud, corruption or other financial irregularity, or otherwise paid other than in accordance with the terms and conditions of the Project Document.

    Note: The term "Project Document" as used in this clause shall be deemed to include any relevant subsidiary agreement further to the Project Document, including those with responsible parties, subcontractors and sub-recipients.

16. Each contract issued by the Implementing Partner in connection with this Project Document shall include a provision representing that no fees, gratuities, rebates, gifts, commissions or other payments, other than those shown in the proposal, have been given, received, or promised in connection with the selection process or in contract execution, and that the recipient of funds from the Implementing Partner shall cooperate with any and all investigations and post-payment audits.

17. Should UNDP refer to the relevant national authorities for appropriate legal action any alleged wrongdoing relating to the project, the Government will ensure that the relevant national authorities shall actively investigate the same and take appropriate legal action against all individuals found to have participated in the wrongdoing, recover and return any recovered funds to UNDP.

18. The Implementing Partner shall ensure that all of its obligations set forth under this section entitled "Risk Management" are passed on to each responsible party, subcontractor and sub-recipient and that all the clauses under this section entitled "Risk Management Standard Clauses" are included, mutatis mutandis, in all sub-contracts or sub-agreements entered into further to this Project Document.

XIII. MANDATORY ANNEXES

A. Multiyear Workplan (see template below)
B. GEF Tracking Tool (s) at baseline
C. Terms of Reference for Project Board, Project Manager, Chief Technical Advisor and other positions as appropriate (see example template below)

D. UNDP Social and Environmental and Social Screening Template (SESP) and Environmental and Social Management Plan (ESMP) for moderate and high risk projects
E. Stakeholder Engagement Plan
F. Gender Analysis and Action Plan
G. Additional agreements: such as cost sharing agreements, project cooperation agreements signed with NGOs (where the NGO is designated as the “executing entity”), letters of financial commitments, GEF OFP letter, GEF PIFs and other templates for all project types, LOA with the government in case DPCs are applied should be attached.

Other Annexes that may be required:

- Procurement Plan (for at least first year of implementation)
- GEF focal area specific annexes (e.g. e.g. GHG calculations, target landscape profile, capacity assessment report, feasibility study, other technical reports)
- List of people consulted during project development

Annex A: Multi Year Work Plan

Table 15: Project execution schedule

Task / Result	Responsible Party	Year 1				Year 2				Year 3				Year 4				Year 5			
		T1	T2	T3	T4	T1	T2	T3	T4	T1	T2	T3	T4	T1	T2	T3	T4	T1	T2	T3	T4
Component 1: Policy and financial instruments and incentive scheme for solar mini-grids and low-carbon bioenergy technologies																					
Output 1.1	MADS																				
Output 1.2	MADS																				
Output 1.3:	MADS																				
Output 1.4	MADS																				
Component 2: Capacity building for mini-grid and low-carbon bioenergy																					
Output 2.1:	MADS																				
Output 2.2:	MADS																				
Output 2.3:	MADS																				
Output 2.4:	MADS																				
Component 3: Mini-grids technologies roll-out																					
Output 3.1:	MADS																				
Output 3.2:	MADS																				
Output 3.3:	MADS																				
Project Reviews and Evaluation																					
Annual review of project implementation.	UNDP																				
Mid-term review.	UNDP																				
Final evaluation.	UNDP																				

Annex B: EF 7 Core Indicator Worksheet

Core Indicator 6	**Greenhouse gas emission mitigated**					***(Metric tons of CO_2e)***
			Expected metric tons of CO_2e (6.1+6.2)			
			PIF stage	Endorsement	MTR	TE
	Expected CO2e (direct)		*190,288*	93,900		
	Expected CO2e (indirect)			*939,000*		
Indicator 6.1	Carbon sequestered or emissions avoided in the AFOLU sector					
			Expected metric tons of CO_2e			
			PIF stage	Endorsement	MTR	TE
	Expected CO2e (direct)					
	Expected CO2e (indirect)					
	Anticipated start year of accounting					
	Duration of accounting					
Indicator 6.2	Emissions avoided Outside AFOLU					
			Expected metric tons of CO_2e			
			Expected		Achieved	
			PIF stage	Endorsement	MTR	TE
	Expected CO2e (direct)					
	Expected CO2e (indirect)					
	Anticipated start year of accounting					
	Duration of accounting					
Indicator 6.3	Energy saved					
			MJ			
			Expected		Achieved	

			PIF stage	Endorsement	MTR	TE
Indicator 6.4	Increase in installed renewable energy capacity per technology					
		Technology	Capacity (MW)			
			Expected		Achieved	
			PIF stage	Endorsement	MTR	TE
		(select)	*2*	*1.19*		
		(select)				
Core Indicator 11	**Number of direct beneficiaries disaggregated by gender as co-benefit of GEF investment**					***(Number)***
			Number			
			Expected		Achieved	
			PIF stage	Endorsement	MTR	TE
		Female	*12,500*	*12,500*		
		Male	*12,500*	*12,500*		
		Total	*25,000*	*25,000**		

**The overall the breakdown of beneficiaries is estimated as 22,700 from mini-grid systems and 2,300 from cookstoves (2,000 from improved clean stoves and 300 from improved kilns)*

Annex C: Terms of Reference

1. Project Manager

I. Job Information	
Position Title:	Project Manager (full time)
Office :	Project Management Unit (PMU)
Organization:	Ministry of Energy, Industries and Natural Resources (MEIRN)
Duration of employment:	A renewable year as many times as necessary
Duty station:	Bissau, Republic of Guinea-Bissau

II. Tasks

- Lead, manage and coordinate the day-to-day activities of the PMU to be implemented within MEIRN, including administration, accounting, technical expertise, financial expertise and the implementation and establishment of concrete reports;
- Lead the development of the project design, including preparation of terms of reference for consultants and subcontractors, identification and selection of national and international subcontractors / consultants, cost estimation, establishment scheduling, procurement and reporting of project activities and budget;
- Provide ongoing financial advice to mini-grid project developers;
- Provide support to mini-grid project developers for access to finance;
- Monitor and track delivery status by consultants, subcontractors, etc.
- Monitor and track delivery status by consultants, subcontractors, etc.
- Coordinate the activities of the consultants, including contract management, management and supervision of field operations, logistical support, review of technical results;
- Assist in the design, supervision and outreach activities of the project;
- Provide technical support for policy discussions on renewable energy technologies for rural electrification in the country;
- Act as a liaison / facilitator between different stakeholders, including the private sector, international and national partners;
- Take responsibility for the quality and timing of project results;
  Establish and maintain relationships and act as a focal point with the UNDP country office to ensure that all programmatic, financial and administrative issues related to the project are managed in a transparent, timely and efficient manner in accordance with the rules and established UNDP regulations;

Undertake other management tasks that contribute to the effective implementation of the project.	
III. Qualifications and Experience	
Education :	• Master's degree or equivalent in engineering, economics, business administration, finance, international development, social sciences, public administration or other relevant field.
Experience :	• Minimum of 5 years of management experience, preferably in the field of energy. • Proven ability to write, review and produce written proposals and results-based reports. • Demonstrated experience working with government, civil society, international organizations or donors, in combination with knowledge of economic and financial analysis, institutional, regulatory and policy frameworks. • Good knowledge and experience of GEF's climate change issues, its operational modalities and knowledge of UNDP-GEF procedures would be an asset. • Knowledge of UNDP rules, regulations and administrative procedures would be an asset. • Prior knowledge and experience of political, social and environmental factors and issues related to energy development and climate change mitigation in African countries; • Experience using computers and office software (MS Word, Excel, etc.)
Language requirements:	• Excellent in Portuguese, plus additional language in either English or French.

2. Project Assistant

I. Job Information	
Position Title: Office : Organization: Duration of employment: Duty station:	Project Assistant (full time) Project Management Unit (PMU) Ministry of Energy, Industries and Natural Resources (MEIRN) A renewable year Bissau, Republic of Guinea-Bissau
II Tasks	

Under the general supervision of the project manager, the project assistant will:

• Support the activities of international / national experts, potential investors and subcontractors;

• Providing administrative support, typing, filing, obtaining visas for international experts / subcontractors, maintaining project financial records, etc.;

• Administer project accounting according to UNDP procedures;

• Assist the project manager in organizing workshops, project committee meetings and other events;

• Assist in the purchase of goods and services;

• Write invitation letters and agendas for Project Committee / Workshop meetings;

• Prepare background information, backgrounders, reports, etc. as needed;

• Write the minutes of the meetings, follow / follow the required actions.

III. Qualifications and Experience

Education:

- Advanced education in economics, management, accounting, finance or other related field.
- Specialized training in finance is desirable;

Experience:

- 3 years of relevant administrative, accounting and financial experience at national and / or international level.
- Experience in the use of computers and office software (MS Word, Excel, etc.).
- Previous work experience for a program (s) implemented at the national level and funded by bilateral / multilateral organizations.
- Practical procurement experience will be an asset.

Language requirements:

- Excellent in Portuguese, plus additional language in either English or French.

3. Senior Technical Advisor (Non-resident)

I. Job Information	
Position Title: Office : Organization: Duration of employment: Duty station:	Senior Technical Advisor (CTP, Non-resident) Project Management Unit (PMU) Ministry of Energy, Industries and Natural Resources (MEIRN) 15 weeks (over a period of 5 years) (15 days per year including 2 assignments of 5 days each, 12-month contract, renewable on the basis of satisfactory performance)

	Home Office + Bissau, Bissau-Guinean Republic

II Tasks

Under the general supervision of the project manager, the non-resident Senior Technical Advisor will:

• Work closely with the Project Manager to coordinate and facilitate contributions from government agencies, partner organizations, scientific and research institutions, subcontractors and national and international experts in a timely and effective manner;

• Provide advice and assistance to the PM and project staff to ensure that project activities are in accordance with the approved project document;

• Provide advice and assistance on financial support and procurement activities.

• Assist the PM during the first two months of the project to prepare a "start-up report" detailing the project logical framework matrix and planned project activities, the annual work plan and the first year budget, terms of reference for key project staff and a monitoring and evaluation plan;

• Assist the PMU to develop relevant RDTs and to recruit / mobilize qualified experts and organizations nationally and internationally, as appropriate, to provide specific consulting and engineering services;

• Assist the PMU in the development of the call for proposals and selection criteria for promoters of PCH sites;

• Formulate detailed procedures for the implementation of the financial instrument

• In close collaboration with the PMU and the UNDP focal point on energy and the environment, and in consultation with project partner organizations and stakeholders, prepare annual project work plans to be approved by the UNDP. Project council;

• Provide technical advice and on-the-job mentoring to the PMU to strengthen their capacity to effectively implement the technical aspects of the project;

• Assist the PM in reporting to the PR on progress of project implementation and achievement of project results in accordance with the project logical framework matrix;

• Support the PMU in project related meetings as needed;

• Review reports of national and international consultants, project budget revisions and administrative arrangements required by UNDP-GEF procedures;

• Assist the Project Manager in developing a concrete monitoring and evaluation plan from the beginning of the project (in the initial report);

• Support the Project Manager in the preparation and implementation of the mid-term review and independent final evaluation missions (terms of reference, identification and recruitment of appropriate candidates, organization of missions, joint field missions and discussion with the evaluators, etc.);

• Support UNDP country office staff in their annual monitoring visits to the project sites.

III. Qualifications and Experience

Education :	• Graduate degree in energy / renewable energy development or finance

	with energy knowledge.
Experience :	• At least ten years of experience in implementing renewable energy projects in combination with knowledge of economic and financial analysis, institutional, regulatory and policy frameworks; • Good knowledge and experience of GEF climate change issues, operational modalities; • Knowledge of UNDP-GEF rules, regulations and administrative procedures would be an advantage, but not a requirement; • Prior knowledge and experience of political, social and environmental factors and issues related to energy development and climate change mitigation in developing countries in Africa; • Proficiency in computer science, particularly with regard to professional desktop software packages; • Excellent writing and communication skills.
Language requirements	Excellent in Portuguese, plus additional language in either English or French.

Annex D: UNDP Social and Environmental Screening Procedure and plans as needed

Project Information

Project Information	
1. Project Title	Promoting better access to modern energy services through sustainable mini-grids and low-carbon bioenergy technologies among Guinea-Bissau's forest-dependent communities
2. Project Number	5885
3. Location (Global/Region/Country)	Guinea-Bissau

Part A. Integrating Overarching Principles to Strengthen Social and Environmental Sustainability

QUESTION 1: How Does the Project Integrate the Overarching Principles in order to Strengthen Social and Environmental Sustainability?
Briefly describe in the space below how the Project mainstreams the human-rights based approach
The project will ensure social equity and equality. Based on selected sites the project will be implemented in 4 of 9 protected areas (PAs) pertaining to the National System of Protected Areas of Guinea-Bissau, which is managed by IBAP, the Institute of Biodiversity and Protected Areas (Remark: Presently, National System of PAs is comprised of 9 PAs, located in terrestrial, coastal and marine zone. Each unit is a geographical entity of national territory, where co-exists a biological diversity (people, animals and resources), which role consists on conservation of national and international relevant biodiversity values, to contribute for a rational management of valuable renewable resources, and to promote an equitable and sustainable development based on participative management processes). Project sites are located inside or in the periphery of PAs. Regarding human rights, the project will benefit individuals and communities through better access to modern energy services, thus empowering them for an improved life quality. Individuals will see its capacity reinforced for an open and free choice of energy alternatives uses, and communities will have the opportunity to improve institutional and communitarian equipment, such as health centers, schools, public lighting, clean water, and transformation and collective conservation of agricultural and fishing products for a better business doing. The project will also increase the health and safety of population through access to clean potable water extracted from new well by modern PV pumping systems. Same kind of benefit is expected through the use of improved cookstoves of fuelwood and charcoal for food cooking (energy efficiency, time saving for fuelwood harvesting and time for cooking, low level of smoke and reduced exposition time to smoke, reduced pressure on forest for satisfaction of domestic energy demand, among others).

Briefly describe in the space below how the Project is likely to improve gender equality and women's empowerment

Mainstreaming gender in renewable energy policies and programming can enhance the efficiency of energy policies. Through the incorporation of gender perspectives, the effectiveness and gender equality not just of energy programs and policies, but of all development activities involving energy use can be ensured. Thus, this UNDP-GEF project has a focus on gender sensitive planning and implementation, particularly for women-headed households and businesses, including younger women and children. A Gender Action Plan (GAP), inspired on ECOWAS Program on Gender Mainstreaming in Energy Access (ECOW-GEN), the key program of ECREEE (ECOWAS Centre for Renewable Energy and Energy Efficiency) working at the intersection of gender and energy, is specifically developed for this project. It is focused on capturing and expressing the way in which gender can be mainstreamed within this project to promote better access to modern energy services through sustainable off-grid and low-carbon bioenergy technologies. While electrification will benefit both women and men by enhancing their engagement in more productive activities, gender gains in the first phase are derived mainly from reducing the workload of women and girls.

By rehabilitating one or more existing UNDP Multifunctional Platform, the project will boost agro-processing activities (grinding, milling, stocking), mainly managed by women through their communitarian associations/groups, thus enabling women farmers and women led-households to increase and diversify their income. The renewable energy systems provided will also strengthen capacities for service delivery of institutional/public infrastructures (health center, school, administration, water pumping, and public lightning) in rural communities.

Furthermore, it is estimated from past experiences (EU/INITA project "*Rational utilization of forestry asset*", 2005) that the time spent for cooking with charcoal in urban and semi-urban areas varies between 2 to 3 hours per woman per day in Guinea-Bissau with a consumption of 3 to 4 small plastic bag of 0.5 kg (1.5 to 2 kg). By using improved cookstoves, this time window can be reduced to only 45 to 60 minutes per day for a total consumption of 0.5 to 1 kg of charcoal. Additional advantage is a lesser pollution by toxic smoke.

It is expected that approximately 8,600 women will directly benefit from project activities.

Briefly describe in the space below how the Project mainstreams environmental sustainability

The project is expected to deliver significant long lasting environmental benefits as it will reduce emissions of GHGs; (i) through reduction of electricity production from fossil fuels in Cacheu, Cacine, Pitche and Jemberém, and (ii) through dissemination and use in all project sites of clean/efficient technologies (improved kilns and cookstoves) and PV systems for conservation chambers that will allow for lower pressure over forest. For electricity generation it is estimated an avoidance of total reduction of 1,420 tones CO_2 per year and for efficient technologies a total of 10,837 tones CO_2. Traditional cooking appliances will be replaced by improved cookstoves (improved three-stone fuelwood cookstoves and improved charcoal cookstoves) which are energetically more efficient.

Part B. Identifying and Managing Social and Environmental Risks

QUESTION 2: What are the Potential Social and Environmental Risks? *Note: Describe briefly potential social and environmental risks identified in Attachment 1 – Risk Screening Checklist (based on any "Yes" responses). If no risks have been identified in Attachment 1 then note "No Risks Identified" and skip to Question 4 and Select "Low Risk". Questions 5 and 6 not required for Low Risk Projects.*	**QUESTION 3: What is the level of significance of the potential social and environmental risks?** *Note: Respond to Questions 4 and 5 below before proceeding to Question 6*			**QUESTION 6: What social and environmental assessment and management measures have been conducted and/or are required to address potential risks (for Risks with Moderate and High Significance)?**
Risk Description	***Impact and Probability (1-5)***	***Significance (Low, Moderate, High)***	***Comments***	***Description of assessment and management measures as reflected in the Project design. If ESIA or SESA is required note that the assessment should consider all potential impacts and risks.***
Risk 1: There is likelihood that the Project would have inequitable or discriminatory adverse impacts on affected populations, particularly people living in poverty or marginalized or excluded individuals or groups.	I = 2 P = 3	**Moderate**	The magnitude or intensity of the impact would be moderate on gender, poor families and marginalized groups. It is limited in scale (site-specific) and has medium-term duration, but can be avoided, managed and/or mitigated with	In Guinea-Bissau, women, poor families and marginalized groups continue to be outside the decision-making sphere and their human rights are often bypassed when addressing energy access issues. Albeit undeniable contribution to economic, social and political development, these groups have not equally reaped its benefits. A detailed study is required to understand if the impacts can

			relatively uncomplicated accepted measures.	be managed and which management measures are required to reverse or mitigate the impact on affected populations.
Risk 2: The Project potentially reproduce discriminations against women based on gender, especially regarding participation in design and implementation or access to opportunities and benefits.	I = 3 P = 3	**Moderate**	Although the impact would occur in a PA, its magnitude would be moderate on gender, limited in scale (site-specific) and with medium-term duration, but can be easily avoided, managed and/or mitigated with relatively uncomplicated accepted measures.	There is the risk that energy supply and electrification could increase social inequalities, particularly for women, giving access to a broad range of opportunities to those who can afford to connect, and leaving the others still further behind. A detailed study based on gender disaggregated data is required to understand if the impacts can be managed and which management measures are needed. Especially the capacity/ability of low income group of women to afford electricity must be assessed and an adjusted tariff proposed, approved and applied accordingly.
Risk 3: Women's groups/leaders raised gender equality concerns regarding the Project during the stakeholder engagement process.	I = 3 P = 3	**Moderate**	The magnitude of the impact would be moderate on gender, limited in scale (site-specific) and with medium-term duration, but can be easily avoided, managed and/or mitigated with relatively uncomplicated accepted measures.	This has been considered and included in the overall Project proposal and in the risk assessment, and a specific Gender Plan has been designed thoroughly covering related issues.
Risk 4: Project activities proposed within or adjacent to critical habitats and/or environmentally sensitive areas, including legally protected areas (e.g. nature reserve, national park), areas proposed for protection, or recognized as such by authoritative sources and/or indigenous peoples or local	I = 3 P = 3	**Moderate**	Impact is sure on biodiversity conservation and sustainable natural resource management but is limited in scale (project sites). This issue can be addressed through application of standard best practice, mitigation measures and stakeholder involvement during Project implementation.	Project activities are planned to take place inside protected areas (Tchetche, Duta Djara, Lamane and Jemberém) and in their periphery (Cacheu, Cacine and Pitche). This would potentially generate social and environmental risks and adverse impact, that are limited in scale but lasting from short to medium-term can be easily identified with a reasonable degree of certainty, and can be mitigated. For example, capacity building, alternative roads and pathways may serve as mitigation measures. It must be also pointed out that restrictions of access to the installation or equipment field in the project sites of

communities.				concerned PAs will not have potential adverse impacts on livelihoods of local communities. Co-management plans coordinated by IBAP with the implication of local communities and other stakeholders (technical and financial partners, NGOs) represent a powerful tool for risk mitigation with relatively uncomplicated ground for mutually accepted measures.
Risk 5: The Project involve large-scale infrastructure development (PV solar hybrid power plants, communitarian PV ice making chamber, PV water pumping).	I = 2 P = 2	**Low**		1. For the Project Sites located inside PAs (Tchetche, Duta Djara, Lamane and Jemberém), although impact would occur inside a PA in terms of Safety and Working Conditions, it will be very limited in magnitude (small affected area) with a short duration limited to the period needed for equipment installation. This impact may be easily managed and mitigated with the involvement of the local community. 2. For the Project Sites located outside the PAs (Cacheu, Cacine and Pitche) which are bigger in capacity/population and energy demand, projected power plants are of bigger size, similarly impacting Safety and Working Conditions of peoples, but impact will also be very limited in magnitude (bigger open space and available area) with a short duration limited to the period needed for equipment installation. This impact may also be easily managed and mitigated with the involvement of the local community.
	QUESTION 4: What is the overall Project risk categorization?			
	Select one (see SESP for guidance)			**Comments**
	Low Risk	☐		
	Moderate Risk	**X**		It is a CLIMATE CHANGE MITIGATION Project based on the use of small and medium-scale PV solar technology systems for decentralized rural electrification (powering household, communitarian equipment, productive use/business and water pumping), associated with a strong showcase on Energy Efficiency through the use of improved kilns an

			cookstoves in making and using forestry biomass (fuelwood and charcoal) for food cooking. This risk categorization is acceptable for the type of planned interventions.
	High Risk	☐	
	QUESTION 5: Based on the identified risks and risk categorization, what requirements of the SES are relevant?		
	Check all that apply		**Comments**
	Principle 1: Human Rights	**X**	There is likelihood that the Project would have inequitable or discriminatory adverse impacts on affected populations, particularly people living in poverty or marginalized or excluded individuals or groups. A detailed study is required to understand if the impacts can be managed and which management measures are required to reverse or mitigate the impact on affected populations.
	Principle 2: Gender Equality and Women's Empowerment	**X**	The Project potentially reproduces discriminations against women based on gender, especially regarding participation in design and implementation or access to opportunities and benefits. Specific study on Gender is carried out and a Gender Plan designed and included in the project design for risk mitigation.
	1. Biodiversity Conservation and Natural Resource Management	**X**	Project activities proposed within or adjacent to critical habitats and/or environmentally sensitive areas, including legally protected areas (e.g. nature reserve, national park), areas proposed for protection, or recognized as such by authoritative sources and/or indigenous peoples or local communities. Co-management plans coordinated by IBAP with the implication of local communities and other stakeholders

			(technical and financial partners, NGOs) represent a powerful tool for risk mitigation with relatively uncomplicated ground for mutually accepted measures.
	2. ***Climate Change Mitigation and Adaptation***	☐	
	3. ***Community Health, Safety and Working Conditions***	☐	
	4. ***Cultural Heritage***	☐	
	5. ***Displacement and Resettlement***	☐	
	6. ***Indigenous Peoples***	☐	
	7. ***Pollution Prevention and Resource Efficiency***	☐	

Final Sign Off

Signature	***Date***	***Description***
QA Assessor Dauda SAU, Program Specialist / Head of Sustainable Development Cluster		UNDP staff member responsible for the Project, typically a UNDP Programme Officer. Final signature confirms they have "checked" to ensure that the SESP is adequately conducted.
QA Approver Gabriel DAVA, UNDP Resident Representative a.i.		UNDP senior manager, typically the UNDP Deputy Country Director (DCD), Country Director (CD), Deputy Resident Representative (DRR), or Resident Representative (RR). The QA Approver cannot also be the QA Assessor. Final signature confirms they have "cleared" the SESP prior to submittal to the PAC.
PAC Chair Gabriel DAVA, UNDP Resident Representative a.i.		UNDP chair of the PAC. In some cases PAC Chair may also be the QA Approver. Final signature confirms that the SESP was considered as part of the project appraisal and considered in recommendations of the PAC.

SESP Attachment 1. Social and Environmental Risk Screening Checklist

Checklist Potential Social and Environmental Risks		
Principles 1: Human Rights		**Answer (Yes/ No)**
1.	Could the Project lead to adverse impacts on enjoyment of the human rights (civil, political, economic, social or cultural) of the affected population and particularly of marginalized groups?	No
2.	Is there likelihood that the Project would have inequitable or discriminatory adverse impacts on affected populations, particularly people living in poverty or marginalized or excluded individuals or groups? [13]	**Yes**
3.	Could the Project potentially restrict availability, quality of and access to resources or basic services, in particular to marginalized individuals or groups?	No
4.	Is there likelihood that the Project would exclude any potentially affected stakeholders, in particular marginalized groups, from fully participating in decisions that may affect them?	No
5.	Is there a risk that duty-bearers do not have the capacity to meet their obligations in the Project?	No
6.	Is there a risk that rights-holders do not have the capacity to claim their rights?	No
7.	Have local communities or individuals, given the opportunity, raised human rights concerns regarding the Project during the stakeholder engagement process?	No
8.	Is there a risk that the Project would exacerbate conflicts among and/or the risk of violence to project-affected communities and individuals?	No
Principle 2: Gender Equality and Women's Empowerment		
1.	Is there likelihood that the proposed Project would have adverse impacts on gender equality and/or the situation of women and girls?	No
2.	Would the Project potentially reproduce discriminations against women based on gender, especially regarding participation in design and implementation or access to opportunities and benefits?	**Yes**
3.	Have women's groups/leaders raised gender equality concerns regarding the Project during the stakeholder engagement process and has this been included in the overall Project proposal and in the risk assessment?	**Yes**

[13] Prohibited grounds of discrimination include race, ethnicity, gender, age, language, disability, sexual orientation, religion, political or other opinion, national or social or geographical origin, property, birth or other status including as an indigenous person or as a member of a minority. References to "women and men" or similar is understood to include women and men, boys and girls, and other groups discriminated against based on their gender identities, such as transgender people and transsexuals.

4.	Would the Project potentially limit women's ability to use, develop and protect natural resources, taking into account different roles and positions of women and men in accessing environmental goods and services? *For example, activities that could lead to natural resources degradation or depletion in communities who depend on these resources for their livelihoods and well-being.*	No
Principle 3: Environmental Sustainability: Screening questions regarding environmental risks are encompassed by the specific Standard-related questions below		
Standard 1: Biodiversity Conservation and Sustainable Natural Resource Management		
1.1	Would the Project potentially cause adverse impacts to habitats (e.g. modified, natural, and critical habitats) and/or ecosystems and ecosystem services? *For example, through habitat loss, conversion or degradation, fragmentation, hydrological changes*	No
1.2	Are any Project activities proposed within or adjacent to critical habitats and/or environmentally sensitive areas, including legally protected areas (e.g. nature reserve, national park), areas proposed for protection, or recognized as such by authoritative sources and/or indigenous peoples or local communities?	**Yes**
1.3	Does the Project involve changes to the use of lands and resources that may have adverse impacts on habitats, ecosystems, and/or livelihoods? (Note: if restrictions and/or limitations of access to lands would apply, refer to Standard 5)	No
1.4	Would Project activities pose risks to endangered species?	No
1.5	Would the Project pose a risk of introducing invasive alien species?	No
1.6	Does the Project involve harvesting of natural forests, plantation development, or reforestation?	No
1.7	Does the Project involve the production and/or harvesting of fish populations or other aquatic species?	No
1.8	Does the Project involve significant extraction, diversion or containment of surface or ground water? *For example, construction of dams, reservoirs, river basin developments, groundwater extraction*	No
1.9	Does the Project involve utilization of genetic resources? (e.g. collection and/or harvesting, commercial development)	No
1.10	Would the Project generate potential adverse transboundary or global environmental concerns?	No
1.11	Would the Project result in secondary or consequential development activities which could lead to adverse social and environmental effects, or would it generate cumulative impacts with other known existing or planned activities in the area? *For example, a new road through forested lands will generate direct environmental and*	No

	social impacts (e.g. felling of trees, earthworks, potential relocation of inhabitants). The new road may also facilitate encroachment on lands by illegal settlers or generate unplanned commercial development along the route, potentially in sensitive areas. These are indirect, secondary, or induced impacts that need to be considered. Also, if similar developments in the same forested area are planned, then cumulative impacts of multiple activities (even if not part of the same Project) need to be considered.	
Standard 2: Climate Change Mitigation and Adaptation		
2.1	Will the proposed Project result in significant[14] greenhouse gas emissions or may exacerbate climate change?	No
2.2	Would the potential outcomes of the Project be sensitive or vulnerable to potential impacts of climate change?	No
2.3	Is the proposed Project likely to directly or indirectly increase social and environmental vulnerability to climate change now or in the future (also known as maladaptive practices)? *For example, changes to land use planning may encourage further development of floodplains, potentially increasing the population's vulnerability to climate change, specifically flooding*	No
Standard 3: Community Health, Safety and Working Conditions		
3.1	Would elements of Project construction, operation, or decommissioning pose potential safety risks to local communities?	No
3.2	Would the Project pose potential risks to community health and safety due to the transport, storage, and use and/or disposal of hazardous or dangerous materials (e.g. explosives, fuel and other chemicals during construction and operation)?	No
3.3	Does the Project involve large-scale infrastructure development (e.g. dams, roads, buildings)?	**Yes**
3.4	Would failure of structural elements of the Project pose risks to communities? (e.g. collapse of buildings or infrastructure)	No
3.5	Would the proposed Project be susceptible to or lead to increased vulnerability to earthquakes, subsidence, landslides, erosion, flooding or extreme climatic conditions?	No
3.6	Would the Project result in potential increased health risks (e.g. from water-borne or other vector-borne diseases or communicable infections such as HIV/AIDS)?	No
3.7	Does the Project pose potential risks and vulnerabilities related to occupational health and safety due to physical, chemical, biological, and radiological hazards during Project construction, operation, or decommissioning?	No
3.8	Does the Project involve support for employment or livelihoods that may fail to comply with national and international labour standards (i.e. principles and standards of ILO	No

[14] In regards to CO_2, 'significant emissions' corresponds generally to more than 25,000 tons per year (from both direct and indirect sources). [The Guidance Note on Climate Change Mitigation and Adaptation provides additional information on GHG emissions.]

	fundamental conventions)?	
3.9	Does the Project engage security personnel that may pose a potential risk to health and safety of communities and/or individuals (e.g. due to a lack of adequate training or accountability)?	No
Standard 4: Cultural Heritage		
4.1	Will the proposed Project result in interventions that would potentially adversely impact sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture (e.g. knowledge, innovations, practices)? (Note: Projects intended to protect and conserve Cultural Heritage may also have inadvertent adverse impacts)	No
4.2	Does the Project propose utilizing tangible and/or intangible forms of cultural heritage for commercial or other purposes?	No
Standard 5: Displacement and Resettlement		
5.1	Would the Project potentially involve temporary or permanent and full or partial physical displacement?	No
5.2	Would the Project possibly result in economic displacement (e.g. loss of assets or access to resources due to land acquisition or access restrictions – even in the absence of physical relocation)?	No
5.3	Is there a risk that the Project would lead to forced evictions?[15]	No
5.4	Would the proposed Project possibly affect land tenure arrangements and/or community based property rights/customary rights to land, territories and/or resources?	No
Standard 6: Indigenous Peoples		
6.1	Are indigenous peoples present in the Project area (including Project area of influence)?	No
6.2	Is it likely that the Project or portions of the Project will be located on lands and territories claimed by indigenous peoples?	No
6.3	Would the proposed Project potentially affect the human rights, lands, natural resources, territories, and traditional livelihoods of indigenous peoples (regardless of whether indigenous peoples possess the legal titles to such areas, whether the Project is located within or outside of the lands and territories inhabited by the affected peoples, or whether the indigenous peoples are recognized as indigenous peoples by the country in question)?	No

[15] Forced evictions include acts and/or omissions involving the coerced or involuntary displacement of individuals, groups, or communities from homes and/or lands and common property resources that were occupied or depended upon, thus eliminating the ability of an individual, group, or community to reside or work in a particular dwelling, residence, or location without the provision of, and access to, appropriate forms of legal or other protections.

	If the answer to the screening question 6.3 is "yes" the potential risk impacts are considered potentially severe and/or critical and the Project would be categorized as either Moderate or High Risk.	
6.4	Has there been an absence of culturally appropriate consultations carried out with the objective of achieving FPIC on matters that may affect the rights and interests, lands, resources, territories and traditional livelihoods of the indigenous peoples concerned?	No
6.5	Does the proposed Project involve the utilization and/or commercial development of natural resources on lands and territories claimed by indigenous peoples?	No
6.6	Is there a potential for forced eviction or the whole or partial physical or economic displacement of indigenous peoples, including through access restrictions to lands, territories, and resources?	No
6.7	Would the Project adversely affect the development priorities of indigenous peoples as defined by them?	No
6.8	Would the Project potentially affect the physical and cultural survival of indigenous peoples?	No
6.9	Would the Project potentially affect the Cultural Heritage of indigenous peoples, including through the commercialization or use of their traditional knowledge and practices?	No
Standard 7: Pollution Prevention and Resource Efficiency		
7.1	Would the Project potentially result in the release of pollutants to the environment due to routine or non-routine circumstances with the potential for adverse local, regional, and/or transboundary impacts?	No
7.2	Would the proposed Project potentially result in the generation of waste (both hazardous and non-hazardous)?	No
7.3	Will the proposed Project potentially involve the manufacture, trade, release, and/or use of hazardous chemicals and/or materials? Does the Project propose use of chemicals or materials subject to international bans or phase-outs? *For example, DDT, PCBs and other chemicals listed in international conventions such as the Stockholm Conventions on Persistent Organic Pollutants or the Montreal Protocol*	No
7.4	Will the proposed Project involve the application of pesticides that may have a negative effect on the environment or human health?	No
7.5	Does the Project include activities that require significant consumption of raw materials, energy, and/or water?	No

Annex E: Stakeholder Engagement Plan

Stakeholders consulted during the PPG

The table below represents the baseline of stakeholders engaged during the PPG process. They are grouped in two broad categories: Category A) Community members in project Sites; Category B) Institutions and ordered chronologically for the sites and alphabetically for the institutions

Table 6 : Current baseline of stakeholder involvement

Number of entities	Entity ordered (see note on order)	Stakeholder / Position / Note	Gender	Approximate baseline headcount and gender balance status
Category A) Project Sites	Site Level Consultations with local communities, and community-based groups / associations and NGOs	All stakeholders directly consulted on sites: Includes community members directly consulted during the PPG, of which approximately 25% women ([16])	Gender balance varied, but overall it was unbalanced	At least 180, and no more than 220, community members directly (29% women) (226 total & 67 female)
1	Bambadinca Sta Claro Bambadinca Community Energy Service (SCEB) Community based Management & Administration of Photovoltaic-Hybrid Plant	Technicians and administrative personnel of SCEB on the site, 7 persons, 1 female	f & m	7 persons, 1 female
2	TESE Bafata, NGO, international	Representatives consulted on site, Energy Sector Chief Georgios Xenakis, Project Coordinator Lara Espirito Santo & other staff , 5 persons, 1 female	f & m	5 persons, 1 female
3	Community of Pitche,	Representatives consulted on site, Administator Adulaizinho Embalo & administrative and technical energy staff Local Women Association, Ms. Fanta Mané, community members male & female	f & m	20 persons, 25% women
4	Community of TcheTche	Regulo Corca Djalo, village chief & other local authorities, school	f & m	25 persons, 10% women

[16] Evidence of community level stakeholder consultation

Number of entities	Entity ordered (see note on order)	Stakeholder / Position / Note	Gender	Approximate baseline headcount and gender balance status
		director (Sana Suane) Head (Mamadu Boi Djalo) +members of Village Association of TcheTche Head of Women Association and Women Horticulture Organization, Fatumatu Djalo + members of Women Association		
5	Gabu NGO ADAC NAFAIA	Head (Mamasamba Cande + members of local NGO consulted, sustainable rural development Gabu region, gender /FGM, participation of technician (Braima Barri) in our trip to Tche Tche and Sintcha Morreira consulted	m	5 persons
6	Community of Sintcha Morreira - Gabu	Community members and local traditional authority / Chief of village consulted on charcoal production (at least 20)	f & m	25 persons, 35% women
7	Buba IBAP HQ of PA Cufada Lake Region	Director of Cufada Natural Park Joazinho Mané Technician + Administration (2)	m	3 persons
9	Community of Bakar Konte, Cufada Lake Region	Village Chief Rashid Sané, Community members consulted, village association has 87 members (at least 20 women)	f & m	20 persons, 13% female
10	Community of Tchandanca, Cufada Lake Region	Head of Women Association, Safiata Indjai + community members consulted,	f & m	15 persons, 80% female
11	Community of Duta Djara, Cufada Lake Region	Vice President of Village Association Djibril Turé and community members consulted	f & m	12 persons, 8% female
12	Community of Lamane, Cufada Lake Region	Fishermen's Association of Rio Grande de Buba – Lamane (APRGB), legalized local fishermen & women organization consulted, responsible Sadjá Mané, + 24	f & m	40 persons, 40% female

Number of entities	Entity ordered (see note on order)	Stakeholder / Position / Note	Gender	Approximate baseline headcount and gender balance status
		members; village teacher + other community members; some households		
13	Community of Cacine/ Cantanhez National Park	Government / local authorities /administrator / FISCAP/ hospital staff consulted Fishermen at Cacine Port consulted South Korean Fish enterprise President of Women Association (legalized, 100 members), Aissata Cassama + community members	f & m m m f	8 persons, 2 women 10 persons 2 persons 4 persons
14	Community of Iemberem/ Cantanhez National Park (virtual visit as road was closed)	Director of NGO AD in Iemberem, Abubacar Sere, telephone interview; Local Associations of Women, Youth, Fishermen and Community members consulted	f & m	10 persons, 3 female
15	Communities of Cacheu Urban	Community members consulted (at least 20), through IBAP director Dr. Justino Biai	f & m	20 persons, 7 female
	FINISHED	**PART A**		
Category B) Institutional	Government (national), research institutes, NGOs and development agencies + others	Stakeholders consulted during the field missions, both through direct contact (focused meetings) and/or through their participation in the PPG Workshops (Inception, in July 2018, and Validation in October 2018)	Gender balance headcount: 51 (88%) are men and only 7 (12%) are women	Baseline headcount: 58 Bissau-based 'decision-makers'
16	Action for Development (AD), NGO, Bissau	Jorgito Handem, Director of Vocational School (Escola de Artes e Ofícios de Quelele), Bissau 2 RE + electricity teachers	m	3

Number of entities	Entity ordered (see note on order)	Stakeholder / Position / Note	Gender	Approximate baseline headcount and gender balance status
17	Action for Development from People to People (ADPP), NGO, Bissau	Project coordinator Renewable Energy David Kerkholz, Fernando Binhafa	m	2
18	African Development Bank (AfDB)	Responsible consulted	m	1
19	AMAE Association of Women of Economic Activities (AMAE)	Antónia Adama Djalo (Vice-president) and 3 more women from the directive board Technician workshop	f m	4f 1
20	Director General of Agriculture (DGA)	Casimiro J. Sanca, Administrative and Financial Assistant	m	1
21	Director General of Forests DGF	Representative consulted	m	1
22	Director General of Public Works	Representative Consulted	m	1
23	Directorate General of Territorial Administration	Buli Dabo, Técnico	m	1
24	Secretariat of State of Environment / Ministry of environment and Sustainable Environment (SEA/MADS) Directorate General for the Environment DGA, MADS / SEA (Project Executive / Implementing Partner)	Sra. Quite Djata, Secretary of State of Environment and GEF Political Focal Point Laurenço Vaz, Head Office SEA	f	13, 5 women
		Guilherme da Costa, General Inspector of the Environment	m	
		Edilson S.V. Camará, DGDS, PFEA/RCA	m	
		Viriato Cassamá, Director General of the Environment	m	
		Mangla Nantchia, Perito	m	
		Mário S. N.M. Correia, Administrative Assist.	m	
		Nélida Gomes Teixeira, Accountant	f	

Number of entities	Entity ordered (see note on order)	Stakeholder / Position / Note	Gender	Approximate baseline headcount and gender balance status
		Titina Pereira, Administrative Assistant	f	
		João Raimundo Lopes, PFO/GEF	m	
		Mário Batista Camalá, Environmental Lawyer	m	
		Maria da Conceição G. Lopes, Sustainable Development Advisor	f	
		Aua Zahia Sow, Coordinator N-PB	f	
		Julio M.L. Bá, Director General of Sustainable Development	m	
		Albert Calde, Technician	m	
25	DGE - General Directorate of Energy (MEIRN)	Lamberto Soares Camará, Director / Coordinator of Rural ElectrificationJúlio António Raul, Technical, Renewable Energy, GEF Project Coordinator - UNIDOCarlos Adriano Conduct, Director of Renewable Energies Biabe Siga, ECREEE Focal Point	m m m m	4
26	Directorate General of Water Resources (MEIRN)	Inussa Baldé, Director General of Water Resources, Focal Point OMVGCrisóstomo Alvarenga, Director Planning Water infrastructures	m m	2
27	Research Center Applied Research - CIPA	Albano Indafá (Laboratory Manager)	m	1
28	Gestão Durável e Valorização de Recursos-GDVR, NGO Bissau	Bernardino Santos - President of the Executive Council	m	1
29	INITA (National Institute for Applied Technology)	José Carlitos Ialá, President	m	1
30	IBAP Institute of Biodiversity and Protected Areas (IBAP)	Justino Biai, PhD, General Manager Maurício Insumbo, Project ManagerLeonildo Cardoso, Responsavel SA	m m m	5, 1 woman

Number of entities	Entity ordered (see note on order)	Stakeholder / Position / Note	Gender	Approximate baseline headcount and gender balance status
		Fernanto Biay, PNTC Director of Mangrove National Park of Cacheu (PNTC) Udimila Cadija Keita, Communication Officer	m m	
31	INEP National Institute of Studies and Research INEP	Bucar Indjai, environmental specialist Dr. Paulina Mendes, socio-economics & gender expert	m + f	2
32	INE National Institute of Statistics	Braima Manafá, Director General	m	1
33	IGA Institute of Environmental Management / MADS	Fred F. Cabi, Technician	m	1
34	Ministry Economy Finance / General Direction of the Plan	Responsible consulted	m	1
35	Environmental Competence Assessment Authority (AAAC)	Quintino Imbadji, Técnico,	m	1
36	Ministry of Energy, Industry and Natural Resources (MEIRN)	António Serifo Embaló, Minister	m	1
37	UNDP – UNDP	Gabriel Dava, Deputy Resident Coordinator Dawda Sau, Responsible Cluster Environment and Sustainable Development Dam-Vieira da Costa, Responsible of Communication	m	3
38	PROSOLIA	Riogand Kandety, Responsible Finance	m	1
39	SEPIR / MOP	A. João Gomes, Technician	m	1

Number of entities	Entity ordered (see note on order)	Stakeholder / Position / Note	Gender	Approximate baseline headcount and gender balance status
40	TESE, NGO	Georgis Xenakis, Sustainable Energy Expert	m	1
41	TEDITRONIC	Alano Lurenço Lopes, Responsible	m	1
42	UICN - International Union for Conservation of Nature	Alfredo da Silva, Program Manager and Representative	m	1
43	UNDP Istanbul Regional Hub, Regional Bureau for Europe and the CIS	Mr. Saliou Touré, Regional Technical Adviser, Climate Change and Sustainable Energy	m	1
44	West African Development Bank	Responsible consulted	m	1
45	World Bank	Responsible consulted	m	1
46	JRL-ENERGIA SOLAR	Rogério Kafett Costa, Representative	m	1
47	PP-Energy Sarl	Pedro Jesus Oliveira, Director	m	1

[END OF STAKEHOLDER COUNT: n total = 284 / 74 female]

Participatory Stakeholder Engagement Approach

The approach to stakeholder engagement and participation during project implementation will be based on the principles outlined in the in the table below:

Principle	Stakeholder Participation will
Value adding	Be an essential means of adding value to the project
Inclusivity	Include all relevant stakeholders
Accessibility and access	Be accessible and promote access to the process
Transparency	Be based on information transparency and fair access to it
Fairness	Ensure that all stakeholders are treated in a fair and unbiased way
Accountability	Be based on a commitment to accountability by all stakeholders
Constructiveness	Seek to manage potential conflicts and promote the public interest
Redressing	Seek to redress inequality and injustice
Capacitating	Seek to develop the capacity of all stakeholders
Needs-basing	Be based on the needs of all stakeholders
Flexibility	Be flexibly designed and implemented
Rationality and coordination	Be rationally coordinated and not be improvised
Excellence	Be subject to ongoing reflection and improvement

The project design incorporates several features and activities to ensure ongoing and effective stakeholder participation. The project implementation mechanisms should facilitate involvement and active participation of different elements:

Project inception workshop

- enables stakeholder awareness at the beginning of the project implementation;
- provides an opportunity for all stakeholders to be provided and share the most updated information on the project and work plan;
- provides a forum for further consultation at the start of the project implementation;
- addresses many key issues including assisting all implementation partners to fully understand and take ownership of the project, its outputs and activities;
- discussion of the roles, functions and responsibilities of all implementation partners with the project structure, including reporting and communication lines and conflict resolution mechanisms;
- allows review the project budget and first annual work plan, revision of indicators, targets and their means of verification, recheck assumptions and risks;
- overview of reporting, monitoring and evaluation (M&E) requirements.

Project Management Team to be established that will oversee stakeholder engagement processes during the project implementation.

Project communications to facilitate ongoing awareness of project.

The project will develop, implement and maintain a communications strategy to ensure that all stakeholders are informed on an ongoing basis about: the project's objectives, the projects activities, overall project progress, and the opportunities for involvement in various aspects of the project implementation.

Stakeholder consultations and participation in project implementation:

A comprehensive stakeholder consultations and participation process will be developed and implemented for all project outputs/activities, building on the process already started during the project conception.

A participatory approach will be adopted to facilitate the continued involvement of local stakeholders, mainly women and youth.

Wherever possible, opportunities will be created to train and employ the population of the targeted sites and proximate villages for project intervention with at least 50% women participation.

Capacity building:

All project activities are strategically focused on building the capacity at the systemic, institutional and individual level -- to ensure sustainability of initial project investments. The project will, wherever possible, use the services and facilities of existing local training and skills development institutions.

Stakeholders ***of the Renewable Energy & Low Carbon Technologies /Management*** in Guinea-Bissau have visible and important capacity deficits[17]. At the same time, it is important to recognize that **capacity problems in Guinea Bissau are of systemic nature** and directly linked to the very low levels of adult literacy that still persist in the country, the fact that less than 0.1 % of the population has had access to tertiary education and the limited governmental spending.

The project will focus quite specifically on RE and Clean Energy/Low Carbon Bioenergy Technologies with respect to capacities. Capacities have to be built locally and nationally with the help of this and other projects (Outcome 1 and 2). Therefore a superficial and generic capacity assessment of **key institutions** may end up not being fully satisfactory and would not help the project scoping. It is considered more effective to determine the (future) tasks and responsibilities for key stakeholders under the project first and then try to increase their capacity accordingly.

Due to the lack of higher education institutions and national industries in the energy sector, there are not many entities implementing research and development projects. The implemented projects use known and mature technologies that have already been developed and applied in other countries. The only national institution that operates in this area is INITA, which has implemented projects with innovative technologies at national level since the 1980s (Baseline study, p. 128).

[17] see Baseline Study

Expected Stakeholders engagement per output

Outcome/ output	Stakeholders	Engagement Activities
I. **Component 1**		
Output 1.1	Ministries, particularly MEIRN + MADS+ MEF + Customs+ representatives of local communities; community development committees, at least 50% women	Drafting of policy package at government level; Participatory research with communities; Advocacy &Political will; Voices of community committees raised and included, particularly women
Output 1.2	MEIRN + UNDP+ MEF, EAGB?, commercial banks, private investors, local associations, particularly economic activities, AMAE, Local Women Associations	Cooperation between UNDP, Ministries, EAGB & banks for development of standard nation-wide electricity cost system, tariff criteria, including poor community and women+ private entrepreneurs needs and coordination
Output 1.3	MEF, MEIRN, MADS, INITA, IBAP, GEF small grants program (please refer to the table below), +private enterprise, lessons learned from other projects + NGOs, AMAE, EU, Local associations, mainly women	Support private investment, women entrepreneurship, Free capacity building and training for marketing and use of improved cooking stoves
Output 1.4	INITA, IBAP, INEP, local community representatives, Ministries of Energy + Environment	Joint development and demonstration of alternative solutions, including community proposals by women, for high efficient electric cooking devices, solar cooker etc. beyond wood fuel
Component 2		
Output 2.1	INEP, INITA, Ministries, ADPP, AD-Escola de Arte, local communities with 50% women, Others Technical and Education specialized schools	Based on existing guidebooks introduce specific country components, capable to respond effectively to low knowledge and technical level of targeted communities, participation of targeted communities and women in guidebook development
Output 2.2	INITA + AD+ ADPP,, construction/civil engineering MEI, Private Sector Enterprises + other RE business, new women RE business initiatives	On the job training in electricity & RE for women and men equally, mostly from local communities; general capacity building for project site population (promoters and consumers) and on RE use, economic benefits and environmental advantages.

Output 2.3	MEI, involved NGOs, Chambers of commerce ??, women community committees or similar	"Help Desk" established to provide commercial and technical advisory services to potential mini-grid developers, at MEIRN (Rural Electrification Agencies). Trained staff assists and advices mini-grid developers, including on preparation of feasibility / business plans and interpretation of agreements involving them as developers, rural electrification agencies and EAGB.
Output 2.4	MEIRN, IBAP, MADS, INITA, ADPP, AD, women community committees, artisans, farmers or similar	Design and implementation of capacity building program on fuelwood value chain based on tripartite partnerships: ministries- NGOS and women community committees; Capacities developed within the IBAP, artisans, farmers, loggers, and key national actors such as the Ministry of Agriculture and the Ministry in charge of natural resources (MEIRN) on best practices and opportunities in the sustainable use of firewood. Awareness of deforestation and capacity building / competence in sustainable & efficient cutting and use of wood as well as improved stove making techniques; Development of viable business models for the production, marketing and maintenance of improved stoves.
Component 3		
Output 3.1	ASCENT/UNDP, MEIRN, GEF +Co-financing partners	7 pilot sites identified, assessed and investment model as hybrid solar-diesel mini grid defined; further (financial) assessment in implementation phase;
Output 3.2	Banks, National/International Investors operating in RE in GB & Region, Female green business, MEIRN, MADS)	Competitive and transparent bidding process; hybrid power plant sites / concessions will be awarded to potential developers; feasibility study, construction and operation of the plants will be the sole responsibility of the potential developers for 25 years (renewable). 1. Project will provide financial incentives to investors.

		2. Proponents with least amount of funding required will be selected for the next step in the process, namely: feasibility study and business plans, including technical considerations, economic parameters, financial parameters and environmental considerations (environmental impact assessment). 3. Signing of the partnership agreements.
Output 3.3	MEIRN, MADS, IBAP, INITA, private enterprises for production, commercialization and regional distribution (at targeted sites, in main urban centers) of improved cookstoves and kilns; cycle built on women entrepreneurship.	-Awareness and information campaign on the impact of deforestation and climate change for rural communities to understand the notion of sustainability of the source/wood. -During this campaign, alternatives will also be proposed. -The organizations (associations, producers) will then be trained on the methodology of manufacturing and operating improved stoves and kilns. -Legal devices will frame these craftsmen/women and allow them to market their production.

Relevant SGP projects:

Project Title	Country	Area Of Work	GEF Amount	Objective
Environmental Education and Clean Energy	Guinea-bissau	Biodiversity Climate Change Mitigation Land Degradation	20,000	Contribute to changing the mentality of society in general in the relationship with nature, as well as promoting access to clean energy.
Women's Empowerment for Economic, Social and Environmental...	Guinea-bissau	Biodiversity Climate Change Mitigation Land Degradation	33,000	Reduce the pressure on the use of renewable natural resources (forestry and fish), horticultural valorization as an alternative income-generating activity and ecologically correct in the perspective of guaranteeing social
GHG emission reduction	Guinea-bissau	Climate Change Mitigation	25,000	Phase 5 of the project. Promote Renewable Energies, and Clean energy (biogas, improved cookstoves)

Annex F: Gender Analysis and Action Plan

Among the 17 SDGs is the goal of ensuring "access to affordable, reliable, and sustainable modern energy for all" (SDG 7).

Access to modern energy is also a key enabler for women's empowerment (SDG 5) – because rural women and girls are predominantly responsible for the bulk of household work, access to energy makes a significant difference to their health and well-being.

While access to energy services would not necessarily guarantee gender equality, it would go a long way in relieving women and girls of the drudgery associated with their daily tasks and providing them time for income-generating opportunities and education.

(UNDP/CCGA, 2016)

Frame of Gender Action Plan

This section sets out the rationale for gender mainstreaming and presents the background to the process

Women's empowerment is crucial for all-round social development and environmental sustainability. Mainstreaming gender in renewable energy policies and programming can enhance the efficiency of energy policies. Through the incorporation of gender perspectives the effectiveness and gender equality not just of energy programs and policies, but of all development activities involving energy use can be ensured (UNDP 2012).

Modern energy poverty and pollution from extensive biomass use affect men and women, boys and girls, the old and marginalized populations in rural and remote areas all over the developing world (UNIDO& UN-Women, p. 6). It is the time to improve their energy access with sustainable clean (renewable) energy, where beneficiaries are recognized as men and women with different energy needs and priorities. The goal of this project is to enhance the welfare of both women and men in Guinea-Bissau as their distinct energy needs are taken into account.

According to "SE4ALL Guinea-Bissau Investment Plan" (p. 38), the electricity coverage rate in 2010 was 29.1% in the city of Bissau, 4.3% in other major cities and only 1% in rural areas. Thus, only a very small portion of rural population have standalone power connections, run mainly on small conventional diesel generators (WB 2017). PV based electricity by private households is a rare exception in Guinea-Bissau´s rural areas (see image from the project site, the village of Dutu Djara in the Cufada Lake Region, Community Report, p. 16).

Cooking, heating, conservation of fish & other food (through smoking etc.) are all predominantly women activities and based up to 95% on biomass use, i.e. mainly solid fuel (wood) and charcoal to a lesser extent. Men and women, children and the elderly, especially in remote rural locations, suffer from energy poverty (UNIDO & UN-Women, p. 8). Much female labor is unremunerated and overlooked in the national account. Women in Guinea-Bissau continue to be outside the decision-making sphere and their human rights are often comprised when addressing energy access issues. Albeit undeniable contribution to economic, social and political development, these women have not equally reaped its benefits.

Energy poverty is defined as the lack of adequate modern energy for the most basic needs of human life and development, such as cooking, heating, lighting, income generation, community schools and health centers. The lack of appropriate, sufficient and clean energy limits human development in many spheres

such as education, nutrition and food security, water (ECREEE 2015, Situation Analysis, p. 31). The energy divide is gendered with women in most developing countries experiencing energy poverty differently and more severely than men.

In Guinea-Bissau a considerable part of rural areas is located outside the main energy grid installations planned to be carried out as energy mix in the next 5-10 years (see areas covered in SE4All GB IP). The only alternative to cover their energy needs is through renewable energy/hybrid off grid / standalone mini grid solutions. The current GEF project works in this direction and aims to demonstrate the scope of mini and micro RE off grid solutions for Guinea-Bissau´s forest dependent communities with focus on gender equality.

The Gender Action Plan (GAP) of the present project is orientated by the recent "ECOWAS Program on Gender Mainstreaming in Energy Access (ECOW-GEN)", the key program of ECREEE (ECOWAS Centre for Renewable Energy and Energy Efficiency) working at the intersection of gender and energy. Achieving gender equality and transitioning to clean energy are efforts that are interconnected and mutually reinforcing in surprising and complex ways. Complementing the regional effort to achieve access to sustainable energy for all, ECOWAS Regional Gender Mainstreaming Policy on Energy Access aims to harmonize and strengthen legislation and policies on gender equality,[18] increase women's participation in decision-making and expand the economic opportunities for women. It contains a number of specific provisions relevant to women`s energy needs and work burden:

> *Access to Water and Sanitation -- Member States shall take the necessary measures for the improvement of sanitation and access of people to safe drinking water through: (b) the improvement of renewable energy (use of solar) and rural electrification in a bid to reduce the burden of work on women and improve the competitiveness of businesses managed by women;*

> *Environmental Management and Protection: -- Member States shall take measures to promote the use of alternative energy for domestic purposes in order to mitigate the negative impact of firewood use, such as the disappearance of certain plant and animal species, the degradation of forest resources, soil impoverishment, and the occurrence of bush fires;*

> *Access to Energy Services -- Member States shall undertake to promote access to energy services for all, and particularly to increase access to energy services for rural and peri-urban populations in an equitable manner and without gender related discriminations enable Development Goals, Climate Policies, ECOWAS RE and EE policy, etc.*

(ECREEE, 2015: Situation Analysis, p. 58).

[18] Program areas are the **ECOWAS Women's Business Fund** to support the establishment and expansion of women-led energy businesses; **Women's Technical Exchange Program** to facilitate knowledge and technology transfer among women groups; **Women's Economic Empowerment through Energy for Productive Uses** to provide business development and capacity building for rural women etc. (ECREEE 2015, p.57ff.) In the same line we find the **African Development Bank Gender Strategy (2014-2018) which intends** to strengthen gender mainstreaming in all of the Bank's country and regional operations and strategies, including a focus on Women's legal status and property rights; Women's economic empowerment; and Knowledge management and capacity building. In 2018, ECREEE partnered with AfDB to launch a regional workshop to increase women participation in the renewable energy sector – only 2% of energy sector entrepreneurs in West Africa today are women. The joint initiative ultimately seeks to develop a pipeline of investment-ready, women-owned energy businesses across the region.

Another important regional initiative is The West African Clean Cooking Alliance (WACCA), launched in 2012. It is led by ECREEE to ensure that by 2030 the entire ECOWAS population has access to efficient, sustainable, and modern cooking fuels and devices (ECREEE, 2015, p. 59).

Objectives of the Gender Action Plan (GAP)

Based on accurate gender analysis, the ECOWAS´ and other regional gender initiatives emphasize that women are increasingly and rightly perceived and viewed as ***key actors of change in the energy sector*** and not only as end-users or clients of energy solutions.

It is in this line of thought that the present project is conceptualized. It places the economic activities of women in rural areas / forest dependent communities of Guinea-Bissau at the center of renewable energy provision and related business models to be developed within the project.

The Gender Action Plan captures and expresses the way in which gender can be mainstreamed within this project to promote better access to modern energy services through sustainable mini-grids and low-carbon bioenergy technologies. The gender action plan will touch on all components of the project.

Its objectives are:

- integrating gender as an overall goal by promoting gender equality and empowering women in community-based renewable energy systems;
- ensuring effective participation of women in all project activities and lead position of women in the management of community based renewable energy systems;
- capacity building of women in the RE sector and low-carbon bioenergy technologies in technical aspects and management alike;
- mainstreaming women's energy needs and concerns as identified through the gender analysis and consultations with all relevant stakeholders into policies, planning and implementation of RE solutions;
- continuous assessment of equal need-responsiveness of the energy project for women and men through gender indicators and monitoring and evaluation (M&E) from a gender perspective.

Gender Responsive Project Design

The field visits for the identification of seven project sites of forest dependent communities in the region of Gabu, Cufada Lake in Buba region, Cantanhez National Park and Cacheu Protected Areas include a careful need assessment and a gendered understanding of energy needs and priorities which will be dealt with in detail in the following section (gender assessment, situation analysis).

Reality on the ground from previous projects has important lessons for us to be learned. Although policies should guide the implementation of gender mainstreaming into energy projects, experiences from projects in Benin, Senegal and Guinea-Conakry demonstrate that gender is integrated at the project level first while policies are taking cues from activities already implemented. Only a few gender policies establish links between gender equality and energy poverty; vice versa the energy policies lack of gender considerations (ECREEE 2015: Situation Analysis, p. 62).

In view of the prevailing energy priorities and needs that we are able to identify at the community level, the RE supply model designed by the project not only will impact most positively on women as beneficiaries and consumers of energy but will put the productive economic activities of women at the center of power supply design and participatory energy management structures (business model).

The gender responsive project design is based on three core elements with a history of evidenced successful implementation

First core element: Multifunctional Platform

Inspired by the UNDP Multifunctional Platform Program[19], the project aims at focusing the energy installation on agro-processing machines (existing or partly existing in the visited communities) which enable women farmers and women led households to increase and diversify their income. The RE/hybrid energy provided will also strengthen capacities for service delivery of public infrastructures (health center, school, administration, water pumping, public lightning) in rural communities.

Women in isolated rural communities in Guinea-Bissau with no access to electricity are the primary customers for the energy services, purchasing grinding and milling services, and also the entrepreneurs who would own and operate these machines through local Women Associations/ Groups. While electrification will benefit both women and men by enhancing their engagement in more productive activities, gender gains in the first phase are derived mainly from reducing the workload of women and girls.

In addition to processing the local crops, pressing the oils etc., households and public buildings receive electricity, drinking quality water can be pumped by solar systems and energy can be provided for local private business, handicraft, battery chargers, welding stations and carpentry equipment.

While the main focus is to alleviate women`s hardship of manual processing of crops etc., the energy also creates jobs for men (carpentry, welding, machine maintenance and repairs), increases agriculture production for the entire village, and, most importantly, involves both men and women in decision making processes.

Second Core Element: Cooling and storage systems

Beside agriculture, fishing is the second main activity of rural communities in Guinea-Bissau, particularly in the coastal zone where 5 of 7 project sites are located. Renewable / hybrid energy supply would focus on the provision of energy for cooling and storage of fish as well as of other products of seasonal character (different types of fruits and vegetables, cold water, ice, meat, chicken). As fish is traditionally commercialized by women, the management of cooling and storage systems could be with the Women Association who buy all fish of the community (mostly fished by men) or in combination with the Fisher Association.

RE/hybrid energy plants based fish conservation will increase the revenue of the whole community as the value of fresh fish is higher than of smoked fish. It further reduces the need to smoking fish by biomass (and consequently deforestation and health issues) and increase eventually overall fish catch if sales market is guaranteed. Good experience has been made with improved fish smoking facilities, some already existing in the visited villages which further reduce wood consumption (see ECREEE, 2015: Situation Analysis, p. 68).[20]

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[19] Lead countries of the UNDP MFP program in West Africa are Mali, Senegal and Burkina Faso with about 3000 platforms installed as of 2012, benefitting over 3.5 million people, ECREEE, 2015: Situation Analysis, p. 66.

[20] See on-the-ground projects such as a women's fishing cooperative in Senegal that received support to modernize kilns used for smoke-drying fish. In addition to healthier working and air quality conditions. The eco-friendly kilns have boosted productivity, product quality and overall income levels for women. https://www.seforall.org/content/empowering-women-west-africa-advance-climate-and-energy-access-goals, 13.11.2017.

Third Core Element: Low carbon bioenergy technologies

This third element of the project design is by itself gender-responsive and empowers women.
Fuelwood use for domestic purposes is synonymous with women in Guinea-Bissau. Women are active agents in biomass management: they are the primary household cooks and they participate in many biomass-intensive processing sectors (on the subsistence and income generating level). As such they will play a key role in shifting the country onto a cleaner development pathway.

It is estimated that the time spent for wood collection varies between 2 to 3 hours per woman per day in Guinea-Bissau (PIF). With adequate management of firewood and improved cook stoves, this time window can be reduced to only 2 or 3 hours per week, while significantly reducing the firewood and toxic smoke due more energy efficiency (more than 50%).

Women most likely have a majority share in cook stoves` production and commercialization, if properly trained. The production and selling of improved charcoal is a business that can be of equal benefit for women and men small producers and traders.[21]

Gender Assessment
This section presents the results of the gender assessment, identifies gender gaps and issues that need to be addressed by the project.

Lack of gender disaggregated data
Lack of or limited availability and poor accuracy / authenticity of gender disaggregated data for nearly all social, economic and political issues in Guinea-Bissau hinder a profound gender analysis (www.genderindex.org/country/guinea-bissau). Systematic collection of comprehensive data, disaggregated by sex, and measurable indicators to assess trends in the situation of women and the progress towards gender equality are still missing. It is a challenge and an urgency to gather more energy need and consumption related gender disaggregated data within this project´s frame.

Gender Context and Policies
The cultural and economic context in Guinea-Bissau dictates structural inequalities between men and women, added by specific social and political circumstances. Exclusion and discrimination of women in Guinea-Bissau are based on the logics of patriarchal power, reflected in phenomena such as female mutilation, domestic violence, early marriages, limited access to resources, land and credit, weak institutional representation, inequalities in the judicial and economic fields as well as in education, especially in rural areas. Less literate and less educated, more exposed to health risk factors and with less access to means of production and decision making, women in Guinea-Bissau are more exposed to poverty and not able to fully participate in the development of the country.

[21] The World Bank's Africa Renewable Energy and Access (AFREA) Gender and Energy program was implemented in Mali, Senegal and Benin. In Senegal, the program promoted the sustainable supply of wood fuel, sustainable management of forests as well as dissemination of improved cook stoves and installation of bio-digesters for cooking. AFREA's activities to integrate gender included a gender-sensitive rural appraisal to establish the baseline for the forest resource management; evaluation of the capacity of female and male charcoal producers to access urban markets; small business management training for men and women charcoal traders (ECREEE, 2015: Situation Analysis, p.67)

Awareness on gender issues was raised systematically, for the first time, in ***DENARP II (2011)***, the Second Poverty Reduction Strategy, which links structural gender inequalities to the economic development of the country, and through the ***PNIEG (2012***/2017), the "National Policy for Gender Equity and Equality".

The DENARP II sets (i) a specific objective for the promotion of gender equality and equity, which aims to "eliminate structural inequalities between men and women"; (ii) genders most of the specific objectives, explicitly differentiating men and women as targets of the proposed action; (iii) integrates the gender dimension into planning and follow-up indicators; and (iv) the project`s budgeting takes into account gender structural disparities and inequalities.

The "National Policy on Gender Equality and Equity" (PNIEG) was elaborated with participation of all stakeholders over a period of two years and nationally validated in 2014. On one side, it is the product of the work of the *National Institute for Women and Children*, created as an operational structure for the development and coordination of policies on women's rights, gender equality and the advancement of women in Guinea-Bissau, and on the other site, the fruit of tireless engagement of strong civil society women's organizations, supported by committed international donors.

The PNIEG for the first time takes into account the role of women in the different sectors of the society as well as in the decision-making spheres. The document which also emphasizes the priority of sustainable development is the most important roadmap for gender equality in Guinea-Bissau, providing thorough and comprehensive analysis and recommendations.

The PNIEG finds that women and girls have been especially disadvantaged by the years of crisis since they are allocated by gender to a secondary status in all spheres of household, community and national life. They face gender-based restrictions on their access to scarce resources (such as credit and land) and to education, and the double burden of household work to care for and feed their families along with market work to contribute to family income. Additionally, girls and women in Guinea-Bissau face the gender-specific risk of maternal mortality, and gender-specific abuses such as domestic violence, female genital mutilation (FGM), and early/forced marriage (PNIEG, 2012/2017).

The 2015 national development plan "Terra Ranka" (A Fresh Start), a guiding document despite the political impasse lived in Guinea-Bissau since 2015, does not pronounce much about gender and has only one specific gender program (34c). Gender policies in GB are a relatively new issue: In 2015 the African Development Bank and UN-Women Guinea-Bissau presented the first "Gender Profile" of Guinea-Bissau; only some national policy documents, for example the Security Sector Reform and Justice Reform, mainstreamed gender into their plans in recent years.

Strategic Links between Gender, Environment and the Energy Sector (Renewable Energies)

It comes to no surprise that so far national documents on Climate Change and Plans for Energy and Electrification (via Renewable Energy/ energy mix) contain very little on gender, such as gender differentiated vulnerabilities and resilience or gender specific energy needs and priorities.

The Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW), explicitly refers to women's electricity rights and obligates state parties to the convention to "take all appropriate measures to eliminate discrimination against women in rural areas […] [and to] ensure to such women the right […] to enjoy adequate living conditions, particularly in relation to housing, sanitation, electricity and water supply, transport and communications" (UNIDO& UN-Women, p.8).

Whereas the overall effects of energy poverty are largely consistent for both women and men and include low standards of living, suboptimal health, reduced productivity, and missed opportunities, women in poor settings often experience excess vulnerability to conditions of energy poverty above and beyond that faced by men (UNDP, 2016, p. 2-3).

Women are responsible for ferrying of goods, water, children, fuel, market goods, and household supplies. They do all manual labor at the household level beside farm work and manual agro-processing. They are exhausted physically by this workload and also exposed to physical hazards such as pollution, burns, back pain and poorly lit childbearing experiences (ECREEE, 2015: Situation Analysis, p. 33).

A clear linkage was found between energy and gender, once modern energy is provided, namely the differential access, use and effects of energy sources and appliances in the home, community and wider society. Socio-economic and socio-cultural dynamics interact to create distinct – engendered – patterns of use and benefits of the same energy source or appliance. (ECREEE, 2015, p. 18).

In the Guinea-Bissau NAPA (2007) the gender criteria are part of the multi-criteria analysis of intervention options for the increase of food security without women being focused on as particular vulnerable group (NAPA, Table 9, p. 46). The ACCC NAPA follow-up project (2011-2015) goes beyond this approach on the implementation level by consulting about 40% women out of a total number of 1120 persons and defining their specific needs. The project applies a meaningful participatory process for engaging women's voices and making them targeted beneficiaries of project interventions. Its success concerning mainstreaming gender into the project implementation is significant.

On the other hand, The Initial National Communication (2004) and the Second National Communication (2011) do not refer to gender issues beyond the notion of general vulnerability and gender biased climate change impact in particular. The same appears to be the case regarding basic environment laws and regulations.[22]

As described above for the regional level, ECOWAS in "ECOWAS Regional Policy for Gender Mainstreaming in Access to Energy" stresses the importance of incorporating gender considerations in the planning and execution of energy interventions to ensure socio-economic development. The dire energy poverty in the ECOWAS region remains a severe concern as well as the gender dimensions that characterize it. ECOWAS´ innovative gender policy (see above) will support women and bring clean energy access to remote population across West Africa. Through contextualization of gender aspects to the realities and development priorities of the ECOWAS countries, the ECOWAS gender policy intends to address existing barriers that hinder the equal participation of women and men in expanding energy access in the region and empower women and girls by improved conditions (ECREEE, 2015, p. 17).

A National Validation Workshop for the "ECOWAS Regional Policy for Gender Mainstreaming in Access to Energy" was held in 2017 in Bissau. The workshop concluded that the Guinea-Bissau energy authorities need external support, i.e. technical and financial assistance for the integration of gender into energy programmes, particularly gender focal points in the lead ministries (energy, environment,

[22] See the "Basic Legislation on Environment" (2011), the recent regulations of 2017, such as the "Regulations of Environmental impact" (No 7/2017), the "Civil Protection Law" (Decree 09/2011), the "Land Law" (1998) and the "Plan of Investment in Agriculture" (2013).

agriculture) as well as the creation of a Regulatory Authority in the Energy Sector (see ECREEE, 2017: Situation of gender mainstreaming in access to energy in Guinea-Bissau).

None of the main three policy documents/ Guinea-Bissau National Action Plans of Energy,

- Renewable Energy (PANER),
- Energy Efficiency (PANEE) and
- SEforAll Action Agenda (SEforAll AA) for Guinea-Bissau and Investment Plan

Developed in recent years (2013-2017,) refers explicitly to gender divide in energy demand and consumption schemes nor to the gendered dimension of biomass fuel burning on the household level. In its calculations of energy demand and satisfaction rates, in the description of types of renewable energy and energy mix models to be provided, the main grid and off grid solutions for rural population and the roll-out of many different projects - in all these areas the gender perspective is missing.

The PANER, the New National Plan on Renewable Energy (2015 until 2030 / (*Plano de Ação Nacional no Setor das Energias Renovaveis*) makes some references to gender, but does not contain a specific analysis of women's energy needs and specific expected results of energy supply for women. It is the only document that mentions the lack of gender *disaggregated data* on energy consumption. The chapters on improved cooking stoves and awareness campaigns for renewable energy solutions (p. 28, 41-42) do not refer explicitly to women but only to population in general and primary and secondary school students. The paragraph on the Energy Information System which consists in a systematic and exhaustive statistical data compilation also is not gendered. It only mentions that the relevant indicators should consider socio-cultural factors.

All in all, gender is hidden and nearly completely invisible in these plans; it is referred to directly only once (at the margin of the biomass consume – health link for school manuals, p.42). ***Gender equality and gender gaps are so far poorly addressed in the relevant Guinea-Bissau National Energy Policy documents: such as gender differentiated energy needs and use; the impact of energy use on women; the resource ownership question and female participation in the energy sector as technicians, employees and women entrepreneurship***.

The lack of explicit gender references in energy policies of Guinea-Bissau and vice versa of energy references in gender policies, is undoubtedly a big gap. Guinea-Bissau´s energy policies follow that of the region with no clear gender linkages and references to gender equality prior to the recent ECOWAS-GEN.

Progress in practical terms on the national level is constrained by limited human and financial resources available. The modern energy sector is very much male dominated, both at the decision and technical level, suffering from the absence of many would be participants and their efforts and talents.

In any case, some of the targets formulated in PANER have strong gender implications which should be considered as indicators at the implementation level: 9% of Guinea-Bissau`s population (ca. 220.000 inhabitants), living in rural areas, should benefit from RE/hybrid minigrid solutions until 2030; 35% of population should use improved (traditional) cooking stoves and 40% use modern cooking equipment (solar, biogas, GPL) until 2030; 80% of health care facilities/hospitals should be electrified until 2030 (targeting primarily rural areas).

Legal Status of Women in Guinea-Bissau

The reality of living conditions in Guinea-Bissau indicate severe discrimination of women and violation of their human rights. Articles 24 and 25 of the Constitution of Guinea-Bissau (1984, 1996) prohibit all forms of discrimination on the grounds of sex, race or religion. The country ratified the "Convention on the Elimination of All Forms of Discrimination against Women (CEDAW)" in 1985 and "The African Charter on Human and Peoples' Rights" in 1986 which obligates states to ensure the elimination of every discrimination against women and the protection of women and children rights as per international declarations and conventions. Guinea-Bissau also signed the UN-Resolutions 1325 (2000), 1820 (2008) and 1899 (2009) on the participation of women and their protection in conflict and post-conflict situations. It further ratified "The Protocol to the African Charter on Human and Peoples' Rights on the Rights of Women in Africa" of Maputo (2003), and the "Optional Protocol on Violence against Women" in 2009. In all these declarations, member states have reaffirmed commitment to gender equality.

The Maputo Declaration sets out parameters for the right of women to sustainable development, calling for realistic participation of women in development planning; guaranteed land and property rights; and access to credit, training, skills development and extension services to promote economic rights and combat poverty (Article 19).

Finally, the SDGs, adopted in September 2015 by the United Nations General Assembly, include an independent goal on gender equality (5), expanding the human rights-based development framework of the MDGs, under the theme of "leaving no one behind". The most recent relevant policy, the African Union's "Gender Strategy 2018-2027" provides for a continent where women, girls, men and boys have equal voice, choice and control over their lives. The strategy addresses as a priority: A conceptual shift from practical needs to women's strategic needs and a structural framework of gender equality, women's empowerment and women's rights.

It has to be admitted that there is still a lack of legislation and enforcement capacity in place that would enable the realization of the national and international commitments to gender equality and equity and the rights of women in Guinea-Bissau (www.genderindex.org/ country/guinea-bissau). Women are still waiting for these documents to become national standards that can be effectively applied at the local level for their benefit.

With respect to human rights baselines, the country's performance has improved substantially over the last years. Some positive developments in terms of women's fundamental rights and gender mainstreaming have been achieved in recent time, including the adoption of the law against Female Genital Mutilation in 2011 and Domestic Violence[23] in 2014, whose drafting and advocacy among stakeholders were strongly supported by UNDP Guinea-Bissau.

Nevertheless, in many rural communities of Guinea-Bissau customary law is still followed, and women do not always chose when and whom to marry. Marriages often are not registered. The legal age of marriage in Guinea-Bissau is 18 for women and men, but there are no penalties in place to punish those who force children into marriage (www.genderindex.org/ country/guinea-bissau/).

[23] Domestic violence is still a reality and legitimized by the victims themselves. In fact, 41.8% of women against 28.7% of men between the ages of 15-49 declare that a husband is justified in beating his wife in at least five circumstances. The 2014 law establishes sentences of up to 12 years in prison. Although police intervene in domestic disputes if requested to do so, women are often reluctant to report abuse due to stigma and social pressure (www.genderindex.org/country/guinea-bissau/).

While article 25 of the Constitution establishes the principle of equality between men and women, some provisions of the Civil Code and Family Code, inherited from the colonial period, discriminate against women. Although under civil law, both parents have equal parental authority within marriage and following divorce, article 1674 of the Civil Code states that the husband is the head of family, thus providing him the right to represent his wife and make decisions of behalf of entire family (www.genderindex.org/country/ guineabissau/).

From the cultural point of view, early marriage continues to be a recurrent practice in almost all communities. This custom of forced marriage and use of bride price to settle family debts, can disempower women and deprive them of opportunities to complete school education. Early marriage is a predominantly rural phenomenon, with 47% of girls between the ages of 20 and 49 marrying before age 18 against 26.8% of married women in urban areas. Polygamy as one of the factors that generates gender inequality, which is still present in the Guinean society (MICS 2014). Among individuals aged 15 - 49 years who live in a polygamous union, 44% are women and 25.8% are men. Female Genital Mutilation (FGM), according to the MICS 2014, affects 45% of women aged 15-49. Among girls aged 0-14 years, 29.6% were the target of this practice (MICS, 2010).

Property rights of women, inheritance & assets

Legally, women have the same rights as men to ownership, but among certain ethnic groups ***women have no access to land or non-land assets***, largely because of discriminatory customary laws relating to inheritance (www.genderindex.org/country/guinea-bissau/). Women's access to property other than land and their access to bank loans are heavily restricted because as heads of households, men hold sole authority over most family matters. In addition, according to article 1686 of the Civil Code, a woman cannot run a business without her husband's consent unless she is an owner of all of household's property or if couple has a separate property (FIDH, 2013).

Laws governing inheritance are also discriminating. According to Article 1678 of the Civil Code, the couple's assets belong to husband, but woman can take them over in case for some reasons husband is unable to do so. Customary laws that govern some ethnic groups prohibit women from inheriting property which is passed on to a male heir. Some communities practice widow inheritance, whereby a widowed woman is forced to marry one of her husband's male relatives. If she refuses, she and her children can be evicted from the family's land (www.genderindex.org/country/guinea-bissau/).

Women in Guinea-Bissau do not have full control over productive assets, and the question of ownership remains unclear because formal jurisdiction partially contradicts customary laws. In any case, women´s ownership and access to / control over productive assets (household, inheritance, financing capital) can determine development as it is critical to securing collateral for a loan or strengthening resilience against climate change threats etc.

Land is a crucial and critical asset with economic, social and cultural functions.[24] It is central in fighting poverty, hunger and deprivation as it provides families in rural communities the means to livelihoods through the production of farm products for sale. Women who have rights to secure access to, ownership and control over land are able to manage their lives better and are less vulnerable.

[24] Property rights that create or protect assets are vital, particularly inasmuch as they influence the start-up capital available to women to open new businesses, which can have lingering effects on productivity, returns and growth prospects (ECREEE, 2015, p. 80).

Gender disparities in land access remain significant in most states of ECOWAS, regardless of their level of development, confining women especially in the rural areas to economic insecurity and dependency. The percentage of women landholders is very small (Senegal 10%, Mali 5%, for Guinea-Bissau figures are not available, probably less than 1% (see ECREEE, 2015, p. 23, according to FAO "Gender and Land Rights database").[25]

Women are decidedly disadvantaged when it comes to land ownership. Land ownership impacts on business success in many ways: by providing appropriate premises from which to operate or sufficient area for agriculture production; also by functioning as collateral for loans (ECREEE, 2015, p. 79).

Underrepresentation of women in political life and government positions

Men hold 69% of government positions and are over-represented in key ministries related to women such as Agriculture, Education or Energy, where women hold just 14%, 26% and less than 10% of positions, respectively (GP, PNIEG). In the last elected Parliament (2014) women won only 14 seats, corresponding to 13.7% of participation.

The right of women to participate in political life is recognized in the Constitution of Guinea-Bissau. There is currently no legislation in place (such as ***quotas***) to ensure a more equal representation of women in politics. Women are underrepresented in decision-making positions as is obvious from the small number of women parliament delegates and the very few women who hold high functions in the political parties (UNIOGBIS, 2014).

In a recent research on women in Guinea-Bissau it was noted that there is strong appreciation of gender stereotypes by both sexes. Women are relegated to a domestic role while men are valued in the public and political spheres. The social preference of having men in decision-making bodies is particularly reinforced by women themselves who accept and contribute to strengthening their role away from decision-making. Many understand their role in supporting men`s ascension to their own detriment. This attitude of submission and self-exclusion is highly valued in society, mirroring all the complexity of gender relations and their transposition in all domains of society, including in decision-making spheres (Voice of Peace: 2018).

Some data indicate that representation, at the national level, had been better before the millennium turn due to the then influence of the first generation of women freedom fighters on active politics. For example, Carmen Pereira, one of the most important women fighters for independence, became the first woman National Assembly President (1984-1989).

The low representation of women in political life is all the more astonishing as women played an outstanding role in the Liberation war. Women "supported PAIGC because they saw in it the potential for their own liberation" (Urdang, 1979, p. 123). While at first, women grew and cooked food for the movement, they later began to transport supplies for the fighters or worked as nurses in the liberated zones. Others eventually took up arms as part of the regular army or local militias. The women were

[25] Women's right to land is buried in a complex cultural web of norms that deny them outright ownership. Women's traditional rights to use tribal land and retain the agricultural production from that land were disrupted by the combined pressures of population growth making land scarce and private tenure patterns imposed during colonial times. Men began to predominantly purchase and hold title to land.

extremely capable in mobilizing the population, and likewise some were made political commissars or were elected to village committees and the
People's Tribunals. Without the participation of women the liberation of the country would not have been achieved. At the party level, women were quickly integrated into leadership positions and a quota system was in place for the [PAIGC] tribunals and councils, in an effort to ensure that at least two of the five elected members were women" (Urdung, 1979, p. 275).

There is a strong network of NGOs, involved in women's issues, such as Women's Political Platform (PPM); Network on Peace and Security for Women in the Space of ECOWAS (REMSECAO); National Network for Combating Violence Based on Gender and Children (RENLUV); Association of Women with Economic Activities (AMAE); National Committee for the Abandonment of Nefarious Practices (CNPN); Association of Agricultural Producers and the Fight Against Hunger (APALCOF, Contubuel).

There are also associations for specific areas of women's work such as women fish-vendors, who are linked as members of the umbrella association of economically active women AMAE.

The PNIEG now needs to be operationalized and prioritized, with resources and targets. Some progress has been made to incorporate a gender perspective in new policies and priorities. Cases of FGM were brought to justice for the very first time in 2015, and debates to establish an up to 40% share of female representation in Parliament unfolded (Canchungo Declaration of October 2014). UNDP in its current Country Programme also established specific outputs to foster women political leadership and train them in relevant skills (UNDAF Guinea-Bissau 2016-20202, Outcome 1).

Socio-economic situation of women
A key characteristic of gender roles in the economy of Guinea-Bissau is the high percentage of women engaged in economic activity. About 80% of female household members over 15 are economically active compared to 90% of men (ILAP II, GP).

The economy of Guinea- Bissau is based on small farmers subsistence agriculture combined with cashew as the key export crop, involving all rural labor force (seasonally) and providing 90% or higher of exports. The quality of life of the most vulnerable segments of the population, especially women and young people, is dependent almost entirely on the cashew export: poor harvests such as occurred in 2012, 2013 and lately low producer prices, as happened in 2016 and 2018, have a negative impact on rural poverty and particularly female poverty, plunging significant part of the population into a state of under-nutrition, (African Development Bank: 2015-2019 Strategy, p. 5).

It is important to note that there is no social safety net to protect against social marginalization, extreme poverty and unemployment. The country also still lacks a modernized system for civil protection of citizens.

A small portion of the total labor force, estimated at about 10 percent and primarily male, are salaried employees in the private or public sector (GP). The few women who are educated hold government positions, own small businesses, work in hotels or restaurants, head local non-profit organizations or work for international development technical or funding partners. The private sector is composed of a few finance, trade, transport and service enterprises, primarily in Bissau. Industrial production is small-scale and largely limited to the processing of agricultural products with men as owners and managers and women at the bottom of the value chain.

As the majority of Bissau-Guineans and women (ca. 65%) live in rural areas with poor education and services, ethnic group traditions have an especially strong hold. The principal ethnic groups are Fula, Balanta, Mandinga, Manjaco and Papel. All of them have traditional beliefs and practices that restrict women's roles and rights. The Bijagó group of the Bijagos Archipelago in the Bolama region are an exception as they have a uniquely matrilineal structure (Fernandes 2012).

Gender roles and gendered division of labor

Women and girls in Guinea-Bissau, especially in the rural areas, are certainly constrained and disadvantaged by many gender norms. They are expected to perform caregiving for children, the sick and the elderly family members and household duties almost exclusively, making it more difficult for them to pursue opportunities elsewhere. Despite the fact that female household related activities have a huge value in terms of human development, health and family wellbeing, there is no monetary value placed on this work.

Without access to modern energy services, women and girls spend most of their day performing basic subsistence tasks, including time-consuming and physically draining tasks of collecting biomass fuels and exposing themselves to the grievous health effects of biomass energy. Cooking from biomass is particularly detrimental to the health – respiratory system - of women and children.

75% of households in Guinea-Bissau do not have on-site drinking water, and women over 15 years old typically spend at least 30 minutes per day to get water, according to the Multiple Indicator Cluster Survey (MICS 4). 95% cook with wood or charcoal (PNIEG 2012/2017), only 18% of the population has improved sanitation facilities for human waste; in rural areas the number is as low as 3%. The ILAP 2 survey (2010) found that a majority of Bissau-Guinean households (65%) uses candles for lightning at night.

Resource collection tasks are traditionally gendered in West Africa, but sometimes in surprising ways. Water collection is the most gendered task, with women, but also children, most involved in its collection. The picture on cooking fuels is more nuanced, with men participating in some countries where refilling LPG cylinders is a predominantly male activity. Unsurprisingly, there is evidence of less collection burden, more gender parity, and decreased involvement of children in the urban areas, as opposed to rural sites (ECREEE, 2015, p.41ff).

As 95% of cooking in Guinea-Bissau is based on biomass, it is almost exclusively women and children who have to collect *lenha* (wood for cooking) in the rural areas of Guinea-Bissau. Men participate in this task only with the objective to sell the wood in the local markets. Deforestation obligates women to overcome increasing distances from their home village to collect firewood and "illegal cutting" of wood frequently occurs in the Protected Areas, the project sites (Cufada, Cantanhez, Cacheu Natural Parks) because need for *lenha* is increasing every day with population growth.

Data from 13 countries showed that girls in sub-Saharan African homes with polluting cook stoves spend about 18 hours a week collecting fuel or water, while boys spend 15 hours. In homes mainly using cleaner stoves and fuels, girls spend only 5 hours a week collecting fuel or water, and boys just 2 hours (ECREEE, 2015, p.41ff).[26]

[26] It was found, only to give one impressive example, that improving public infrastructure for water and fuel collection as well as for food preparation (e.g. grain-milling facilities) could free women in Tanzania from a burden that represents 8 billion hours of unpaid work per year, which is equivalent to the hours required for 4.6 million

Solid fuel use exacerbates income and gender inequality and constraints women from accessing decent wage employment, educational opportunities and livelihood enhancing options. It also limits their options for social and political interaction outside the household (UNIDO-UN-Women, 2016).

The role and responsibilities of women in Guinea-Bissau have increased at the productive level, without parallel decrease in domestic obligations. This is due to the scarcity of social services and infrastructures, the emigration of men who often abandon their home once abroad, early widowhood of women and the absence of traditional mechanisms of social regulation.

Women end up like this, being responsible for large part of the family's living obligations, cooking and domestic work, with no real counterpart in terms of access to and management of household goods, but contributing more and more to family income. They usually resort to informal, low-income work as a farmer, merchant woman (*bideira*) and / or domestic worker (PNIEG, 2012/2017).

Bissau-Guinean women bear on average five children (6.8 in rural areas, according to UNICEF country statistics) for whom they care - all without access to basic infrastructure of water, sanitation, electricity or transport to save time, ease physical burdens, and increase productivity.

The informal sector of the economy is very large, with women in both Bissau and the regions engaged in small-scale trade, market sales and services. During economic crisis it is expanding even further. Rigorous data are lacking, but the PNIEG estimates that women are 51.6% of those engaged in the informal sector and particularly women heads of households (62.2%). (GP, p. 32). High degree of hardship is induced by the type of work undertaken in the informal sector and combined with the pressures linked to household duties are responsible for the physical exhaustion of many women.

Although both women and men suffer from the effects of poverty, the feminization of poverty in Guinea-Bissau is highly visible. Men control whatever scarce resources are available and women's income-producing labor becomes more arduous and less productive.

Women's gender specific roles are also more burdensome under poverty: domestic labor, which keeps women from income-producing activities and girls from school,[27] and women's reproductive roles, for which poverty increases risks of malnutrition, illness and maternal mortality. Young girls are also at greater risk of early/forced marriage to relieve family economic burdens.

Farming and gender

Women in sub-Saharan Africa have the highest average agricultural labor-force participation rates in the world. They produce up to 80% of basic foodstuffs for home consumption and sale. Cultural norms in the region have long encouraged women to be economically self-reliant and traditionally give women substantial responsibility for agricultural production in their own right. This is also the case in Guinea-Bissau where women farm on their own fields (or have user rights for these fields if they do not own them) and sell their own agricultural products; normally to invest in and for the family household, in food, school, medicine, and other purchases.

full-time jobs (see PRODOC Tanzania).

[27] There are weaker expectations that girls and women will study the so-called hard sciences at a high level. Professions typically thought of as "female" are generally not as lucrative, and women are often paid less for the same work performed by men.

Cashew is grown exclusively for export; rice and horticulture crops as well as vegetables and fruit are grown for consumption and national market sales. Despite the importance of agriculture to the economy, the majority of farmers – especially women – work at primitive levels, with little or no tools, equipment, training, access to water or decent roads. Since men control decisions over land and resources and often allocate these first to their own work, women farmers are especially disadvantaged (GP).

Labor division in agriculture is widely gender biased. Farm roles tend to be gendered with men performing much of the land clearing while women primarily process crops, transport them, hoe and weed (ECREEE, 2015, p. 38-39). Women's roles are concentrated at the low end of the value chain. For example, the farming and harvesting of the cashew nuts is done largely by women (peak season March to May), whereas the land is owned by their husbands or fathers. The nuts are sold by male farmers to a trader, exporter or processor (also male), or stored for later sale or exchange for family income and consumption items as decided by men. (GP). Women also work at peeling and sorting nuts. The cashew stem or fruit ("apple") is only processed on a very small scale, almost entirely by women.
Some crops in West Africa are culturally determined to be "male" or "female," others are logistically difficult for women to grow because of the expensive inputs needed, particularly irrigation, seeds, and fertilizer, or larger fields. Peanuts for example in the east of Guinea-Bissau are only cultivated by men as well as millet (Embaló, 2008).

Rice, the staple base of household consumption, is farmed by both women and men. Among most ethnic groups clear labor division is the rule. Rice farming is labor intensive and most work burden lays on women´s shoulders. In rice cultivation, men dig the fields, while the women sow the seedlings and do most of the harvest. They are also the ones who must peel the rice, usually done entirely by hand. Families have converted fields to the low labor-intensive cashew crop, trading cashews for rice through a bartering system rather than producing it, with the result that families often do not have enough rice throughout the year. The continued reliance on mono-culture cashew crop bears high economic and food risks.

Horticulture (vegetables) is a smaller sector of Guinean agriculture, directed to subsistence production and small business on the regional markets. It is exclusively done by women with an overall weak level of organization and lack of marketing strategies. *Badjiki* (roselle leaves), *djagatu* (African eggplant), *kandja* (okra), *malagueta* (chilli) and *sukulbembe* (West African pepper) – as well as palm oil and groundnuts – are used to prepare the sauces for Guinean *bianda* (rice meal). A few larger production units exist with some NGO projects and associations. This activity could significantly support the economic empowerment of rural women, if technical and financial means, irrigation by solar pump systems, storage by cooling and marketing strategies were improved.

Trade with local agricultural and forest products, such as *yam/cassava*, palm oil, fruits (orange, lemon, banana, mango, ananas), dried fruits from the forest, such as Guinea gumvine (*fole*), baobab fruit (*kabasera*), néré (*faroba*) and velvet tamarind (*veludo*) and their transformation on the local, national and regional level are practiced by women and men. The sector is underdeveloped due to dominance of commercialists from neighboring countries and lack of appropriate technical equipment, primarily electric oil pressing and conservation through electric cooling systems.

The Bideira, woman vendor and merchant

“Bideira” refers not only to women farmers or vendors of small foodstuffs in markets, but to women who search for or make life (*fasi bida*), i.e. gain some independent income for their household and improve their living conditions What is more, *bideiras* are not necessarily only women farmers, but also intermediaries who buy the products from the farmers and resell them in the food markets of Bissau, other larger markets in the provincial capitals or export it to neighboring countries. There even exists an export chain to the large Guinean community in Lisbon, organized completely through informal channels on the basis of traditional African market exchange structures by these *bideiras*. Women`s small but regular income from commercial farming and trading is mainly invested in basic supplies for children and to help each other and the mothers, following practices of female solidarity commonly found in West Africa (Abranches, 2013).
These contemporary merchant women face several hierarchical hurdles, such as social inequality based on the homogeneity of the formal markets and political rules mainly inherited from colonial and patriarchal norms. *Bideiras* transcend these structures by building new forms of everyday exchange. They take advantage of the decentralization and multi-dimensionality of their practices of daily entrepreneurship. Because of these efforts, *Bideiras* help create networks and relationships not only within Guinea-Bissau, but across borders traversing West Africa (Lundy et. al, 2016).

Fishing and gender
Guinea-Bissau has highly productive fishing waters, and fish and other seafood are a second principal export good, mostly through international commercial fishing, i.e. the payment of licenses to the Guinean state. Fishing is also the second leading occupation after agriculture, although political instability over the last years increased illegal fishing (by international companies and neighbors) as well as corruption, and as consequence of this diminishing fishing activity of the population.

The coastal ecosystem of Guinea-Bissau is covered by Mangrove which is the origin of an alimentary chain that reveals all the vulnerability of this system. Rich zooplankton that emerges there feeds young shrimps, crabs, oysters and some fish, before they become adults and move into high sea. In turn fish predators, birds and men feed from these species.

Besides mangrove, the coastal ecosystem offers, at low tide, a favorable soil where shrimps, clams - *combé*, *lingron* - and crustaceans - *cacri*, crabs -, oysters and some fish species are collected. It is here that most artisan fish capture is done. Most of the *moranças* (households) of the coastal area practice subsistence fishing, only the surpluses are marketed.

Women fish mainly in mangrove areas, in river estuaries and its tributaries and lakes, in *bolanhas* and in slimy lands between two low tides. This particularity, the presence of women in all phases of fishery, so far has not been properly analyzed; studies are more concentrated in the activities of processing and marketing fish as women activity (Fernandes 2012). The non-recognition of fish capture by women means to undervalue a feminine know-how essential to the sustainability of this ecosystem from which adult species depend in the high sea, where men exert adult fish capture.

Bideiras also work in fish and molluscs transport and dominate the fish selling and commerce. Fishing is an important source of economic activity for women who work conserving, cleaning and selling fish at the markets, particularly in Bissau.

Severe gender gaps in health and education
Poor health service conditions in rural areas of Guinea-Bissau impact on women disproportionally. Their severity is reflected in the following statistical data: one woman dies in 126 births (UNICEF, 2017) which

indicates that the maternal mortality rate in Guinea-Bissau is one of the highest in the world. Many children do not celebrate their fifth birthday: 129 out of 1000 children die prior to five years of age, many during childbirth, making Guinea Bissau one of the worst countries in the world to be born (UNICEF, 2017).

Improving basic health services for mothers, pregnant women and small children in rural areas is a task that is directly linked to sufficient and continuous energy provision. To reduce maternal and under 5 mortality are priority concerns for ensuring women's sexual and reproductive health rights (SDG goal 5.6), together with approaches to effectively combat the feminization of HIV / AIDS observed in Guinea-Bissau, with a rate of 4.5% in pregnant women compared to 1.5% in men of the same age range (National Institute of Public Health, 2014).

Among the extreme poor, educational levels are proportionally the lowest with a significant gender gap (WB 2016). Current statistical data on formal education show that girls on average attend school only 1.4 years, less than half the time of Guinean men, who average 3.4 years of schooling (MICS, 2014). The general high illiteracy rate of women, with extreme values of 90% in rural areas, particularly in Oio and Tombali regions, demonstrates the magnitude of the challenge to create a skilled labor force in the country (DENARP II, p.22). The primary (19.3%) and lower secondary (8.9%) school completion rates for women are abysmal and worse than their male counterparts, with only 38.6% and 5.9% of girls even enrolling in primary and secondary schools respectively (2014 est.; World Bank, 2016).

Summary of main constraints and barriers against women equity and empowerment

Inclusive participation of women in Guinea-Bissau remains an ongoing challenge. Gender inequality persists, as women are under-educated and generally have a lower socio-economic status, with inadequate access to basic social services and reduced economic opportunities compared to men. Gender inequality persists throughout the nation especially in rural areas where customary and religious practices tend to supersede state policies and laws.

As we have seen, there are several main constraints and barriers that contribute to the general status of gender inequity and hinder the empowerment of women in Guinea Bissau. All of them play a role in the disadvantaged position of women in socio-economic and political terms as well as in relation to women`s energy needs, consumption and provision of energy for women:

- Lack of sufficient sex disaggregated data in all relevant sectors for this project, beyond lack of gender differentiated data on energy need, priorities, access etc.
- Lack of institutional capacity and legal and policy gaps for gender mainstreaming;
- Underrepresentation of women in decisive positions, government where men are holding almost 70% of positions in key ministries;
- Traditional gender-specific roles, where a man decides for the family;
- Lack of education, high illiteracy rates among rural women;
- Women land ownership rights are very much limited, the inheritance laws are discriminating, women face much more difficulties than men to access loans or other form of financial support;
- Women are main food producers for the households, they are responsible for housework, taking care of children, elderly and sick.
- Women are not decision-makers in terms of access to and management of household goods, even though their contribution to family income is generally growing.

- The efficiency of women work is very low in light of lack of basic sanitary infrastructure and electricity;
- Lack of basic health care leads to high rates of maternal and under 5-year mortality;

- Women are underrepresented in higher levels of value chains – they occupy the low end of value chains, namely positions such us harvesters. The men traditionally work in farming for export, bringing higher income. Traders and processors of export products are male. Also, more labor-intensive farming (like rice) lays on women's shoulders.

A Gendered Understanding of Energy Needs &Priorities and Access to Energy in Guinea-Bissau´s forest dependent communities

NOTE: *Data on the electrification benefits for women in the ECOWAS region are still poor. Compelling scientific studies exist on the gender impacts of electrification only with regards to: 1) women's market work and/or leisure time, 2) decreased time spent preparing meals, and 3) lowered fertility. There is a smaller body of research that supports some effect of electrification on school enrolment. This in no way indicates that many other positive impacts do not exist but more research is needed to credibly establish their causal chains.* (cf. ECREEE, 2015).

Women in Guinea-Bissau, particularly in rural / forest areas addressed by this project, belong to the poorest segments of society. Poverty in Guinea-Bissau is, as we have seen, feminized, and shortage or complete lack of modern forms of energy is intimately tied up with poverty. Rural women and girls are primarily responsible for the bulk of household work, as is explained by the women we had the opportunity to speak to during field visit at the project sites: Pitche, Tchetche, Tchandanca (DutaDjara neighbour village), Lamane and Cacine (see field visit notes). These women overwhelmingly support the introduction of improved cook stoves and, more generally, access to modern, clean and renewable energy. Women in Lamane, Tchetche and Cacine, for example, started to imagine how this energy would make a significant difference to their quality of life, foremost possibilities to improve and increase economic activities and their health. They talked about preservation of fresh fish by cooling systems instead of exclusively smoking fish and selling for much better prices at the local markets; the possibilities of continuous irrigation of horticulture fields with solar water pumping and the immense need they have for agro-processing of different crops, peanuts, palm oil etc. which all would be possible with the multifunctional platforms, run by Solar or hybrid mini grids (see field notes).

In-door pollution from the use of traditional three-stone-fire places, normally used in the *tabancas* (villages, is at the origin of serious health problems for women and girls. Improved cook stoves have been distributed to several of selected villages in former projects (EU and NGOs) but not in sufficient numbers. Women informed that they cook with them and confirmed their comfort with these improved cook stoves in terms of energy efficiency and less smoke. But they simultaneously have to use the three-stone fire places due to insufficient supply with improved cook stoves. The Responsible of the Women Association in Tchetche reported that women of her association have gained some experience with the production of improved cook stoves with local materials and are very interested in training for production and commercialization of these improved cook stoves.

Clean cooking

With biomass accounting for as much as 95 percent of the total energy consumed for cooking in Guinea-Bissau, women could be liberated from 2-3 hours daily collection of wood, and **50% or more** of toxic smoke inhalation per day. For the important health gains related to clean cooking see statistics from ECOWAS countries that are collected in ECREEE 2015, p. 29-30.
Poor women tend to participate in the informal economy, more particularly in the food transforming and production sector which also relies strongly on biomass as its main energy source. Provision of clean renewable energy would therefore mean additional gains in terms of time and health for the impoverished women as well as a significant reduction in the use of forest resources. The number of female-headed households is increasing everywhere in Guinea-Bissau, and these tend to be among the poorest and most vulnerable households in rural communities. Although there are no actual statistical data available at national level on the number of men migrating from the rural areas to the urban centers or abroad, we know through other recent studies (IOM/INEP 2018) that the number is huge. Women stay behind with all the responsibilities for their households, according to some women of the villages selected as intervention sites in the Cufada Natural Park.

The major gender dimension of cooking stem from who is doing it, how and in what context. The specific kitchen practices and preferences for stove technology are important to understand and incorporate into clean cooking interventions. In the meetings we conveyed with women at the project sites, we noted a quite high acceptance for improved cook stoves. Those women are well aware about the difficulties to collect enough biomass cooking fuels and the health consequences. According to the technical possibilities *in loci*, energy provided by sustainable mini-grids and low-carbon bioenergy technologies in the frame of this project will likely be able to reduce significantly the amount of biomass cooking fuels or even replace it completely by solar cookers or electric induction plates at later stage.

Solar-pumping systems for access to clean water

Water is the first resource gap in all visited project sites. An intensive demand for clean and drinkable water has been articulated by all interviewed women. Men in the villages also show great concern over lack of water quality and /or broken/ failing pumping-systems, most of them manual. There are very few solar pumps, but with need to be scaled up, for example in TcheTche (see Community Field Report).

At all project sites women and men alike are deeply concerned about clean water. In each village, we have been shown the existing wells or waterholes and the population complained about the many problems related to water: mainly too small quantities of water because there are mechanical or no pumping systems in place and even worse, the inadequate water quality. In some villages, like Tchetche, Bakar Konte and Tchandanca, women collect drinking water from springs far away in the forest or from nearby rivers and spend a lot of their day with this arduous work (see images in the Community Field Report).

Access to clean, secure, and sufficient quantities of water is vital to achieving adequate standards of food and goods production, sanitation, and health, and all these have direct links to energy. As the population in Guinea-Bissau as well as in the region continues to rise and the development pressures increase, access to water will have to be handled in tandem with access to energy (ECREE 2015).

In terms of standard household activities electrical pumps for domestic water supply are very important to reduce women's manual labor in hand pumping or hauling. Easier access to clean water, for households, health posts and school kitchens as well as to water for horticultural gardening/irrigation of fields are mentioned as top priority by all communities.

The inhabitants of the Northern Cufada Lake Region, DutaDjara and neighbouring villages, undertake the hardship to cross the nearby river Corubal in risky passages with primitive boats (we visited the jetty for the small boats) to reach the village of Xitole at the other site of the river for medical treatment and any other issue related to their needs. The health posts and schools are not functioning in their villages without electricity and water supply (as observed in loci and informed in the meeting with village women, see field notes).

Water collection is a gendered activity and the installation of reservoirs, rain water harvesting systems, and foremost pumps, including solar PV-driven ones, can reduce the human female energy and time spent on water collection and transport. Creating access to fresh and clean water by solar pumping systems within this project will help significantly to build resilience into the Guinea-Bissau forest dependent communities, particularly at the remote sites within the Protected Areas which are very much threatened by climate variability, climate change and the degradation of fresh water sources.

Solar/hybrid agro-processing / multifunctional platforms

Agro-processing is another very time and energy intensive gendered women daily activity. Producing palm-oil manually, as it is common practice in Guinea-Bissau´s southern parts, for example in the Cantanhez Forrest region, at the project sites Iembérem and Cacine, needs extreme heat and is very labor intensive. The pounding of grains can take hours every day, including the time-consuming threshing and milling; these are activities of daily routine in nearly all Guinea-Bissau´s villages.

At household subsistence level as well as at market orientated production level, electric grinding mills or multifunctional platforms can replace hand pounding of grains and of oil seeds, both backbreaking activities that are carried out by many women daily and that are very time consuming. Estimation go to up to 6 hours per day for these activities (Embaló, 2009).

The communities of Pitche, Tchetche, DutaDjara and Iembérem possess some agro-processing machines run on conventional diesel; Pitché even has a small multifunctional platform (former UNDP project) which currently is out of work due to technical defect. There is huge demand for agro-processing machines and the enlargement of existing systems as per the reports from the project sites where women and men alike manifested much interest in community managed PV-Solar run agro-processing machines. The villagers are well informed about the advantages of these machines and wish very much to benefit from them. Most of them lack of capital for such huge private investment (according to women association in Pitché and private owner of agro-processing machine in Tchetche). To be equipped with solar energy for running agro-processing machines in the frame of community projects is a highly welcome perspective for all villagers who talked to us. These machines will be more effective, clean and have less running and production costs than the current diesel solutions. That could stimulate increasing demand, also from surrounding smaller villages, as the Head of the Village association in Tchetche explains. The Heads of the Women and Inhabitants` Associations in the villages of Pitche, Tchetche and DutuDjara which already have gained some experience with agro-processing machines unanimously highlighted the importance of agro-processing for economic progress, new income generation and sustainable development of their communities.

The following time-use estimates are obtained from a Bissau-Guinean agricultural expert (Embaló, 2009):

- 2-3 hours each day for just preparing grains (rice, millet, sorghum, fonio) for pounding, i.e. the threshing and milling;

- 82 women-hours for processing one drum of oil palm fruits;
- Furthermore, it can take up to 13 hours to pound enough maize for a family to eat over 4-5 days according to data of ECREEE (ECREEE, 2015, p.39).

The time and energy taken by women to collect water and fuelwood and to process crops, grains, oil etc. manually is negatively related to income earning activities.

To the extent that any of the renewable energy technologies frees women from unnecessary labor, it presents an opportunity for development. To the extent that any of these technologies protects the natural resources, it helps to reduce pressure on the forest environment and is beneficial for climate protection.

According to case studies in the ECOWAS sub-region, there is nothing quite so large scale and high profile as that of the Multi-Function Platform (MFP) program. Among the most often quoted benefits of the MFP program, a combined technical-institutional intervention that provides mechanical processing capabilities to women's groups, is the time-saving aspects and reduction in drudgery. In Mali, some female project participants reported saving between two and six hours of work per day, and that extra time was shifted to income generation (ECREEE, 2015, p. 45). Poverty alleviation, gender equity, and environmental sustainability are the three key gains related to MFP which on the other side might be problematic in as much as the machines are too sophisticated and donors insist on female ownership and control (ibd. 46).

Conservation of food and cold-chain via modern renewable energy solutions

Beside the agricultural sector, modern energy services can also play a transforming role in the fishing sector, the second subsistence and income activity of rural population in Guinea-Bissau. Fish conservation and fish selling at the local and national markets are an exclusive women business among all coastal communities that we visited, namely in Cacine, Cacheu, Lamane, Dutu Djara and Iembérem. The women contacted in these project sites give cooling facilities and ice production high privilege and express the energy need for cooling / storage as a major demand and concern. Traditional fish smoking is not only harmful (equivalent to cooking) but sufficient wood is often not available with the result that fish rots and has to be thrown away, reports the President of the Women Association in Cacine. Fresh fish has a higher value chain; fishing capacities could increase if storage and conservation is guaranteed. There is huge demand for more fresh fish at the regional and national markets according to the fish business women at project sites. Fish selling could be transformed in a much more lucrative and less arduous business for village women, say the women in Lamane who are active members of the local Fisher Association. Beside the women also the fishermen in Lamane and Cacheu show great interest in the cold chain and consequent value increase of fish.

Cooling devices are multifunctional: beside fish seasonal fruits, meat, cold water/ice etc. can be stored safely to be sold in larger (more economic) quantities at the local or town markets. The huge potential of cooling equipment for diversification of the villagers` alimentation and increase of market opportunities have been evidenced by women in all project sites. Similar to agro-processing, the lack of affordable and reliable energy in the forest dependent communities and the high investment costs have so far hindered individual purchasing of cooling devices. Some women in Cacine and Cacheu have their individual refrigerators but are not able to use them economically due to lack of or irregularity of electricity supply.

The introduction and use of the "Chorkor Smoker" in fishing communities has the potential to reduce costs, pollution and fuelwood consumption (ECREEE, 2015, p. 40). This technology has been adopted by some countries in the region; the project site Lamane in the Cufada Lake Region has some installation of this kind. The women of the local Fisher Association do enjoy the significant progress in comparison to traditional smoking facilities and will be very happy to further improve the smoker in terms of output and carbon emission reduction.[28]

The new technology has an oven and mechanical drier that is well suited for small-scale fish processors. It is a simple technology that can be used to upgrade traditional kilns and leads to significant reductions in carcinogenic emissions and uses much less wood fuel. Many women are fish processors, so the new fish smoking technology will have immediate impacts on their health and incomes. The working environment is much cleaner and there is less exposure to the fumes. The shorter processing times with the systems reduces the fuel usage, directly contributing to their profits, as well as frees up time for their families and household work (ECREEE, 2015, p. 47).

Household electrification

There are non-traditional activities that women may take up when electricity is present in the household which may change their lives and their social position significantly towards gender equality.

Because of the gendered roles that exist at home, with men and women often responsible for different tasks at different times of the day, it is reasonable to assume that they have varied preferences for types of electrical devices in use, their features, and also potentially, their placement within the house.

Secondly, it is also assumed that household decision-making practices are not monolithic, but a reflection of combined individual preferences as well as the relative bargaining strength of each of those individuals.
Data on the advantages of household electrification in Guinea-Bissau and gendered preferences for what and how to use this energy at home are very scare, if not none available for Guinean towns and villages. Even concerning the capital, we do not dispose of any meaningful statistical data how exactly household electricity is used.

What is sure is that household electricity can help women a lot in accomplishing household duties, such as cooking, cooling, ironing etc. with less time, effort and hazard. *Gender differentials are noted within households with women using electricity for storage of food, freezing of fish and ice, used to sell cold water while men invest in the purchase of television sets for entertainment or occupy all space in the refrigerator for their beverages.* Men also use electricity for powering machines for car repair etc. at the household level. This are general results from the region, probably also applicable in the Guinean context (ECREEE, 2015, p. 42ff.)

Electric light and evening courses enable women to participate in literacy or other educational or social programs, be this in the local school or at the own home, as was emphasized by the Women Association in Tchetche, Lamane and Iembérem. Through reliable and affordable provision of electricity for

[28] Some development projects have focused on providing solar lighting solutions for fishing boats and improved smokers, making sure to address the distinct needs of both the male and female workers and maintain a sense of fairness in the dynamic between the two economically dependent groups (ECREEE, p. 40).

households, schools, community centers and the like at the project sites women can have easier access to TV and Internet (smartphones) with enormous benefits for them:

They can became much more aware of what is going on in the world, thus giving them confidence to speak on such matters as well as opening their horizon to new ideas in general, including health, nutrition and business issues, i.e. a process of changing the world view is initiated. Media and television, in particular, is shown to have a transformative effect on gender relations, especially in rural areas, according to research in the ECOWAS region (ECREEE, 2015, p. 32-34). As economic and social development as well as the educational level in Guinea-Bissau are belated in comparison to other countries of the region the exact effects of media on the women at the project sites are not very predictable at present. We need practical experience being made before the women in remote regions of Guinea-Bissau can better imagine and understand about the many benefits of TV and Internet.

Electricity can also bring women potential opportunities to start income earning activities that weren´t even considered yet, without leaving their house or village, i.e. with the possibility of taking care for their children and household simultaneously, for example, as *solar bakeries*, *cold drink sellers* or through *small enterprises in food processing*. This will need some capacity training for village women who never before in their life have had access to clean and reliable electricity. Therefore, the women at project sites think first in selling raw products at local or nearby town markets; food processing and cold juice/water selling come as next step as that take more logistic effort and sufficient demand according to the women interviewed at Cacine, Lamane, Tchetche, and the villages of the Northern Cufada Lake region.

There is only limited experience with off grid rural electrification in Guinea-Bissau. Outstanding is the project of the NGO ADPP "Renewable energy for local development in Bissora, Guinea-Bissau with community agro-processing centers, solar water-pumping and community clubs for community outreach activities beside direct intervention in agriculture training. Also the Bambadinca Sta Claro project, implemented by the NGO TESE, has impacted positively on supply with household electricity and the energy needs for women in the broader trading and service sector in the small town of Bambadinca (shops, beauty center, small homewares, and textiles). The conception of this project is very similar to the Bambadinca approach regarding household electrification.

Further, in the value chain of renewable energy supply, there are economic opportunities opening up for small scale enterprises in the electricity production itself as well as in allied services which could be promoted as new women profession. In the frame of the ADPP project, equipment and maintenance services are placed in the hand of local providers, as far as possible. These have been trained for that effect by the project.

It is in this line that the current project aims at motiving also female participation in the energy supply and services sector. Positive experience with new women profession in the electricity business have been made in other African countries (ECREEE, 2015, p.40) which indicates that replication in Guinea-Bissau should be possible, even if it seems quite an ambitious objective seen from the present baseline in women entrepreneurship and technical knowledge. Women certainly could trade energy products like cook stoves, solar kits, fuels, etc. and engage in services like phone charging, cold purified water and the like.

Health and energy services for women & children

One of GEF´s objective is to electrify 50% of health centers in rural Africa until 2030. Health Energy services are essential for providing safe, affordable and effective health services. Energy is part of the solution to improve maternal and children`s health and prevent disease among young children. Energy's role in lighting, sterilization, powering equipment, and cooling is vital in this regard. In addition, the

devastation of childhood diseases can be reduced through interventions such as vaccinations, which require cold chain storage.[29]

Modern medical facilities with access to electricity for lighting, mobile phone charging and refrigeration will also be able to attract and retain doctors in rural areas.

None of the visited communities has a functioning health center due to lack of energy and repair issues as well as lack of human resources and basic emergency equipment. The ill and pregnant, women, children and men alike, at the project sites are all complaining the health situation. They have to overcome distances up to 20km (case of Dutu Djara, Lamane and surrounding villages), or even 30km (Tchetche), or to cross the river Corubal by unsecure small boats to reach very basic health services at bigger villages (Xitole) or small towns (Buba, Cacine, Fulakunda, Boé). The small hospital of Cacine that has to serve 12.250 inhabitants of the region has a few solar panels, far from being enough to satisfy all energy needs of that health center. The hospital of Pitche, also serving about 10.000 population of that region, is without any electricity supply and beds for delivery.

The health situation at the project sites is really dramatic, and access to energy is catalytic to expand and maintain health care facilities in rural areas in Guinea-Bissau. It is hoped very much that the project interventions can close this gap in favor for the poorest and most vulnerable women and children.

Education and Energy

There is a clear linkage between energy poverty and education. 90 percent of children in sub-Saharan Africa go to primary schools that lack electricity or any form of energy services, hampering their chances of receiving adequate and quality education. This concern is raised for girls and boys equally. It is much easier to recruit and retain teachers in the rural areas if the schools are well-equipped and well-maintained and have access to electricity, clean water, and sanitation facilities (ECREEE, 2015, p. 25-26)

In the visited communities, some schools have limited energy access. This is the case of the primary school of Tchetche where UNICEF has implemented WASH and installed some solar electricity; WFP built up the “Cantina Escolar” and improved cook stoves. The primary and secondary schools in Iemberem run on diesel generators; the public school in Cacine has no electricity supply. There is one private school run by Brazilian church with solar energy access.

All in all, the panels are too small and insufficient to guarantee lightning at night and electricity supply for the attached teacher´s room/apartment. The current electricity installations are too expensive and not reliable; at the other project sites the schools are without any electricity connection.

Public investments in the education sector will be most effective if energy access is considered at the same time. Schools benefit from reliable electricity and basic heating/cooling, water and sanitation systems; all these measures add to the physical health and educative development of the students. Furthermore, well equipped schools may supply drinking water, hot lunches, communications facilities and civic meeting spaces.

[29] With improved nutrition, enhanced health care during pregnancy and delivery, and access to currently available technologies and medicines, 80% of maternal and new-born deaths would be preventable. Additionally, the prevention or prompt treatment of pneumonia and diarrheal diseases are vital to children’s health (ECREEE, 2015, p.24-25).

It is envisaged that these schools in the curse of the project will serve as normal schools and simultaneously as training centers which will offer local electricity courses & solar system maintenance and repair for women and men, functional alphabetization of women, and teaching in marketing strategies. The schools will also function as project coordination point and administrative center of community management of energy provision.

Women entrepreneurship in the renewable energy sector

Access to sustainable energy can provide women with green employment opportunities by freeing up their time, enhancing their productivity and unleashing new economic potential (UNIDO/UN-Women, 2016).

Provision of lighting, for example, extends the working day and thereby the time available for engaging in income-generating activities for women who often face substantial time constraints due to domestic work obligations. Lighting can extend business hours for traders of textiles, clothing, beauty supply, vegetables, ambulant street food producers and sellers, small homewares, and more, who disproportionately tend to be women. There is also the issue of women trading energy products, such as cook stoves, solar kits, fuels, and doing business in energy related services, such as phone charging, cold purified water, computer repair etc.

There is only limited experience with off grid rural electrification in Guinea-Bissau. Outstanding is the project of the NGO ADPP “Renewable energy for local development in Bissora, Guinea-Bissau with community agro-processing centers, solar water-pumping and community clubs for community outreach activities beside direct intervention in agriculture training. Also the Bambadinca Sta Claro project, implemented by the NGO TESE, has impacted positively on supply with household electricity and the energy needs for women in the broader trading and service sector in the small town of Bambadinca. The conception of this project is very similar to the Bambadinca approach regarding household electrification.

Further, in the value chain of renewable energy supply, there are economic opportunities opening up for small scale enterprises in the electricity production itself as well as in allied services which could be promoted as new women profession. In the frame of the ADPP project, equipment and maintenance services have been successfully placed in the hand of local providers, as far as possible. These have been trained for that effect by the project. The electricity and solar installation teachers from “Escola de Arte de Quelele”, the Vocational Training School of the NGO AD (Action for Development) in Bissau have trained technicians for the Bambadinca Sta Claro project. It is a real challenge to motivate women to participate in such trainings and decide for the electricity business, says the school director, Jorgito Handem, but nevertheless it should be possible to overcome gender barriers. AD has succeeded in orienting the teaching at the level of education of its students with the methodology of learning better and more by doing and by errors. The school is optimistic to implement these modules successfully with female students as well (meeting with AD, field notes).

That technical capacity building in the RE energy sector with a gender-based approach is possible, is demonstrated by the experience of the NGO Tostan. This NGO partnered with the Barefoot Project Initiative, training women in setting up and maintaining small PV systems. In this context, two groups went to India to do a 6-month specialization, and then replicate with neighboring communities. TOSTAN was able to train women for the installation of small solar systems and the creation of management committees for the assembly, maintenance and operation of the systems. The equipment was financed by the University of India and the beneficiaries themselves decided the tariffs to be paid by the systems

(for domestic lighting). The members of the management committees themselves started to set up and maintain the systems. Small facilities in schools and community health centers were also built. The project had most of its activities in the regions of Oio, Bafata and Gabu (see Baseline Study, p. 114).

It is in this line that the current project aims at motiving massive female interest and participation in the energy supply and services sector. Positive experience with new women profession in the electricity business have been made in other African countries (ECREEE, 2015, p.40) which indicates that replication in Guinea-Bissau should be possible, even if it seems quite an ambitious objective seen from the present baseline in women entrepreneurship and technical knowledge in Guinea-Bissau.

Barriers, such as lack of access to credit and training, limit the possibilities for women to develop and use energy-based technologies: it has to be invested quite a lot into the poorly educated and often in traditional gender role fixed Bissau-Guinean women to enable them to benefit from the opportunities that energy provision and the green business sector could provide for them. The project intends to foster the potential of women in the energy sector, as well as the value of their work. It will aim at engaging private companies together with AMAE (nationwide acting NGO "Association of Women with Economic Activities") or other NGOs to promote women entrepreneurship in energy business. Simultaneously, the project intends promoting the design of gender- equitable energy policies and legal frameworks that support green employment creation.

In 2018, ECREEE partnered with AfDB to launch a regional workshop to increase women participation in the renewable energy sector – only 2% of energy sector entrepreneurs in West Africa today are women. The joint initiative ultimately seeks to develop a pipeline of investment-ready, women-owned energy businesses across the region.[30] Business owned and operated by women often includes farming, agro-processing, restaurant operation, petty trading and craft production; these are all mostly activities in the low value-added sectors. Value additions could be created by the energy services provided. It seems that socio-cultural and economic variables push women into particular sectors, especially in the informal sector. According to experiences from the ECOWAS region, this also appears to be the case in the energy sector where certain subsectors which are less capital intensive at start-up, less advanced technology complex and more orientated to the final customer are privileged by women entrepreneurs (ECREEE, 2015, p. 71-73).

The Solar Sisters project in Sub-Saharan Africa promotes women's entrepreneurship by providing women with a start-up kit to equip them to operate, maintain and sell solar technology, such as solar lamps, in their communities. Evaluations reveal multiple benefits of the initiative: The solar business allows the entrepreneurs to double their household income; the income generated by the entrepreneurs is reinvested 90% back into their families, thus providing benefits for the next generation; and the women who buy the solar lamps can reduce household expenses by 30% when the solar energy replaces expensive kerosene (www.solarsister.org; ECREEE, 2015, p. 75).

Overcome barriers for women to access and use energy

The specific design of a rural energy project can greatly affect the degree to which women and men benefit from it. Therefore this project aims at a gender responsive design which does not leave women behind.

[30] "FEASIBILITY STUDY PROMOTES WOMEN'S PARTICIPATION IN ENERGY TRANSITION," ESI AFRICA, (MAY 7, 2018): HTTPS://WWW.ESI-AFRICA.COM/FEASIBILITY-STUDY-PROMOTES-WOMENS-PARTICIPATION-IN-ENERGY-TRANSITION/

Women access mini grid electricity in lower numbers than men because women are overrepresented in poor households and marginalized communities and may not be able to afford connection fees. Social tariffs or some affordable flat rate for small consumers should be applied. This was one key demand at the recent ECREEE workshop in Guinea-Bissau "Situation of gender mainstreaming in access to energy in Guinea-Bissau "(ECREEE 2017).[31] The fact that electricity is common to households covers up important gender dynamics. Whether electrification programs succeed at incorporating the gender dimension depends partly on the extent to which they understand and account for different energy needs of women and men (ECREEE, 2015, p. 33).

Gender disaggregated data regarding connections to the electrical grid in the ECOWAS region are rare; for Guinea-Bissau no such data are known. When the business strategy primarily focuses on increasing connections from the utility´s perspective "a customer is a customer" with the intention of reducing losses and recovering costs and no gender disaggregated data is collected, gender can easily be overlooked.

Reducing the barriers for women to access and use electricity for the identified purposes, within the household or outside, will depend on two key factors:

- the provision of the essential complementary infrastructure and a supply of potential appliances;
- the ability of people to pay both for the supply and the end use equipment which rests on availability of credit, as well as of training and extension.

Lack of female participation in the energy ministry and other lead ministries (environment, agriculture) is another very serious barrier. As soon as women share ideas and project design, even if they don´t have the complete technical qualification, the agenda setting and project selection will change to more accurately reflecting the whole population.

The creation of Gender Focal Points and units within the public sector and significant increase of women at the political and economic leadership level are a condition *sine qua non* for successfully gender mainstreaming into the energy sector and renewable alternatives.

Gendered Risks in access to energy

Social tariff: There is the risk that energy supply and electrification increase social inequalities, particularly for women. This can occur through giving access to a broad range of opportunities to those who can afford to connect while leaving the others still further behind.

Women´s needs, priorities, constraints and potential have to be carefully analyzed to mitigate access problems (by gender disaggregated data in the selected communities). It has to be considered also that women are not always one homogeneous group. Even in rural settings, they may be quite differentiated by income and social group (via consultations with women and men in separate groups).

[31] Access and affordability of RE: The Ghana Electrification project shows that In terms of access, women especially those in female headed households, face difficulties in accessing electricity connections due to their lack of adequate cash incomes. Even though there was a government subsidy to facilitate the rate of connections by poorer communities and households, many such households could still not benefit from the available electricity resource. Ghana electrification project, see ECREEE ,2015, p. 34.

Another risk is linked to ***gender stereotypes*** that can lead to the exclusion of women from discussions about energy plans and policies and project activities. Women play a pivotal role in energy production, distribution and utilization, especially in poor communities. Thus, the lack of recognition of their role could undermine the effectiveness and sustainability of much-needed energy projects and policies.

Gender and Energy – Key results

At least 12.000 women will benefit directly from project interventions (= Female Population at project sites and surrounding villages with less than 5 km distance) and approximately 30.000 indirectly.

The "ECOWAS Policy for Gender Mainstreaming in Energy Access" is the leading strategy at government and national level. Various efforts are being made to implement measures under this regional framework.

Gender mainstreaming in Guinea-Bissau's energy policy requires capacity building of staff and the implementation of gender management systems at the institutional level to provide guidance on gender responsive leadership and decision making. As part of this process, the Government has established a Gender Focal Point at the Ministry of Energy to promote inclusive participation for women in the energy sector.

Recruiting, training, financial resources and active involvement of women in all energy decision making processes are the greatest gaps needing immediate attention. The Energy and other lead Ministries as well as local government structures are male dominated. Since women are not significantly represented in those structures nor on the community level, it means that critical decisions at the local level about the electrification programs cannot adequately tap into women's knowledge, needs and interests.

Energy poor individuals suffer in myriad ways, robbed of time, robbed of income, physically exhausted, struggling to access education and information. Women and men require modern energy as a means to accomplish distinct sets of tasks within their spheres of activity, whether domestic, agricultural, community-based, or commercial.

Despite many commonalities, women and men exhibit systematic differences in the ways they experience energy poverty. These differences are derived from socially determined gender roles and norms that in turn drive patterns of energy usage in the region. If these differences are overlooked, the new renewable energy off grid installations in rural areas in Guinea-Bissau can end up propagating existing injustices.

Minigrid/offgrid smaller-scale, renewable energy systems, as stand-alone systems of solar energy or solar-diesel hybrids, can provide rural isolated communities with affordable energy, promote productivity and help in creating employment by empowering small /medium enterprises for the benefit of both the rural and urban poor, men and women alike.

Women, as active agents in biomass management, as the primary household cooks, and as participants in many biomass-intensive processing sectors, will play an integral role in shifting the region onto a cleaner development pathway. Modern RE and sustainable energy and low carbon bioenergy technologies can close a significant part of the gender gap observed relative to women and energy needs and use in Guinea-Bissau, given a participative and inclusive approach.

The project creates a strong link between the forest and basic energy needs: by promoting the use of sustainable charcoal and firewood the pressure on biomass resources will be reduced and GHG emissions decrease. At the same time women will be relieved from the burden of wood collection for cooking and the traditional method of fish smoking (improved and / or replaced through cooling) and related health risks can be significantly reduced.

Access to renewable clean energy can provide women with green employment opportunities by freeing up their time, enhancing their productivity and unleashing new economic potential. Provision of lighting extends the working day and thereby the time available for engaging in income-generating activities for women who often face substantial time constraints due to domestic work obligations. Agro-processing, cooling and storage systems based on RE increase women´s business perspectives, foremost in trade of raw agricultural products and fish and in food transforming.

The majority of the beneficiaries of better access to energy needs in rural areas under this project will be women and children. Providing energy access to these most often poor households adds value to agricultural production and to micro, small and medium enterprises. It generates high positive impacts on women as consumers of electricity and managers of RE community systems. While electrification will benefit both women and men by enhancing their engagement in more productive activities, gender gains are derived mainly from reducing the workload of women and girls related to biomass use.

Reducing the barriers for women to access and use RE electricity, within the household or outside, will depend on two key factors:

- the provision of the essential complementary infrastructure and a supply with potential appliances;
- the ability of people to pay both for the supply and the end use equipment which rests on availability of credit, as well as of training and extension.

Gender norms can impact a women's ability to access clean energy, to work in the energy sector, or to have a voice in the public discourse, but they are neither inevitable nor immune from change. The gender gaps and barriers that hinder women to equal accessing and using RE electricity will be addressed in this project through concrete gender affirmative measures:

- energy supply primarily for female dominated domestic and economic activities ;
- development of *business models that privilege community / women management of new RE energy* infrastructures and improved cooking stoves etc. with local women organization at the forefront (bottom-up approach).

GENDER ACTION PLAN
Gender-responsive ground-level strategies / gender related activities per outcomes

The ranking of gender equality outcomes goes from the first step of counting (the number of men and women who were involved in the program) to the second level of "gender-sensitive" outcomes, which means results that address the different needs of women and men in terms of equal distribution or benefits, resources and entitlements, to the third and highest level of transformative outcomes.

Transversal: ***Promote collection of gender-disaggregated data, with key inputs from women: throughout the entire project, in all outcomes and outputs***

Project Objective: To promote integrated investment in sustainable mini-grids and low-carbon bioenergy technologies, and develop an appropriate business model for the sustainability of the system

Table 7 : Gender Action Plan

Project OUTCOME	GENDER ACTION	INDICA-TORS	IMPOR-TANCE
Component 1 Policy and financial instruments and incentive scheme for solar mini-grids and low-carbon bioenergy technologies	Integration of gender-sensitive and gender-responsive energy needs, priorities and management assessments into national policy, planning and development decisions on renewable energy and minigrid solutions Including as much women as possible in decision-making positions of the energy sector, in the line ministries as well as in sector/regional administration and community decision making; Indicators:	Baseline: 0 Target: 4 Baseline: less 5%; Target: 30 – 40%	MEDIUM
	Ensure financial instruments and incentive schemes are gender sensitive: equal participation and access for female and male beneficiaries of minigrid RE infrastructure to the tools and financing that project will provide; Special social tariff for poor female headed households	Indicators: baseline: N/A Target: 2000 poor women receive special social tariffs	HIGH
	Ensure financial instruments and incentive schemes are gender sensitive: equal participation and access for female and male	Indicators: baseline: N/A Target: 500 women receive	HIGH

	beneficiaries of minigrid RE infrastructure to the tools and financing that project will provide: special credit line for poor women & women entrepreneurs,	special credits	
	Gender-sensitive capacity training designed and modules implemented for the production and utilization of improved kilns and cookstoves; study of charcoal market and marketing strategies; New kitchen & cooking techniques overcoming cultural and social barriers; sensitizing and training of women for production, commercializing, maintenance & use of kilns and cooking stoves at project sites and beyond; women green business start-up support + credits	Baseline N/A; Target: 10 women trained in each of 7 project sites + total of 20 women for kilns; special credits for up to 30 women start ups	HIGH
	Development and demonstration of alternative solutions, including community proposals by women, for high efficient electric cooking devices, solar cooker etc. beyond wood fuel;	Baseline: N/A Target: 10 women at each project site can manage alternative cooking devices and share their experience	MEDIUM
2. Capacity building for mini-grid and low-carbon bioenergy	Mainstream women´s needs and active role as RE + sustainable energy agents into guidebook, consider illiteracy (picture based); women participate at Ministry and Local Level in guidebook design	Baseline: 0 Target: gender mainstreamed 10.000 copies distributed	MEDIUM
	Special training for women to overcome gender roles in the male dominated energy sector	Baseline: 0 Target: 30-40% women participation	HIGH
	Train local authorities and community groups on RE solutions, including the integration of gender-responsive activities (30%-50%	Baseline: 0 Target: 30-50% women	HIGH

	women trainees)	trainees	
	Design and implementation of capacity building program on fuelwood value chain based on tripartite partnerships: ministries-NGOS and women community committees: At least 150 women; extend gender appropriate capacity-building for women on renewable energy supply solutions and economic perspectives through including training on self-esteem, innovative leadership, entrepreneurship	Baseline: 0 Target: 50% or more women trainees	HIGH
	Development of gender-sensitive responsive RE-based mini-grid and low-carbon bioenergy system management; training of gender focal points and women of management, technical and, administration units of key ministries; Community-based organizations in selected locations, especially women groups, local, fisher productive users, receive training, assistance and advice on potential income-generating activities & security aspects related to electricity use.	Baseline: 0 Target: 30-50% women trainees	HIGH
	Special office for the issuance of licenses and construction permits to small promoters of mini gridsconsiders particular women interests;	National clearing-house mechanism gender mainstreamed	MEDIUM
	National Communication Strategy touches particularly on women energy business´; establish communication line with regional/foreign women investors, example: Solar Sisters	Baseline: 0; Target: all campaign products include women energy business	MEDIUM
3. Mini-grids	Gender-sensitive screening and	Communities	HIGH

and low-carbon bioenergy technologies roll-out	gender analysis of selected project sites, consultations with women in developing energy policies	Report, GAP + further roll-out	
	Training of women in energy management; development of gender responsive business model,	Target: up to 50% women present in decision making body of public-private partnership	HIGH
	Ensure that RE installations deliver equally for women and men men´s identified different needs; ensure that women have equal opportunities to extra income from work performed in temporary contracts from construction companies (50% women contracts);	Gender equity access to energy and installation jobs applied	HIGH
	Elevated percentage of positions held by women in community councils and decision-making spaces related to local energy management and installation maintenance; create a discussion forum to facilitate dialogue on gender issues between the village associations, local authorities, regional and national administration	Up to 50% women participant	HIGH
	Mechanisms for women's economic empowerment through breaking out of traditional roles in the home and in economic activities: gender equality in access to resources, to begin micro or small businesses	Provision of technical and financial support should strive for a 50/50 balance between male and female recipients	HIGH
	Gender sensitive production chain of cookstoves built up; study on the charcoal market completed, including gender sensitive commercialization and management strategy; women entrepreneurs included	At least 4 small /medium green female biomass business created	HIGH

	Replication includes gender needs and priorities mainstreamed, women`s strong participation in energy driven business and management; focal gender points and women groups participate in replication plan	Up to 50% women participation in replication planning	HIGH

GEF Core Gender Indicators (GEF Gender Equality Plan, 2015)

The proposed tools and methodologies for mainstreaming gender in sustainable energy supply follow / will be elaborated considering the objectives of GEF Gender Equality Plan for 2015-2018 and the GEF core gender indicators:

✓ ***Project design fully integrates gender concerns***

- Gender analysis conducted during project preparation;
- Gender responsive project results framework incorporated (gender responsive output, outcome, indicator, budget etc.); development and implementation of project specific gender strategy/ plan.

Verification through Project Document at CEO endorsement

✓ ***Project implementation ensures gender equitable participation in and benefit from project activities***

- Share of women and men as direct beneficiaries of project;
- Share of convention related national reports incorporated gender dimensions

Verification through Project Implementation Reports, Mid-Term Evaluation Reports, and Terminal Evaluation Reports.

✓ ***Project monitoring and evaluation give adequate attention to gender mainstreaming***

- Institutional Capacity for Gender Mainstreaming, measures to minimize adverse impacts
- M & E reports that incorporate gender equality / women empowerment issues and assess results/progress.

Verification through Project Implementation Reports, Mid-Term Evaluation Reports, and Terminal Evaluation Reports.

Specific Monitoring and Evaluation Gender Indicators for project sites

Monitoring and Evaluation Framework

This section further describes the monitoring and evaluation framework to track progress on achieving the goals, objectives and outcomes specified in the GAP.

Table 8 : plan de monitoring et d'évaluation

Indicator	Total in %	Site 1	Site 2	Site 3	Site 4	Site 5	Site 6	Site 7
Number/percentage of women and men involved in RE minigrid and low carbon technologies policy								

consultations								
Amount of time spent by women and men to obtain energy supplies: fuelwood, charcoal								
Number/percentage of women and men benefiting from energy-saving technologies (improved cookstoves, improved fish smokers; rainwater collection, solar water-pumps)								
Female ownership or co-ownership of equipment and tools for production, processing, commercialization and other services associated with fuel wood, charcoal, agriculture products, fruits, horticulture products								
Number /Percentage of women / female led households benefiting from night lightning at school and alphabetization and other trainings at school								
Proportion of poor women, youths and vulnerable groups benefiting from improved health services (energy/ repair of health posts)								
Proportion of men and women accessing and using multifunctional platforms								
Number/percentage of women benefiting from grants or microcredits for energy equipment installation / energy driven business								
Number/percentage of women and men trained and apt to install, maintain and repair RE minigrid and annexed electric equipment & installation, including domestic electrification								
Number/percentage of women performing alternative new income sources in the energy driven business sector								

Gender-responsive budget / check at project start

NOTE: *Because women are more vulnerable to climate change, it is particularly important that any gender disparities are incorporated into budgeting. There is evidence that incorporating gender awareness into budgeting for climate action (for example, in climate finance mechanisms) constitutes "smart economics", as women's empowerment is closely linked with gains in environmental sustainability, productivity and climate change mitigation. These reasons are in addition to rights of women to gender equality as rights-holding individuals.*

A timeline sets milestones for the implementation of the GAP. The budget relates to the actions identified and the monitoring and evaluation described above.

- Analyse the situation—what will the actual distribution of resources between women and men look like? If there are unequal distributions, why?
- Assess resource distribution applying a gendered lens: Should resources be distributed differently to address gender inequality issues? Have gender-differentiated needs been taken into consideration? What are the root causes of any identified disparities? Are there any other objectives to be pursued to reduce the unequal distribution of resources?
- Whose interests will be best represented by this budget? How powerful are these groups achieving their interest? Who needs to be included to address any gender disparity?

Annex G: UNDP Risk Log

#	Description	Date Identified	Type	Impact & Probability	Countermeasures / Mngt response	Owner	Submitted, updated by	Last Update	Status
1	Enter a brief description of the risk *(In Atlas, use the Description field. **Note: This field cannot be modified after first data entry)***	When was the risk first identified *(In Atlas, select date. Note: date cannot be modified after initial entry)*	Environmental Financial Operational Organizational Political Regulatory Strategic Other Subcategories for each risk type should be consulted to understand each risk type (see Deliverable Description for more information) *(In Atlas, select from list)*	Describe the potential effect on the project if this risk were to occur Enter probability on a scale from 1 (low) to 5 (high) P = Enter impact on a scale from 1 (low) to 5 (high) I = *(in Atlas, use the Management Response box. Check "critical" if the impact and probability are high)*	What actions have been taken/will be taken to counter this risk *(in Atlas, use the Management Response box. This field can be modified at any time. Create separate boxes as necessary using "+", for instance to record updates at different times)*	Who has been appointed to keep an eye on this risk *(in Atlas, use the Management Response box)*	Who submitted the risk *(In Atlas, automatically recorded)*	When was the status of the risk last checked *(In Atlas, automatically recorded)*	e.g. dead, reducing, increasing, no change *(in Atlas, use the Management Response box)*

2			Environmental Financial Operational Organizational Political Regulatory Strategic Other	Text P = I =					

Annex H: Additional agreements

ANNEX I: MONITORING PLAN

The project manager will collect the results data according to the following monitoring plan:

Table 19: Project Monitoring Plan

Monitoring	Indicators / sub-indicators	Description	Data Source / Collection Methods	Frequency	Responsible for data collection	Verification means	Assumptions and risks
Project Objective To promote investment in both solar mini-grids and low-carbon bioenergy technologies in an integrated way	**Indicator 1:** Emissions reduction (in tCO_2 over a 25 year project equipment life).	Emissions reduction of 93,900 $tCO2_2$ over a 25 year project equipment life.	Audit reports.	PIRs, MTR and TE	UNDP CO	Annual project reports, GHG monitoring and verification reports.	Ongoing commitment of project partners, including government agencies and investors / developers.
	Indicator 2: Investment in mini networks.	Nearly $ 12 million invested in mini-networks and low-carbon technologies.	Audit reports.	End of project report.	UNDP CO	Final evaluation report of the project.	Ongoing commitment of project partners, including government agencies and investors / developers.
	Indicator 3: Installed Capacity (MW) and Annual Power Generation (MWh) by Mini-Grids	1.23 MWp of solar PV installed. 780 MWh of electrical energy produced / year.	Audit reports.	End of project report.	UNDP CO	Final evaluation report of the project.	Ongoing commitment of project partners, including government agencies and investors / developers.
	Indicator 4: Number of jobs created.	200 jobs created.	Audit reports.	End of project report.	UNDP CO	Final evaluation report of the project.	Ongoing commitment of project partners, including government agencies and investors / developers.

Monitoring	Indicators / sub-indicators	Description	Data Source / Collection Methods	Frequency	Responsible for data collection	Verification means	Assumptions and risks
	Indicator 5: Number of beneficiary households and enterprises in rural areas.	More than 1500 households and beneficiary companies have access to electricity services.	Audit reports.	End of project report.	UNDP CO	Final evaluation report of the project.	Ongoing commitment of project partners, including government agencies and investors / developers.
	Indicator 6: Number of households using improved stoves in rural areas.	More than 2,000 households use improved stoves.	Audit reports.	End of project report.	UNDP CO	Final evaluation report of the project.	Ongoing commitment of project partners, including government agencies and investors / developers.
	Indicator 7: Number of producers using improved kilns.	More than 100 producers use the improved kilns.	Audit reports.	End of project report.	UNDP CO	Final evaluation report of the project.	Ongoing commitment of project partners, including government agencies and investors / developers.
Component 1: 1. Policy and financial instruments and incentive scheme for solar mini-grids	**Indicator 1:** Existence of policies and strategies.	Not available at the moment.	Completed and approved by the government within 9 months of project launch.	Already completed and approved by the government.	Project Documentation.	Commitment of government entities.	Existence of policies and strategies.

Monitoring	Indicators / sub-indicators	Description	Data Source / Collection Methods	Frequency	Responsible for data collection	Verification means	Assumptions and risks
and low-carbon bioenergy technologies	**Sub-indicator 1.1:** Existence of a set of policies for the development of ER-based mini-grids.	Policy package for mini-network development available.	Government, policy / strategy documents and plans.	End of activity report.	UNDP CO	Project documentation.	Cooperation and interest of government entities.
	Sub-indicator 1.2: Existence of a financial viability mechanism for ER-based mini-grids.	Financial viability mechanism for the development of ER-based mini-grids established.	Project reports.	End of activity report.	UNDP CO	Reports on village energy projects completed.	Continued interest of private sector investors.
	Sub-indicator 1.3: Existence of financial incentives and market mechanisms to secure supply and stimulate demand for energy-efficient stoves and firewood / charcoal	Criteria for setting tariffs for the developed mini-grids and improved stoves / kilns. Approved rates for electricity supply through mini-grids.	Project reports.	End of activity report	UNDP CO	Project reports. Project reports	Continued interest of private sector investors.
	Sub-indicator 1.4: Existence of gender-sensitive capacities and modules for the production and use of improved stoves and stoves	Incentive systems for women producers and users of improved stoves / kilns established and approved	Project reports.	End of activity report.	UNDP CO	Project documentation	Expected expansion of the program. Cooperation of government entities and staff.
	Sub-indicator 1.5: Existence of alternatives to the use of wood	Existence of texts encouraging the use of wood	Project reports.	End of activity report.	UNDP CO	Project documentation	Expected expansion of the program. Cooperation of government entities

Monitoring	Indicators / sub-indicators	Description	Data Source / Collection Methods	Frequency	Responsible for data collection	Verification means	Assumptions and risks
							and staff.
Component 2: Capacity building for mini-grid and low-carbon bioenergy	**Indicator 2:** Evidence that the capacity of stakeholders has been developed.	Not available at the moment.	Completed within 12 months of project launch.	Already finished.	Project documentation.	Cooperation of all stakeholders.	Evidence that the capacity of stakeholders has been strengthened.
	Sub-indicator 2.1: Availability of a guide on the development of mini-grids.	Not available at the moment.	Completed within 12 months of project launch and Stakeholder Approved Guide by the end of the first year.	Already completed and Guide validated.	Published documents.	Commitment of different government institutions and NGOs.	Availability of a guide on developing PCH to power mini-grids.
	Sub-indicator 2.2: Availability of the program for on-the-job capacity building.	On-the-job capacity building program.	Project reports	Annual Report	UNDP CO	Project reports.	Ongoing commitment of project promoters.
	Sub-indicator 2.3: Existence of effective business and technical advisory services.	Ease of providing business advisory services and established technical.	Project reports.	End of activity report	UNDP CO	Evidence of a fully operational advisory services unit. Project reports.	Ongoing commitment of various government institutions and project promoters.
	Sub-indicator 2.4: Evidence of the capacity building program provided to	Capacity building program provided to relevant national agencies.	Project reports.	Annual Report.	UNDP CO	Project documentation.	Appointment of staff by the ministerial departments / other institutions

Monitoring	Indicators / sub-indicators	Description	Data Source / Collection Methods	Frequency	Responsible for data collection	Verification means	Assumptions and risks
	the appropriate national agencies.						concerned.
	Sub-indicator 2.5: Proof of the on-site capacity building program for 300 artisans, farmers and other stakeholders in the fuelwood value chain.	Capacity building program provided artisans, farmers and others in the fuelwood value chain	Project reports.	Annual Report.	UNDP CO	Project documentation.	Appointment of staff by the ministerial departments / other institutions concerned.
	Sub-indicator 2.6: *Existence of a window dedicated to national exchanges on renewable energies and natural resources / one-stop shop*	Opening of a window dedicated to *national* exchanges *on renewable energies and natural resources / one-stop-shop*	Project reports.	Annual Report.	UNDP CO	Project documentation.	Appointment of staff by the ministerial departments / other institutions concerned.
	Sub-indicator 2.7: Existence of *public relations and the promotion of available and operational investment*.	Increased investments on mini-grids	Project reports.	Annual Report.	UNDP CO	Project documentation.	Appointment of staff by the ministerial departments / other institutions concerned.
Component 3: Mini-grids technologies roll-out	**Indicator 3:** Availability of the business model.	No such model available now.	Completed within 12 months of project start.	Already finished.	Project reports.	Government entities and the private sector willing to cooperate.	Availability of the business model.

Monitoring	Indicators / sub-indicators	Description	Data Source / Collection Methods	Frequency	Responsible for data collection	Verification means	Assumptions and risks
	Sub-indicator 3.1: Existence of full feasibility studies and business plans or pre-feasibility studies for the 8 identified sites.	No feasibility study / business plan available at the moment.	Completed within 12 months of project start.	Already finished.	Project reports.	Continued interest of government and private sector.	Existence of complete feasibility studies and business plans or pre-feasibility studies for the 8 identified sites.
	Sub-indicator 3.2: Existence of fully executed partnership documents.	Fully executed partnership documents available.	Partnership documents.	End of activity report.	UNDP CO	Contracts confirming the establishment of partnerships.	Continued interest of government entities and private investors.
	Sub-indicator 3.3: Evidence that a production capacity of at least 2 MW of mini-grids is operational.	At least 2 MW of installed capacity of mini operational networks.	Completion reports	Final evaluation report.	UNDP CO	Report indicating that a total of at least 2 MW of hydroelectric capacity has been constructed and is operational.	Continued interest of government entities and private investors.
	Sub-indicator 3.4: Evidence of the selected sustainable model.	The sustainable model is implemented.	Annual Report.	Annual Report.	UNDP CO	Project documentation	Continued interest of government entities and private investors.
	Sub-indicator 3.5: Proof of productive uses of electricity.	Increased consumer purchasing power	Annual Report.	Annual Report.	UNDP CO	Project reports	Interest and willingness of electricity consumers to engage in income generating activities.
	Sub-indicator 3.6:	Availability of	End of activity	End of activity	UNDP CO	Project	Interest of local (and

Monitoring	Indicators / sub-indicators	Description	Data Source / Collection Methods	Frequency	Responsible for data collection	Verification means	Assumptions and risks
	Existence *of Deployment Plan for Kilns and Improved Stoves*.	deployment plans for improved kilns and stoves	report	report		documentation and website	international) stakeholders.
	Sub-indicator 3.7: Existence of products and dissemination tools.	Dissemination products and tools available.	End of activity report	End of activity report	UNDP CO	Project documentation and website	Interest of local (and international) stakeholders.

ANNEX J: MONITORING PLAN EVALUATION PLAN

Title of the evaluation	Expected start date Month year	Expected end date Month year	Included in the evaluation plan of the Country Office	Budget for consultants ($)	Other budget (travel, site visits, etc.) - $)	Budget for translation
Mid-term review	June 2026		Yes			
Terminal Evaluation	June 2028		Yes			
Budget Total Evaluation						

ANNEX K: GHG CALCULATIONS

From Mini-grids

Direct emissions from electricity generation: CO_2 emission reduction attributed to cumulative 1.2 MW solar PV.

For solar PV:
Assumptions: (1) PV system capacity factor = 20%; (2) Useful life of solar systems = 25 years; (3) Average emission factor from diesel generators = 0.786 tCO_2/MWh

Calculations:
Annual power generation from PV systems = 1.2 x 0.2 x 8760 = 2,102 MWh
Annual CO_2 emission reduction = 0.786 x 2,102 = 1,652 tonnes/year
Lifetime CO_2 emission reduction = 1,652 x 25 = 41,3000 tonnes

Total direct emission reductions from mini-grids are 41,300 tonnes CO_2 eq.

Direct emissions from low-carbon bioenergy technologies: CO_2 emission reduction attributed to the deployment of 50 energy-efficient charcoal kilns (Casamance model) and the dissemination of 5,000 improved cook stoves.

IMPROVED COOK STOVES							
Parameter	**Year**						
	1	**2**	**3**	**4**	**5**	**6**	**7**
Number of improved cook stoves disseminated (units)	100	400	1,000	1,500	2,000		
Total number of improved cook stoves in use (units)	100	500	1,500	3,000	5,000	4,600	3,600
Total reduction of CO_2 equivalent per improved cook stove (in tonnes)	0	750	2,250	4,500	7,500	6,900	5,400

Lifetime of the improved cook stove (ICS), years	**3**
CO_2 emission reduction per ICS, tonnes/year[32]	**1.5**
Total CO_2 emission reduction from the ICSs, tonnes	**27,300**

SUSTAINABLE CHARCOAL PRODUCTION

Type of Charcoal Kilns	Number of Units Installed	Annual Production, tonnes	Useful Life, Years	Total Lifetime Production, tonnes	No. of Replacements
Casamance Kiln	50	9.6	5	7,200	2

NOTES:	**Charcoal to Wood Ratio**	**Useful Life, years**
Traditional Kilns	0.208	3
Casamance Kiln	0.25	5

	CO_2 Emission Reduction, tonnes			**Charcoal Production and Operational Losses, tonnes**	**Net CO_2 Emission Reductions, tonnes**
	Pyrolytic CH_4 avoidance	**Use of non-renewable biomass**	**Total**		
Casamance Kiln	0	26,640	26,640	1,332	25,308

- Traditional Kilns have a minimum charcoal yield of 250 kg from 1,200 kg wood.
- Average annual production from Retort kilns is 24 tonnes of charcoal; for Casamance, it is 9.6 tonnes of charcoal

[32] Stockholm Environment Institute, Working Paper (2013), *Assessing the Climate Impacts of Cookstove Projects: Issues in Emissions Accounting*.

- Based on data from a leading study[33], with a conservative estimate of the percentage carbon content in wood, the CO_2 reduction conversion factor savings from avoiding the use of non-renewable biomass represents an emission reduction of roughly 3.7 tCO_2e per tonne of charcoal produced.
- Total losses (i.e. production facility, charcoal transport and distribution to consumers) do not exceed 5%.

Total CO_2 emission reduced from improved kilns and cookstoves: 27,300 + 25,308 = **52,608 tCO_2** *(rounded 52,600)*

[33] Bailis (2009), *Modeling Climate Change Mitigation from Alternative Methods of Charcoal Production in Kenya*.

Total CO2 emission reduction for this project: 41,300 + 52,600 = 93,900 tCO2 eq.
Consequential post-project (bottom-up): with a replication factor of 3: 3*93,900 = 281,700 tCO2
Consequential post-project (top-down) with replication over the next 10 years of influence of the project: 939,000 tCO2.

Time-frame	Direct project without replication (25-year equipment projected life).	Consequential post-project (top-down) with replication over next 10 years of project influence).	Consequential post-project (bottom-up)
Total CO_2 emissions reduced (tonnes)	93,900	939,000	281,700
Unit abatement cost ($/tonne CO_2)	31	3.1	10